

Micro-to-Macro Gains from Regional Trade Agreements: Evidence from RCEP

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Abstract

We assess how the Regional Comprehensive Economic Partnership (RCEP)—the Asia-Pacific’s largest trade agreement—affects trade and welfare when firms differ in productivity. Using a widely applied firm-level trade framework, we quantify how lower tariffs and easier market entry reshape which firms export, the volume of trade, and the gains for member economies. Most benefits arise because more productive firms start exporting and expand, while less efficient producers shrink or exit. This reallocation toward high-productivity firms raises overall efficiency, with especially strong effects in sectors that face limited import competition. Beyond the familiar gains from higher trade volumes and better international prices, RCEP delivers additional benefits from greater product variety and from productivity improvements driven by firm entry and exit. Malaysia, Vietnam, and Cambodia benefit the most from RCEP, with welfare gains of 22.2%, 5.73%, and 2.37%, respectively, while China, Japan, and South Korea gain relatively less, at 0.37%, 0.50%, and 0.49%, respectively. The magnitude of tariff reductions primarily underlies these welfare changes. The shape parameter of productivity distribution and relative productivity heterogeneity between domestic and foreign firms also have non-negligible impact on welfare outcomes. Our results highlight that ignoring firm dynamics tends to understate the agreement’s welfare effects.

Keywords: RCEP, gains from trade, heterogeneous firms, firm entry and exit, computational general equilibrium

JEL Codes: F12; F13; F14; F17; L11

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1 Introduction

Regional trade agreements (RTAs) have become a dominant force reshaping global trade patterns and production networks. To assess their economic impacts, researchers have often relied on multi-country, multi-sector computational general equilibrium (CGE) models (Kehoe et al., 2017). While these frameworks successfully capture aggregate inter-linkages among countries and sectors, they typically treat firms as homogeneous entities. This simplification obscures the micro-level mechanisms through which trade liberalization affects firm behavior, market structure, and welfare distribution.

The Regional Comprehensive Economic Partnership (RCEP) provides an ideal setting to examine these mechanisms and to deepen understanding of the welfare consequences of modern RTAs. As the world’s largest trade agreement by population and GDP, RCEP unites economies with distinct stages of development and varying industrial structures. Its staggered tariff-reduction schedules and differentiated market access arrangements generate observable variation in trade costs and firm responses both across and within member countries. These features make RCEP a natural policy experiment for empirically investigating how heterogeneous firms adjust to trade liberalization and how such adjustments shape aggregate welfare outcomes. We contribute to the literature by developing a quantitative general equilibrium model featuring firm heterogeneity, building on and extending the Melitz–Chaney framework. The model explicitly integrates firm-level selection, entry, and exit dynamics into a multi-country structure, thereby bridging the gap between theoretical predictions and observed firm behavior. It enables a joint analysis of aggregate trade responses and distributional outcomes across firms of varying productivity levels.

In addition, we revisit the well-known “equivalence result” of Arkolakis, Costinot, and Rodríguez-Clare (2012) (henceforth, ACR), which argues that welfare gains from trade depend solely on the trade elasticity and the change in domestic trade shares, implying that introducing firm heterogeneity yields “not much” new gains from trade. The analysis demonstrates that this equivalence crucially depends on the manner in which tariffs are modeled relative to firm markups. When tariffs are levied on prices inclusive of markups, as is the practical case in most economies, tariff reductions enhance exporter profits, promote additional firm entry, and generate “new” welfare gains arising from variety expansion and productivity reallocation.

To quantify these channels, we extend the Melitz–Chaney model by incorporating firm- and market-specific entry costs and demand shocks following the framework of Eaton, Kortum, and Kramarz (2011) (hereafter, EKK). Using Chinese firm-level data, the structural parameters governing heterogeneity are estimated via the method of simulated moments, as implemented in Eaton, Kortum, and Kramarz (2011). Embedding the estimated model within a general equilibrium system allows us to disentangle how reductions in RCEP tariffs transmit through firm behavior to shape both aggregate trade patterns and welfare outcomes.

We find that Malaysia, Vietnam, and Cambodia benefit the most from RCEP, with welfare gains of 22.2%, 5.73%, and 2.37%, respectively, while China, Japan, and South Korea gain relatively less, at 0.37%, 0.50%, and 0.49%, respectively. We explore the factors that determine the relative gains from RCEP, especially these “new” welfare effects, via thorough decomposition of various channels of welfare effects. We find that the magnitude of tariff reductions primarily underlies welfare changes. Tariff cuts bring about welfare gains through creating more trade volume and varieties to consumers at the expense of average productivity loss. The shape parameter of the productivity distribution amplifies the welfare effects of tariff cuts through two channels: it enlarges the trade elasticity, thereby increasing the volume of trade effect, and it reduces productivity heterogeneity, thereby weakening the average productivity loss. The relative productivity heterogeneity between domestic and foreign firms also moderates the productivity effects of tariff cuts — the greater the heterogeneity of domestic firms relative to foreign ones, the smaller the average productivity loss. Using the quantitative results, we examine the role of these determinants and find evidence supporting the theoretical predictions. We also demonstrate that conventional aggregate analyses likely underestimate the total welfare gains from trade liberalization. Integrating firm-level mechanisms reveals additional channels of adjustment, offering a richer understanding of how RTAs such as RCEP influence competitiveness, variety, and welfare across economies.

Overall, we contribute to three strands of research. First, it enriches the literature on quantitative trade models by providing a multi-country, multi-sector framework that reconciles firm heterogeneity with observed export behavior, offering a flexible alternative to conventional CGE models (Balistreri, Hillberry, and Rutherford, 2011; Crozet and Koenig, 2010; Bergstrand, Cray, and Gervais, 2023). Second, it enhances understanding of the welfare implications of endogenous firm selection by clarifying how the placement of tariffs—before or after markups—determines whether ACR’s equivalence result holds in theory and in quantitative simulations (Arkolakis, Costinot, and Rodríguez-Clare, 2012; Felbermayr, Jung, and Larch, 2015). Third, this study provides new structural estimates of the trade and welfare effects of RCEP, combining microdata-based analysis with macroeconomic general equilibrium, thus connecting firm-level dynamics to policy evaluation (Mahadevan and Nugroho, 2019; Li and Li, 2022; Peng, Fan, Hu, and Yuan, 2024; Wu, Yong, and Lee, 2024).

The paper is structured as follows. Section 2 introduces the policy background for RCEP. In Section 3, we present the quantitative framework used to evaluate the impact of RCEP. Section 4 and 5 present the quantified trade and welfare effects of RCEP respectively. Section 6 discusses the “new” welfare implication of the Melitz-Chaney model. We conclude and provide several recommendations to firms and members in Section 7.

2 Policy Background: The Regional Comprehensive Economic Partnership (RCEP)

The Regional Comprehensive Economic Partnership (RCEP) came into effect in January 2022, becoming the largest regional trade agreement in the world by population and GDP. The agreement brings together the ten ASEAN countries along with China, Japan, South Korea, Australia, and New Zealand, aiming to reduce tariffs, harmonize trade regulations, and promote investment and supply-chain integration. RCEP is therefore viewed as a key institutional development in deepening regional economic linkages and fostering Asia’s economic resilience.

According to the RCEP agreement, each member country is required to reduce tariffs to their final committed levels within a specified timeframe, with tariff concession periods typically spanning 20 to 25 years. Figure 1 summarizes the tariff reduction trajectory of each member over a 20-year horizon. Since approximately 12 bilateral or regional free trade agreements had already been concluded and entered into force within the region prior to the signing of RCEP, initial tariff levels among some economies—such as those between ASEAN members, Australia and New Zealand—were already relatively low, leaving limited scope for further tariff cuts. However, China, Japan and South Korea had not yet established free trade cooperation and RCEP will generate substantial reductions among them. As we can see from Figure 1, once RCEP is fully implemented, China’s tariffs on Japan and South Korea will decline by an average of 6.0 and 3.2 percentage points, respectively, while South Korea’s tariffs on China and Japan will fall by an average of 1.5 and 5.0 percentage points. Additionally, ASEAN countries will enact sizable tariff cuts on China and Japan, averaging 3 percentage points, with these reductions concentrated primarily in Malaysia, Vietnam, and Cambodia¹.

Empirical data already suggest substantial realignment in trade flows since the agreement’s implementation. Figure 2 shows the monthly trade volume of China with its primary partners before and after RCEP entered into force. Between 2021 and 2023, China’s total trade with ASEAN members rose by 11.5% in 2022 and a further 5.5% in 2023 relative to the pre-RCEP year of 2021, whereas trade with the European Union and the United States has grown sluggishly during the same period. This indicates emerging trade creation within the RCEP region alongside mild trade diversion from external partners. The delayed realization of RCEP’s full benefits can be attributed to staggered tariff schedules and ongoing global policy uncertainties. Nevertheless, the agreement’s scope and long-term potential make it a natural policy laboratory for investigating how micro-level firm adjustments generate macroeconomic welfare effects.

While many studies have focused on RCEP’s aggregate trade expansion and welfare effects

¹See Figure D.1 - D.5 in Appendix D for the detailed tariff reduction schedules between ASEAN and other members.

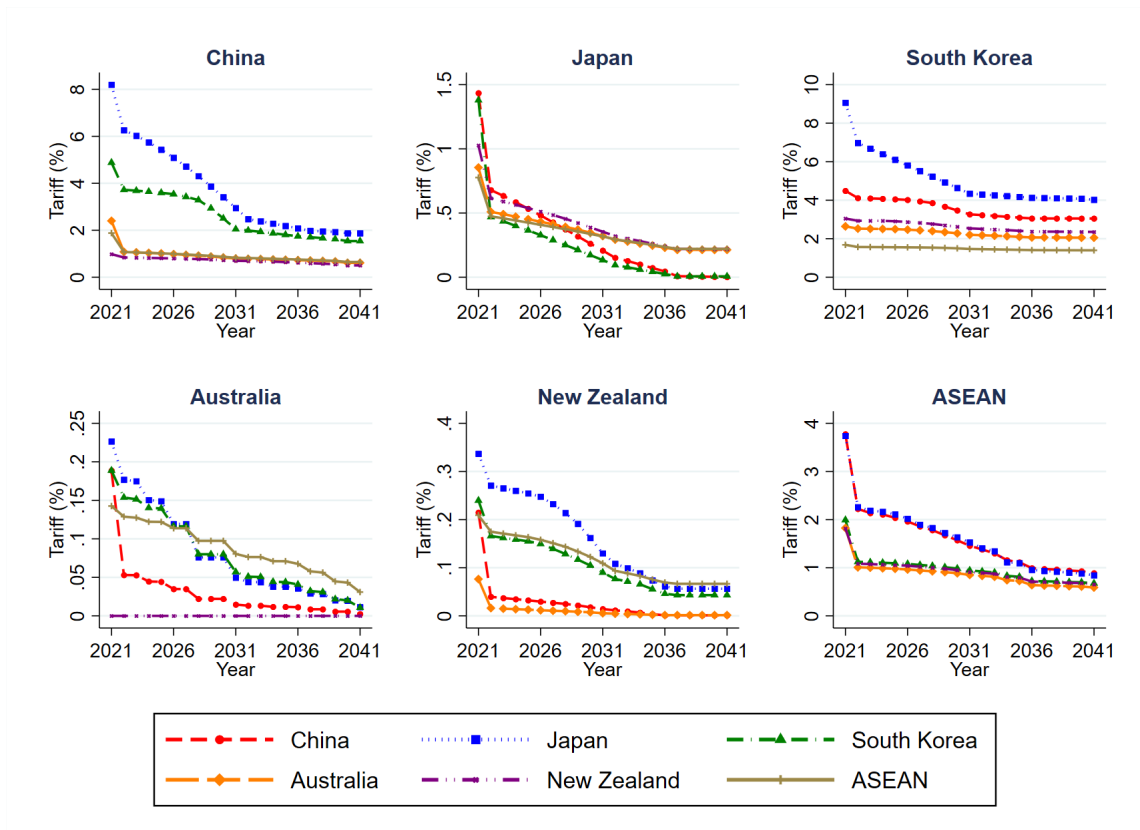


Figure 1 Tariff reduction schedule for RCEP members from 2022 to 2041

Note: The baseline tariffs in 2021 come from Global Tariff Database of Teti (2024) and the commitment levels of tariffs in following 20 years are based on the RCEP agreement text.

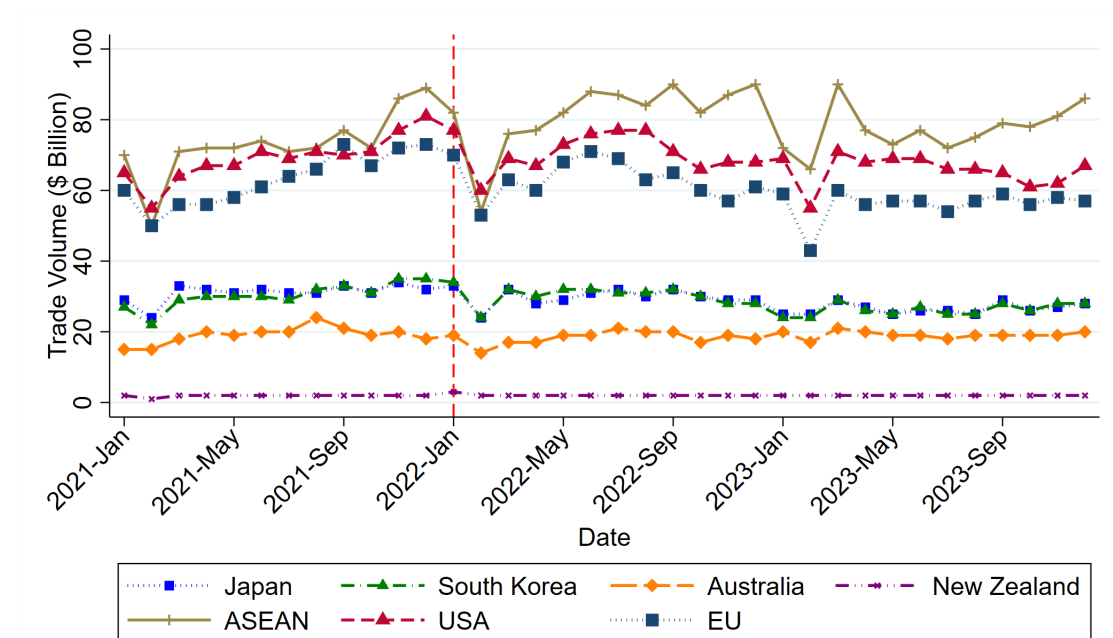


Figure 2 Monthly trade volume of China with primary partners from 2021 to 2023

Note: The trade volume is the sum of China's imports from and exports to partners, with data sourced from General Administration of Customs of China.

(Mahadevan and Nugroho, 2019; Balistreri and Tarr, 2020; Li and Li, 2022; Ding et al., 2022; Jiang and Husin, 2024), fewer have explored its microeconomic consequences. In particular, little attention has been paid to how differential productivity across firms shapes their ability to benefit from tariff reductions and to how these heterogeneous responses influence overall welfare. We address this gap by providing the first structural quantification of RCEP’s firm-level reallocations and “new” welfare effects within an extended Melitz–Chaney framework.

3 Quantitative framework

The quantitative framework builds upon the Melitz–Chaney model proposed by Chaney (2008) and we make the following extensions. Firstly, we formulate market access cost as Arkolakis (2010) and introduce firm and market-specific entry cost and demand heterogeneity following EKK to better capture the firms’ market entry pattern and sales distribution. Secondly, we introduce sectoral heterogeneity in production and input-output linkages to consider the realistic features of global production chains, as Caliendo and Parro (2015) highlighted. Thirdly, our framework incorporates two types of tariffs at the same time, namely tariffs being imposed before or after markups, to explore their implications on the “new” welfare effects.

3.1 The Model

There are N countries and J sectors. We denote countries by i and n and sectors by j and k . Each country is endowed with an exogenous measure L_i of workers or consumers, which is freely mobile across sectors but not mobile across countries. The labor supply is inelastic and the wage in country i is w_i . The intermediate and final goods are both tradable across countries and produced under monopolistic competition market.

Demand The utility of a representative household consuming final goods C_n^j is given by

$$U_n = \prod_{j=1}^J C_n^j \beta_n^j, \text{ where } \sum_{j=1}^J \beta_n^j = 1 \quad (1)$$

where β_n^j is the expenditure share on sector j by consumers in country n .

Following Caliendo and Parro (2015), we define “composite intermediate goods” Q_n^j that are used as materials for the production of intermediate good and as final goods in consumption, which consists of goods ν^j all over the world.

$$Q_n^j = \left[\int_{\Omega_n^j} (\alpha_n(\nu^j) f)^{1/\sigma^j} q_n^j(\nu^j)^{1-1/\sigma^j} d\nu^j \right]^{\sigma^j/(\sigma^j-1)} \quad (2)$$

where Ω_n^j denotes the set of goods around the world available for purchase by country n and $\sigma^j > 1$ is the elasticity of substitution across goods in sector j . The term $\alpha_n(\nu^j)$ reflects exogenous preference shock specific to good ν^j in market n . Depending on supplier’s marketing efforts, any buyer has a chance f of accessing product, thus the f also represents fraction of buyers the supplier reaches. Total demand for good ν^j in country n $q_n^j(\nu^j)$ is given by

$$q_n^j(\nu^j) = \alpha_n(\nu^j) f \left(\frac{p_n^j(\nu^j)}{P_n^j} \right)^{-\sigma^j} \frac{X_n^j}{P_n^j} \quad (3)$$

where X_n^j is total expenditure on sector j for consumption and intermediate goods in country n and P_n^j is the price index which is derived below.

Supply There is a continuum of potential firms in each sector and each firm in sector j produces a distinct good ν^j with productivity $\phi_i^j(\nu^j)$. Production inputs are a Cobb–Douglas combination

of labour and composite intermediate goods from all sectors. The cost of an input bundle is

$$c_i^j = \Upsilon_i^j w_i^{\gamma_i^j} \prod_{k=1}^J P_i^k \gamma_i^{k,j} \quad (4)$$

where Υ_i^j is a constant. The parameter γ_i^j and $\gamma_i^{k,j}$ are respectively the share of value added and intermediate input from sector k used in sector j for production.

All firms face common iceberg trade costs between countries that $\tau_{ni}^j \geq 1$ units of a good in sector j from origin i must be produced in order for one unit to arrive at destination n , with $\tau_{nn}^j = 1$. Besides that, a firm also incurs marketing costs to penetrate a new market. Intuitively, reaching more consumers requires more marketing input such as promoting product, sending advertisement and establishing sales distribution channels. Following [Arkolakis \(2010\)](#), the amount of marketing input required to reach f fraction of consumers is given by

$$M^j(f) = \frac{1 - (1 - f)^{1-1/\lambda^j}}{1 - 1/\lambda^j} \quad (5)$$

where the parameter $\lambda^j > 0$ reflects increasing marginal input of reaching consumers. Larger λ^j implies easier access to consumers and when $\lambda \rightarrow \infty$, $M^j(f) = f$, the marginal input becomes constant and a firm would cover all consumers in destination, which is reduced to entry cost setting in [Melitz \(2003\)](#).

Assuming cost of each bundle of marketing input is specific to pair of origin and destination and is denoted by E_{ni}^j , which is bilateral and reflects destination-market conditions through the wage or other entry cost components. To reach the fraction f of potential consumers in destination n , a firm would incur a fixed marketing cost

$$E_{ni}^j(\nu^j) = \varepsilon_n^j(\nu^j) E_{ni}^j M^j(f) \quad (6)$$

where $\varepsilon_n^j(\nu^j)$ is a marketing cost shock specific to good ν^j in country n .

Tariffs Goods imported by country n from country i have to pay an ad-valorem tariff t_{ni}^j . There are two alternative approaches to modeling tariffs under monopolistic competition market. One is specifying tariffs as being imposed before markups, just as characterized in [Costinot and Rodríguez-Clare \(2014\)](#). Another one is assuming that tariffs are imposed after markups. We choose the latter case as baseline analysis because the tariff base applied in realistic customs practices is contract price negotiated by importer and exporter, which is inclusive of markup. To illustrate differences of these modelling choices and their implications for quantitative results, we consider the two cases in the model at the same time and compare them in section 6.

Firm optimization Conditional on entering market n , a firm producing good ν^j from country i chooses to charge a price p and reach a fraction f of consumers to maximize profit there:²

$$\pi_{ni}^j(p, f; \nu^j) = \left(\frac{p}{1 + t_{ni}^j} - \frac{\tau_{ni}^j c_i^j}{\phi_i^j(\nu^j)} \right) f q_n^j(\nu^j) - \varepsilon_n(\nu^j) E_{ni}^j M^j(f) \quad (7)$$

Optimal price the firm charges is given by

$$p_{ni}^j(\nu^j) = \tilde{\sigma}^j \frac{\tau_{ni}^j (1 + t_{ni}^j) c_i^j}{\phi_i^j(\nu^j)}, \quad \tilde{\sigma}^j = \frac{\sigma^j}{\sigma^j - 1} \quad (8)$$

²The profit function in before-markup tariff case is: $\pi_{ni}^j(p, f; \nu^j) = \left(p - \frac{\tau_{ni}^j (1 + t_{ni}^j) c_i^j}{\phi_i^j(\nu^j)} \right) f q_n^j(\nu^j) - \varepsilon_n(\nu^j) E_{ni}^j M^j(f)$, where the tariff base is production cost of exporter.

and optimal fraction of consumers it reaches is

$$f_{ni}^j(\nu^j) = 1 - \left(\frac{\phi_i^j(\nu^j)}{\bar{\phi}_{ni}^j(\nu^j)} \right)^{\lambda^j(1-\sigma^j)} \quad (9)$$

where $\bar{\phi}_{ni}^j(\nu^j)$ is productivity cutoff for the firm entering market n and is given by

$$\bar{\phi}_{ni}^j(\nu^j) = \left(\frac{\eta_n(\nu^j) X_n^j}{\sigma E_{ni}^j (1 + t_{ni}^j) \zeta} \right)^{-1/(\sigma^j-1)} \frac{\tilde{\sigma}^j \tau_{ni}^j (1 + t_{ni}^j) c_i^j}{P_n^j} \quad (10)$$

where $\eta_n(\nu^j) = \frac{\alpha_n(\nu^j)}{\varepsilon_n(\nu^j)}$ is idiosyncratic entry shock given by ratio of the preference shock to the marketing cost shock. Intuitively, entry into a foreign market is easier for a firm with favorable entry shocks there, such as special preference by consumers due to its product appeal or lower marketing cost resulting from migration relations the firm has established in history. The existence of these shocks explains disordered entry pattern reflected in data that deviates from predictions by Melitz-Chaney model (Eaton et al., 2011).

The parameter ζ is a dummy variable that distinguishes two types of tariff: it is equal to one if tariff is modeled as being imposed after markups and zero if being imposed before. Equation (8) shows that the price charged and quantities sold by a given firm are not affected by the way tariffs are modeled. But the entry cutoff expression (10) presents difference between them. It reflects two possible channels through which tariffs affect market entry. Firstly, tariff reductions decrease import prices consumers have to pay and increase competitiveness of foreign products relative to domestic products, decreasing entry cutoff for foreign firms and attracting them to enter while increasing cutoff for domestic firms and forcing them to exit. We denote this mechanism as competitiveness channel, which operates at both the before and after-markup tariffs case. Secondly, since the after-markup tariffs are imposed based on the sales revenue of traded goods, their reductions would lighten tax burden of foreign firms and increase their profits, further decreasing the cutoff and attracting them to enter, as shown in denominator of the first term in equation (10). We refer the second mechanism existing only in the after-markup tariffs case as tax saving channel. Notably domestic firms who produce non-traded good aren't affected by this channel. As we can see below, it is the tax saving channel that generates net "new" welfare gains.

With equation (3), (8), (9) and (10), sales of a firm producing ν^j in country i to country n can be expressed as

$$X_{ni}^j(\nu^j) = \frac{\alpha_n(\nu^j)}{\eta_n(\nu^j)} \left[1 - \left(\frac{\phi_i^j(\nu^j)}{\bar{\phi}_{ni}^j(\nu^j)} \right)^{-\lambda^j(\sigma^j-1)} \right] \left(\frac{\phi_i^j(\nu^j)}{\bar{\phi}_{ni}^j(\nu^j)} \right)^{\sigma^j-1} \sigma^j E_{ni}^j (1 + t_{ni}^j) \zeta \quad (11)$$

where the term in square bracket governs impact of marketing cost on sales. The less productive firms can only acquire smaller sales due to higher production cost, which in turn impedes marketing investment, contracts consumer fraction and decreases their sales further. In a word, the convex marketing cost erodes sales of the less productive firms disproportionately and accounts for the existence of many small exporters observed in data that Melitz-Chaney model cannot depict (Arkolakis, 2010).

Aggregation Assuming the productivity of producing good ν^j in country n is drawn from a Pareto distribution with dispersion parameter $\theta^j > \sigma^j - 1$ and location parameter T_n^j . With CES demand function, price index for each country and sector is given by:

$$P_n^j = \tilde{\sigma}^j \left\{ \kappa_1^j \sum_{i=1}^N T_i^j \left[\tau_{ni}^j (1 + t_{ni}^j) c_i^j \right]^{-\theta^j} \left[\frac{X_n^j}{\sigma^j E_{ni}^j (1 + t_{ni}^j) \zeta} \right]^{(\tilde{\theta}^j-1)} \right\}^{-1/\theta^j} \quad (12)$$

where $\kappa_1^j = \kappa_0^j \iint \alpha \eta^{\tilde{\theta}^j - 1} g^j(\alpha, \eta) d\alpha d\eta$, $\kappa_0^j = \frac{\tilde{\theta}^j}{\tilde{\theta}^j - 1} - \frac{\tilde{\theta}^j}{\tilde{\theta}^j + (\lambda^j - 1)}$ and $\tilde{\theta}^j = \frac{\theta^j}{\sigma^j - 1}$. $g^j(\alpha, \eta)$ is joint probability density function of preference shock α and entry shock η .

In a similar way, expenditure share by country n on sector j goods from country i is given by

$$\pi_{ni}^j = \frac{T_i^j \left[\tau_{ni}^j (1 + t_{ni}^j) c_i^j \right]^{-\theta^j} \left[E_{ni}^j (1 + t_{ni}^j) \right]^\zeta^{-(\tilde{\theta}^j - 1)}}{\sum_{l=1}^N T_l^j \left[\tau_{nl}^j (1 + t_{nl}^j) c_l^j \right]^{-\theta^j} \left[E_{nl}^j (1 + t_{nl}^j) \right]^\zeta^{-(\tilde{\theta}^j - 1)}} \quad (13)$$

With equation (10), (12) and (13), we can obtain total measure of firms from country i that enters into n :

$$V_{ni}^j = \frac{\kappa_2^j}{\kappa_1^j} \frac{\pi_{ni}^j X_n^j}{\sigma^j E_{ni}^j (1 + t_{ni}^j)^\zeta} \quad (14)$$

where $\kappa_2^j = \int \eta^{\tilde{\theta}^j} g^j(\eta) d\eta$ and $g^j(\eta)$ is the marginal probability density of $g^j(\alpha, \eta)$.

3.2 Model Estimation

Before estimation, we define $u_i^j(\nu^j) = T_i^j \phi_i^j(\nu^j)^{-\theta^j}$ as firm ν^j 's standardized unit cost to isolate productivity heterogeneity. Because productivity levels are drawn from a Pareto distribution, hence the standardized unit cost follows uniform distribution after the transformation. The entry cutoff (10) and sales (11) can thus be reexpressed as

$$\bar{u}_{ni}^j(\nu^j) = \frac{V_{ni}^j}{\kappa_2^j} \eta_m(\nu^j)^{\tilde{\theta}^j} \quad (15)$$

$$X_{ni}^j(\nu^j) = \frac{\alpha_n(\nu^j)}{\eta_n(\nu^j)} \left[1 - \left(\frac{u_i^j(\nu^j)}{\bar{u}_{ni}^j(\nu^j)} \right)^{\lambda^j / \tilde{\theta}^j} \right] \left(\frac{u_i^j(\nu^j)}{\bar{u}_{ni}^j(\nu^j)} \right)^{-1 / \tilde{\theta}^j} \sigma^j E_{ni}^j (1 + t_{ni}^j)^\zeta \quad (16)$$

Then we assume that $g^j(\alpha, \eta)$ is joint log-normally distributed with zero means and standard deviation σ_α^j and σ_η^j and correlation ρ^j . This leaves five parameters to estimate for each sector: $\Theta^j = \{\tilde{\theta}^j, \lambda^j, \sigma_\alpha^j, \sigma_\eta^j, \rho^j\}$. We use Chinese firm-level data in conjunction with a method of simulated moments à la EKK for parameter estimation. The data contains sales revenue of 318,084 Chinese manufacturing firms to each destination (including China) in 2013. As far as we know, this is the most recent firm-level data available before RCEP was active. The model assumes that there is a continuum of firms whose productivity levels, market penetration cost, and country-specific preference and entry shocks follow particular distributions or functional forms, therefore we can simulate sets of artificial firms that operate as the model tells them and search the model parameters that reconcile moments of simulated firms with actual ones.

Simulation Algorithm Given a candidate estimate Θ^j , we simulate the entry and sales behavior for a large number S^j of Chinese firms for each sector. To maintain the stochastic components unchanged as searching over Θ^j , prior to the simulation routine we draw (i) S^j independent realizations of $v^j(s)$ from the standard uniform distribution $U[0, 1]$, and (ii) $S^j \times N^j$ independent realizations of $a_n^j(s)$ and $h_n^j(s)$ from the standard normal distribution $N(0, 1)$, and N^j is the number of destinations where firms in sector j sell.

A given iteration of the model simulation requires a set of candidate parameters Θ^j , the number of Chinese firms selling to each destination N_{nC} , and total sales selling there X_{nC} . An iteration of the simulation proceeds in the following steps.

Step 1. Calculate $\kappa_1^j = \kappa_0^j \iint \alpha \eta^{\tilde{\theta}^j - 1} g^j(\alpha, \eta) d\alpha d\eta$ and $\kappa_2^j = \int \eta^{\tilde{\theta}^j} g^j(\eta) d\eta$. Then use (14) to calculate $\sigma^j E_{ni}^j (1 + t_{ni}^j)^\zeta$.

Step 2. Use the $a_n^j(s)$ and $h_n^j(s)$ to construct $S^j \times N^j$ realizations for preference and entry

shocks $\ln \alpha_n^j(s)$ and $\ln \eta_n^j(s)$ as

$$\begin{bmatrix} \ln \alpha_n^j(s) \\ \ln \eta_n^j(s) \end{bmatrix} = \begin{bmatrix} \sigma_\alpha^j \sqrt{1 - \rho^{j2}} & \sigma_\alpha^j \rho^j \\ 0 & \sigma_\eta^j \end{bmatrix} \begin{bmatrix} a_n^j(s) \\ h_n^j(s) \end{bmatrix} \quad (17)$$

Step 3. Use (15) to construct the $S^j \times N^j$ entry hurdles

$$\bar{u}_{nC}^j(s) = \frac{N_{nC}^j}{\kappa_2^j} \eta_n^j(s)^{\tilde{\theta}^j} \quad (18)$$

Step 4. Calculate the importance weight for each simulated firm

$$\bar{u}^j(s) = \min \left\{ \bar{u}_C^j(s), \max_{n \neq C} \{ \bar{u}_n^j(s) \} \right\} \quad (19)$$

Here $\max_{n \neq C} \{ \bar{u}_n^j(s) \}$ is the foreign market entry cutoff and $\bar{u}^j(s)$ ensures that the simulated exporters can sell in domestic market. Then use $\bar{u}^j(s)$ to construct the realizations of standardized unit cost

$$u^j(s) = v^j(s) \bar{u}^j(s) \quad (20)$$

Step 5. Calculate $\delta_{nC}^j(s)$, which indicates whether the simulated firm s enters market n , as determined by the entry hurdles

$$\delta_{nC}^j(s) = \begin{cases} 1, & \text{if } u^j(s) \leq \bar{u}_n^j(s) \\ 0, & \text{otherwise.} \end{cases} \quad (21)$$

Wherever $\delta_{nC}^j(s) = 1$, calculate sales as

$$X_{nC}^j(s) = \frac{\alpha_n^j(s)}{\eta_n^j(s)} \left[1 - \left(\frac{u^j(s)}{\bar{u}_{nC}^j(s)} \right)^{\lambda^j / \tilde{\theta}^j} \right] \left(\frac{u^j(s)}{\bar{u}_{nC}^j(s)} \right)^{-1 / \tilde{\theta}^j} \sigma^j E_{ni}^j (1 + t_{ni}^j)^\zeta \quad (22)$$

Moments The simulation procedure gives us the entry and sales behavior of each artificial Chinese exporters and next we use these simulated exporters to construct four sets of moments.

The first set of moments is the number of firms shipping to any possible permutation of the seven most popular export destinations. For example, the permutation USA-JPN means selling to USA and Japan but no others; USA-JPN-KOR means selling to USA, Japan and South Korea but no others, and so forth. These moments are informative about the dispersion of entry shocks.

The second set of moments is the share of artificial exporters in each destination with sales there falling into four percentile intervals of the destination sales in the data, that is, $[0, 50]$, $[50, 75]$, $[75, 95]$ and $[95, 100]$. These moments can help pin down the shape parameter of the Pareto distribution and the market penetration function, $\tilde{\theta}^j$ and λ^j respectively. As the entry and preference shock influence sales revenues, the information from these moments can also help identify the dispersion of these shocks.

The third set of moments is the share of artificial exporters in each destination with domestic sales falling into four percentile intervals of the domestic sales in the data, that is, $[0, 50]$, $[50, 75]$, $[75, 95]$ and $[95, 100]$. These moments can also help identify the shape parameter of the Pareto distribution and the market penetration function and the dispersion of entry and preference shocks.

The last set of moments is the share of artificial exporters with the ratio between sales in each destination and sales in China falling into three percentile intervals, that is, $[0, 50]$, $[50, 75]$ and $[75, 100]$. These moments can help identify the dispersion of entry and preference shocks.

Inference We denote the difference between the moments in the data and in the simulated model by $\hat{y}^j(\Theta^j)$ and $E[\hat{y}(\Theta_0^j)] = 0$ is assumed to hold at the true parameter value. The method of

simulated moments selects the model parameters that minimize the following objective function:

$$\hat{\Theta}^j = \arg \min_{\Theta^j} \{y(\Theta^j)' \mathbf{W} y(\Theta^j)\} \quad (23)$$

where $\mathbf{W}$ is a weighting matrix that is the generalized inverse of the estimated variance-covariance matrix of the moments calculated from the data m^j . We compute standard errors using a bootstrap method that repeats the estimation process 25 times.

Estimation Results We conduct the above estimation procedure sector by sector and obtain the estimated parameters and bootstrapped standard errors as displayed in Table 1.

The estimates for $\tilde{\theta}^j$ have the correct sign and the magnitude of them varies across sectors, ranging from 3.41 to 4.48. From equations (16), the fundamental force determining entry and sales is the productivity term of a firm $\left(\frac{u_i^j(\nu^j)}{\bar{u}_{ni}^j(\nu^j)}\right)^{-1/\tilde{\theta}^j}$. The estimates for $\tilde{\theta}^j$ imply that, conditional on entry and preference shocks, the ratios of the 75th to the 25th percentile of the term range from 1.28 to 1.38.

The estimates of the variance of preference shocks σ_α^j range from 1.65 to 2.22 and imply that, conditional on entry shocks and firm productivity, the ratio of the 75th to the 25th percentile of the preference shock range from 9.12 to 19.58. These enormous variations are in stark contrast to the implication of the estimates of the variance of entry shocks η^j , which imply that the ratios of the 75th to the 25th percentile of the entry shock range from 1.27 to 1.54.

The estimates for λ^j are also consistent with the theoretical prediction, ranging from 4.71 to 6.17, and imply that an additional consumer has more than proportionally higher unit marketing costs than any infra-marginal consumers.

Note that EKK found that $\tilde{\theta}$ equals 2.46, which is lower than the values here. And the value of λ was around 1 and the covariance ρ^j was negative in their estimates. The difference between these estimates reflects the difference in the magnitude of underlying economic mechanism between countries and periods, which they use the data of French firms in 1986.

Model Fit To evaluate the model's fit of the data, this section plots simulated data using the estimated parameters alongside the actual data. Figure 3a depicts the close relationship between actual and simulated number of exporters by different destination permutations. The disordered entry pattern is closely captured by the entry shock η . Figure 3b plots the export sales distribution of the simulated and actual exporters. The model also picks up the downward bias in left tail of the distribution, reflecting the role of λ . The correlation coefficients between the actual and simulated data in Figure 3a and 3b are 0.987 and 0.997, respectively, which implies that the model captures very well both the entry pattern and sales distribution of exporters.

3.3 General Equilibrium and Decomposition of Welfare Effects

Market Clearing Conditions Goods market clearing requires equivalence between total revenue of good and total expenditure of all destinations on it. Then, the goods market clearing condition is given by

$$R_n^j = \sum_{i=1}^N \left(1 + \frac{t_{in}^j}{\sigma^j}\right)^{(1-\zeta)} \frac{\pi_{in}^j}{1 + t_{in}^j} X_i^j \quad (24)$$

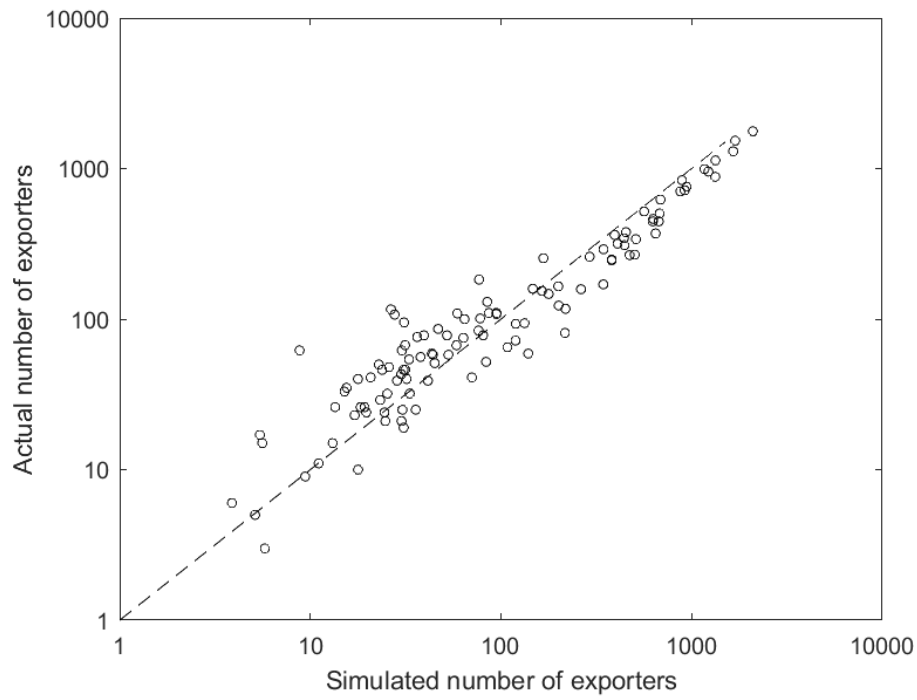
for all sectors j and countries n , where R_n^j denotes the total revenue of sector j in country n . Total expenditure is sum of expenditure for final consumption and production. Then, X_n^j is given by

$$X_n^j = \beta_n^j (w_n L_n + T_n + D_n) + \sum_{k=1}^J \gamma_n^{j,k} R_n^k \quad (25)$$

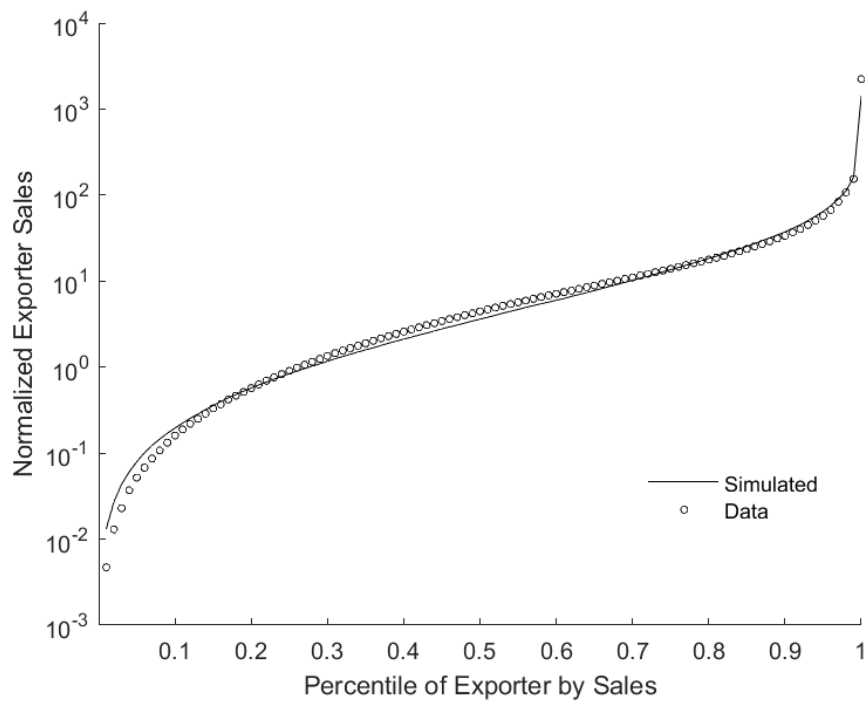
where $T_n = \sum_{i=1}^N \sum_{j=1}^J \left(\frac{\sigma^j - 1}{\sigma^j}\right)^{\delta(1-\zeta)} \frac{t_{ni}^j}{1 + t_{ni}^j} \pi_{ni}^j X_n^j$ denotes total tariff revenues and D_n denotes exogenous trade deficit.

Table 1 Estimated parameters

Sector	θ^j	s.e.	σ_a^j	s.e.	σ_η^j	s.e.	ρ^j	s.e.	λ^j	s.e.
Food	4.16	(0.40)	1.74	(0.03)	0.29	(0.04)	0.83	(0.10)	5.32	(2.02)
Textile	4.19	(0.43)	1.71	(0.04)	0.28	(0.04)	0.57	(0.10)	5.16	(2.01)
Wood	4.31	(0.39)	1.68	(0.03)	0.25	(0.03)	0.55	(0.12)	6.17	(2.09)
Paper	4.40	(0.40)	1.76	(0.03)	0.21	(0.03)	0.75	(0.17)	5.25	(1.73)
Chemicals	4.27	(0.43)	1.81	(0.02)	0.20	(0.03)	0.42	(0.15)	5.17	(2.00)
Plastic	4.10	(0.51)	1.65	(0.03)	0.26	(0.05)	0.66	(0.10)	5.52	(1.63)
Minerals	4.48	(0.27)	1.68	(0.02)	0.20	(0.01)	0.70	(0.18)	4.71	(2.12)
Basic metals	3.55	(0.46)	2.05	(0.04)	0.31	(0.04)	0.40	(0.13)	5.00	(2.42)
Metal products	3.99	(0.46)	1.73	(0.03)	0.25	(0.07)	0.74	(0.11)	5.04	(1.84)
Electronics	3.41	(0.48)	2.22	(0.02)	0.27	(0.04)	0.06	(0.10)	5.96	(2.22)
Electrical	3.74	(0.45)	1.84	(0.04)	0.27	(0.05)	0.59	(0.10)	5.24	(2.08)
Machinery n.e.c	4.27	(0.46)	1.73	(0.03)	0.26	(0.07)	0.51	(0.10)	5.34	(2.22)
Auto	3.96	(0.50)	2.12	(0.03)	0.29	(0.06)	0.56	(0.11)	5.16	(2.30)
Other Transport	3.87	(0.59)	2.03	(0.03)	0.32	(0.08)	0.47	(0.08)	5.36	(2.03)
Other	3.96	(0.45)	1.70	(0.03)	0.18	(0.03)	0.99	(0.02)	5.79	(1.50)



(a) Number of exporters by destination permutations



(b) Export sales distribution

Figure 3 Model fit

Note: The normalized exporter sales are calculated by the ratio of actual sales to average sales of exporters

Factor market clearing condition is given by equivalence between labor supply and demand:

$$w_n L_n = \sum_{k=1}^J \gamma_n^k R_n^k \quad (26)$$

Equilibrium Using Exact Hat Algebra General equilibrium system is given by equation (4), (12), (13), (24), (25) and (26). To quantify consequences of counterfactual changes of tariffs, this section uses exact hat algebra. We denote the counterfactual value of any variable x as x' and define $\hat{x} = x'/x$ as its relative change. The “hat” counterparts of the above general equilibrium system are given by

$$\hat{c}_n^j = \hat{w}_n^{\gamma_n^j} \prod_{k=1}^J \hat{P}_n^k \gamma_n^{k,j} \quad (27)$$

$$\hat{P}_n^j = \left\{ \sum_{i=1}^N \pi_{ni}^j [\hat{\tau}_{ni}^j (1 + \hat{t}_{ni}^j) \hat{c}_i^j]^{-\theta^j} \left[\frac{\hat{X}_n^j}{\hat{w}_n \hat{F}_{ni}^j (1 + \hat{t}_{ni}^j) \zeta} \right]^{\delta(\tilde{\theta}^j - 1)} \right\}^{-1/\theta^j} \quad (28)$$

$$\hat{\pi}_{ni}^j = \frac{\pi_{ni}^j [\hat{\tau}_{ni}^j (1 + \hat{t}_{ni}^j) \hat{c}_i^j]^{-\theta^j} [\hat{F}_{ni}^j (1 + \hat{t}_{ni}^j) \zeta]^{-\delta(\tilde{\theta}^j - 1)}}{\sum_{l=1}^N \pi_{nl}^j [\hat{\tau}_{nl}^j (1 + \hat{t}_{nl}^j) \hat{c}_l^j]^{-\theta^j} [\hat{F}_{nl}^j (1 + \hat{t}_{nl}^j) \zeta]^{-\delta(\tilde{\theta}^j - 1)}} \quad (29)$$

$$R_n^{j'} = \sum_{i=1}^N \left(1 + \frac{t_{in}^{j'}}{\sigma^j} \right)^{\delta(1-\zeta)} \frac{\pi_{in}^{j'}}{1 + t_{in}^{j'}} X_i^{j'} \quad (30)$$

$$X_n^{j'} = \beta_n^j (\hat{w}_n w_n L_n + T_n' + D_n) + \sum_{k=1}^J \gamma_n^{j,k} R_n^{k'} \quad (31)$$

$$\hat{w}_n w_n L_n = \sum_{k=1}^J \gamma_n^k R_n^{k'} \quad (32)$$

where $\delta = 0$ denotes the equilibrium system of the Ricardian model which [Caliendo and Parro \(2015\)](#) builds upon and $\delta = 1$ is the equilibrium system of the Melitz-Chaney models. Moreover, we can compare the implication of the after-markup tariff and before-markup tariff via setting $\zeta = 1$ or $\zeta = 0$, respectively. Besides that, we assume that the market entry cost is paid in destination hence $E_{ni}^j = w_n F_{ni}^j$, where F_{ni}^j is labour required for entry from i to n .

Data and Calibration The initial tariffs t_{ni}^j are actually applied tariffs of 2021 from Global Tariff Database of [Teti \(2024\)](#) and counterfactual tariffs $t_{ni}^{j'}$ are the tariff levels 20 years after the entry into force of the RCEP, extracted from the schedule of tariff commitments in the agreement text. As [Teti \(2024\)](#) documents, standard tariff databases such as WITS report tariffs only for product-country pairs with positive trade flows, discarding zero-trade observations that are often the result of prohibitively high tariffs. Moreover, trade-weighted averages are endogenous: high tariffs suppress trade volumes, so they receive lower weights and are systematically understated in aggregate measures. We therefore use simple average of HS6-digit product level effectively applied tariffs, thus avoiding this positive selection bias and ensuring that high-tariff entries are fully captured in the baseline. As we can see from Figure D.6 - D.8, [Teti \(2024\)](#) keeps high initial tariffs in a few sectors of Malaysia, Vietnam, and Cambodia, which is expected to generate substantial welfare gains for them. We also conduct robustness checks using trade-weighted average tariffs and 95% winsorized tariffs from the [Teti \(2024\)](#) database. The data counterparts of π_{ni}^j , γ_n^j , $\gamma_n^{k,j}$, β_n^j , $w_n L_n$, D_n and R_n^j are computed via input-output table of 2021 from OECD Input-Output Database(2025 edition). The sectoral heterogeneity parameters $\tilde{\theta}^j$ are estimated in Table 1. Following [Bernard et al. \(2003\)](#), we set σ^j equal to 3.79. The shape parameter of Pareto

distribution θ^j can thus be calculated using $\tilde{\theta}^j = \frac{\theta^j}{\sigma^j - 1}$. The equilibrium system can be solved via iterative algorithm in Matlab.

Firm-level Response to Tariff Reductions With the counterfactual changes $\hat{w}_n$, $\hat{P}_n^j$, $\hat{\pi}_{ni}^j$ and $\hat{X}_n^j$, we can calculate changes in real Chinese export value $\hat{X}_{nC}^j = \frac{\hat{\pi}_{nC}^j \hat{X}_n^j}{\hat{P}_C^j}$ and changes in number of Chinese firms $\hat{N}_{nC}^j = \frac{\hat{\pi}_{nC}^j \hat{X}_n^j}{\hat{w}_n \hat{F}_{nC}(1+t_{nC}^j)^\zeta}$ in each destination. Then we can acquire the counterfactual firm-level behavior by running the simulation algorithm in section 3.2 with the estimated parameters and the same stochastic draws as in the baseline simulation. Specifically,

(a) use baseline X_{nC}^j , N_{nC}^j and counterfactual $\hat{X}_{nC}^j X_{nC}^j$, $\hat{N}_{nC}^j N_{nC}^j$ to calculate baseline $\sigma^j E_{ni}^j (1+t_{ni}^j)$, $\bar{u}_{nC}^j(s)$ and counterfactual $\sigma^j E_{ni}^{j'} (1+t_{ni}^{j'})$, $\bar{u}_{nC}^{j'}(s)$ respectively;

(b) set $\bar{u}^j(s) = \max_n \left\{ \bar{u}_n^j(s), \bar{u}_n^{j'}(s) \right\}$. A firm with $u^j(s) \leq \bar{u}_n^j(s)$ can sell in market n in the baseline, while a firm with $u^j(s) \leq \bar{u}_n^{j'}(s)$ can sell there in the counterfactual. Hence the simulation allows for entry, exit and survival;

(c) use equation (21) and (22) to calculate the entry and sales in each destination in the baseline and counterfactual.

Welfare Effects from Tariff Reductions We define welfare of country n as its real income, that is, $W_n = I_n/P_n$, where $I_n = w_n L_n + T_n + D_n$ is total income of country n and P_n is aggregated sectoral price index (12). This section decomposes the welfare effects from tariff changes into four terms of sufficient statistics as follows

$$\begin{aligned}
d \ln W_n = & \frac{1}{I_n} \sum_{j=1}^J \sum_{i=1}^N \underbrace{\left[\left(1 + \frac{t_{in}^j}{\sigma^j} \right)^{\delta(1-\zeta)} E_{ni}^j d \ln c_n^j - M_{ni}^j d \ln c_i^j \right]}_{\text{Terms of trade}} \\
& + \frac{1}{I_n} \sum_{j=1}^J \sum_{i=1}^N \underbrace{t_{ni}^j M_{ni}^j \left[\left(\frac{\sigma^j - 1}{\sigma^j} \right)^{\delta(1-\zeta)} d \ln M_{ni}^j - d \ln c_i^j - \frac{\delta(1-\zeta)}{\sigma^j} d \ln t_{ni}^j \right]}_{\text{Volume of trade}} \\
& + \frac{\delta}{I_n} \sum_{j=1}^J \sum_{i=1}^N \underbrace{\left[\frac{(1+t_{ni}^j) M_{ni}^j}{\sigma^j - 1} d \ln V_{ni}^j \right]}_{\text{Variety}} + \frac{\delta}{I_n} \sum_{j=1}^J \sum_{i=1}^N \underbrace{\left[(1+t_{ni}^j) M_{ni}^j d \ln \bar{\phi}_{ni}^j \right]}_{\text{Productivity}}
\end{aligned} \tag{33}$$

where $E_{ni}^j = \frac{\pi_{in}^j}{1+t_{in}^j} X_i^j$ is country n 's exports of sector j to country i , $M_{ni}^j = \frac{\pi_{ni}^j}{1+t_{ni}^j} X_n^j$ is country n 's imports of sector j from country i . The V_{ni}^j is the number of firms from country i selling to country n , which is given by equation (14). The $\bar{\phi}_{ni}^j$ is the average productivity level of these firms and can be expressed as $\left(\frac{\tilde{\theta}^j}{\sigma^j - 1} \right)^{\frac{1}{\sigma^j - 1}} \int \bar{\phi}_{ni}^j(\eta) g^j(\eta) d\eta$, where $\bar{\phi}_{ni}^j(\eta)$ is given by the cutoff expression (10).

The terms of trade effect reflects the welfare changes due to changes in exporter prices relative to importer prices and the volume of trade effect describes the welfare increase from trade creation as a result of tariff reductions. The first two terms are identical with welfare expression in [Caliendo and Parro \(2015\)](#). Since they are derived from the classical Ricardian model without firm-level adjustment, we define them as “traditional” welfare.

Notably, the expressions of the “traditional” welfare are identical between the Ricardian model and the Melitz-Chaney model with tariff being imposed after markup, but in the before-markup tariff case there are slight differences. The export value E_{ni}^j in the terms of trade effect should be adjusted by $\left(1 + \frac{t_{in}^j}{\sigma^j} \right)$ to reflect the real revenue firms earn in the export market. The volume of trade effect should also consider the impact of the markups on the change of tariff revenues.

Overall, the impact of tariff modelling on the “traditional” welfare depends on the quantitative results but we argue that the “new” welfare effects are the first-order difference between the two types of tariffs.

The variety effect captures welfare changes due to changes in range of products that consumers access to. More kinds of products are consumed the larger is the variety effect. Since firms produce differentiated goods under monopolistic competition market, we measure the number of varieties of products as the number of firms selling in each country V_{ni}^j ³.

The productivity effect represents the price changes due to firm heterogeneity since the firms producing differentiated varieties are also heterogeneous in productivity. This effect is measured as the average productivity level of firms serving one market $\tilde{\phi}_{ni}^j$, which is determined by the productivity cutoff. The higher the cutoff is, the more productive the firms serving that market are and the larger is the productivity effect.

The variety and productivity effects contribute to new channels for gains from trade that cannot be examined by classical trade theories of comparative advantage but have been highlighted in the new generation of trade theories characterized by monopolistic competition and heterogeneous firms; see [Krugman \(1980\)](#) and [Melitz \(2003\)](#). Hence we define these effects as “new” welfare. On the empirical side, [Broda and Weinstein \(2006\)](#) calculates that the increase in varieties of imported products by US between 1972 and 2001 resulted in 2.6 percent decrease in CPI. [Trefler \(2004\)](#) finds that the Canada-U.S. Free Trade Agreement raised Canadian manufacturing aggregate productivity through forcing the domestic least productive plants to exit.

But in a general equilibrium environment, tariff reductions induce turnover of different types of firms at the same time. Entry of foreign exporters brings about import variety gains but exit of domestic firms causes domestic variety losses. Similarly, domestic productivity gains as a result of exit of domestic least productive firms also accompanied by import productivity losses since new foreign exporter has a lower productivity than existing ones. To capture the opposite variety and productivity effects brought by entry and exit of heterogeneous firms, we further decompose the “new” welfare as follows

$$\begin{aligned}
d \ln W_n^{NEW} = & \overbrace{\frac{1}{I_n} \sum_{j=1}^J \left[\frac{(1+t_{nn}^j)M_{nn}^j}{\sigma^j-1} d \ln V_{nn}^j \right]}^{\text{Domestic exit}} + \overbrace{\frac{1}{I_n} \sum_{j=1}^J \sum_{i \neq n}^N \left[\frac{(1+t_{ni}^j)M_{ni}^j}{\sigma^j-1} d \ln V_{ni}^j \right]}^{\text{Foreign entry}} \\
& + \overbrace{\frac{1}{I_n} \sum_{j=1}^J \left[(1+t_{nn}^j)M_{nn}^j d \ln \tilde{\phi}_{nn}^j \right]}^{\text{Domestic exit}} + \overbrace{\frac{1}{I_n} \sum_{j=1}^J \sum_{i \neq n}^N \left[(1+t_{ni}^j)M_{ni}^j d \ln \tilde{\phi}_{ni}^j \right]}^{\text{Foreign entry}}
\end{aligned} \tag{34}$$

3.4 Determinants of Welfare Effects

To open the “black box” of these welfare effects and investigate factors that determine the relative welfare changes across countries, we first define welfare effects by country and sector and further decompose them term by term⁴. The detailed derivation of the decomposition is provided in [Appendix A](#).

In the baseline analysis of monopolistic competition environment and that tariffs are imposed after markup, that is, $\delta = 1$ and $\zeta = 1$, we define changes in terms of trade effect caused by tariff reductions by country and sector as:

$$d \ln tot_n^j = \sum_{i=1}^N \left[\frac{\pi_{in}^j X_i^j}{(1+t_{in}^j) X_n^j} d \ln c_n^j - \frac{\pi_{ni}^j}{(1+t_{ni}^j)} d \ln c_i^j \right] \tag{35}$$

³This equivalence relies on the single-product firm assumption in our analysis. The quantitative case of multi-product firms can be seen in [Arkolakis et al. \(2021\)](#)

⁴We thank the anonymous referee and the editor for reminding us to explore the determinants of welfare effects, in order to enhance transparency and interpretability of the results.

whereas changes in the volume of trade effect by country and sector is:

$$d \ln vot_n^j = \sum_{i=1}^N \left[\frac{t_{ni}^j \pi_{ni}^j}{(1+t_{ni}^j)} (d \ln M_{ni}^j - d \ln c_i^j) \right] \quad (36)$$

Similarly, changes in the variety effect by country and sector is:

$$d \ln var_n^j = \frac{1}{\sigma^j - 1} \sum_{i=1}^N \pi_{ni}^j d \ln V_{ni}^j \quad (37)$$

and changes in the productivity effect by country and sector is:

$$d \ln prod_n^j = \sum_{i=1}^N \pi_{ni}^j d \ln \tilde{\phi}_{ni}^j \quad (38)$$

Terms of Trade The terms of trade effect depends on changes in cost of a country's exports relative to that of its imports, as shown in equation (35). According to the cost function (4), production costs are determined by wages and intermediate goods prices. The relative change in wages hinges on relative magnitude of tariff reductions and relative size of markets across countries. As shown by Bown et al. (2024), in a two-country Eaton-Kortum model without intermediate goods, if country A commits to larger tariff reductions on country B than those B grants to A, country A's terms of trade will deteriorate while B's will improve. Relative market size also influences the terms of trade: even with symmetric tariff cuts, in a trade agreement between a large and a small country, the large country experiences a deterioration in its terms of trade while the small country enjoys an improvement. In our model with intermediate goods, the terms of trade effect is also shaped by relative changes in intermediate goods prices. However, since the magnitude and direction of price index changes differ across sectors, the relative wage change continues to dominate the relative cost changes upon aggregation.

Volume of Trade We decompose the volume of trade effect into two components: changes in import tariffs and other general equilibrium factors. The decomposition is given by:

$$d \ln vot_n^j = -\tilde{\sigma}^j \theta^j \sum_{i=1}^N \left[\frac{t_{ni}^j}{(1+t_{ni}^j)} \pi_{ni}^j d \ln (1+t_{ni}^j) \right] + \epsilon_n^j \quad (39)$$

where ϵ_n^j is residual term that captures the general equilibrium factors such as changes in production cost, price index and market size⁵.

Clearly, import tariff reductions are the first-order determinant of the volume of trade effect. Assuming that the degrees of tariff reduction across origins are uniform, that is, $\forall i \neq n, d \ln(1+t_{ni}^j) = d \ln(1+t_n^j)$, then the derivative of volume of trade effect with respect to tariff change can be expressed as

$$\frac{d \ln \underbrace{vot_n^j}_{(1+t_n^j)}}{d \ln (1+t_n^j)} = - \sum_{i \neq n}^N \frac{t_{ni}^j \pi_{ni}^j}{1+t_{ni}^j} \tilde{\sigma}^j \theta^j. \quad (40)$$

The sign of derivative is negative, indicating that as import tariffs fall, consumers gain access to cheaper foreign goods, import volumes rise, and welfare improves. We can also find that the derivative is determined by shape parameter and initial tariffs, indicating that the volume of trade

⁵Detailed derivation can be found in Appendix A.

$$\epsilon_n^j = \sum_{i=1}^N \left\{ \frac{t_{ni}^j \pi_{ni}^j}{(1+t_{ni}^j)} \left[-(\theta^j + 1) d \ln c_i^j + d \ln X_n^j - (\tilde{\theta}^j - 1) d \ln (w_n F_{ni}^j) - d \ln \Phi_n^j \right] \right\}$$

where $\Phi_n^j = \sum_{l=1}^N T_l^j \kappa_1^j \left[\tilde{\sigma}^j \tau_{nl}^j (1+t_{nl}^j) c_l^j \right]^{-\theta^j} \left[\sigma^j E_{nl}^j (1+t_{nl}^j)^\zeta \right]^{-(\tilde{\theta}^j - 1)}$.

effect of tariff reductions is moderated by them: for a given tariff cut, a larger shape parameter or higher initial tariffs imply a larger volume of trade effect. The reason is that, in the Melitz-Chaney model, trade elasticity depends directly on the shape parameter of the productivity distribution. A larger shape parameter implies lower firm heterogeneity and more concentrated exporters, thereby amplifying the volume of trade effect. As for the initial tariffs, the volume of trade effect reflects increase of tariff revenue due to more trade value. In this respect, a country with higher initial tariffs can reap larger tariff revenue when trade is created, thus benefiting more from the volume of trade effect. In a word, the shape parameter of productivity distribution and initial tariff level both enhance the volume of trade effect of tariff reductions.

Variety We decompose the variety effect by country and sector via aggregating changes in imported variety across origins. The decomposition is as follows:

$$d \ln var_n^j = \frac{1}{\sigma^j - 1} \left[- \sum_{i=1}^N \pi_{ni}^j d \ln (1 + t_{ni}^j) + d \ln \frac{X_n^j}{w_n} \right]^6 \quad (41)$$

It can be seen that tariff reductions are also the primary determinant of the variety effect. Conditional on the changes in market size and wage, the derivative of variety effect with respect to tariff reductions is

$$\frac{d \ln var_n^j}{d \ln (1 + t_n^j)} = - \frac{1 - \pi_{nn}^j}{\sigma^j - 1} \quad (42)$$

The sign of the derivative is negative. The intuition is that when tariffs are imposed after markups, tariff reductions are isomorphic to a decline in entry costs: number of foreign entrants exceeds number of exiting domestic firms, yielding a positive net variety effect. Notice that the parameter of elasticity of substitution is identical across all sectors and countries in our model ($\sigma^j = 3.79$), so there exists no moderating role of it in the results. In addition, the variety effect is also influenced by changes in the importing country's market size and wage level. A larger market can accommodate more firms; a higher wage, however, raises entry costs and reduces the number of entering firms. In general, market size and wages tend to move in the same direction, so their net impact on the variety effect is relatively small.

Productivity We decompose the productivity effect by country and sector via aggregating changes in average productivity of imported goods across origins, which is given by :

$$d \ln prod_n^j = \frac{1}{\theta^j} \left[\sum_{i=1}^N \pi_{ni}^j d \ln (1 + t_{ni}^j) - d \ln \frac{X_n^j}{w_n} \right] \quad (43)$$

Interestingly, the determinants of the productivity effect operate in the opposite direction to those of the variety effect. Conditional on the changes in market size and wage, the derivative of productivity effect with respect to tariff reductions is given by

$$\frac{d \ln prod_n^j}{d \ln (1 + t_n^j)} = \frac{1 - \pi_{nn}^j}{\theta^j} \quad (44)$$

The sign of the derivative is positive, indicating that tariff reductions lower average productivity of firms entering a country, though bring about positive variety effects. Intuitively, import tariff reductions induce exit of domestic firms and entry of foreign firms. Since the productivity levels of both marginally entering and exiting firms lie below incumbents', the exit of domestic firms raises average productivity—as predicted by Melitz (2003)—while the entry of foreign firms lowers it. These two opposing forces jointly determine the direction of the change in average productivity. Because the number of foreign entrants exceeds the number of domestic exits, entry of these extra

⁶In the baseline analysis we set the $\hat{F}_{ni}^j = 1$, so the change in entry cost only comes from change in wage, that is $\hat{E}_{ni}^j = \hat{w}_n$.

less productive foreign firms thus leads to a decline in average productivity. In addition, the shape parameter of the productivity distribution also influences this effect. A larger shape parameter implies lower productivity heterogeneity and, consequently, a smaller impact of firm entry on the net productivity loss.

In summary, the welfare effects of tariff reductions are shaped by a combination of factors that influence terms of trade, volume of trade, variety, and productivity, which can be summarized as following propositions:

Proposition 1 (*Magnitude of tariff reductions*) *The volume of trade effect and the variety effect are increasing in the magnitude of tariff reductions, while the productivity effect declines with tariff reductions.*

Proposition 2 (*Shape parameter of productivity distribution*) *The shape parameter of the productivity distribution moderates the volume of trade effect and the productivity effect of tariff reductions: a larger shape parameter amplifies the volume of trade effect while attenuating the productivity-loss effect, thereby enhancing the welfare effects of tariff cuts.*

Proposition 3 (*Initial tariff*) *The level of initial tariff also moderates the volume of trade effect of tariff reductions, that is, higher initial tariff amplifies the volume of trade effect.*

Proposition 4 (*Relative magnitude of tariff reductions*) *The relative magnitude of tariff reductions directly shapes the terms of trade effect. The larger the tariff cuts a country receives on its exports relative to those it grants on its imports, the greater the improvement in its terms of trade.*

Proposition 5 (*Relative market size*) *Relative market size moderates the terms of trade effect of tariff reductions: for a given relative tariff cut, the larger a country's market size relative to its trading partners, the more limited the improvement in its terms of trade.*

3.5 Extension: Differentiating Shape Parameters of Productivity Distribution by Country

RCEP covers various countries, including both least developed and developed countries whose productivity levels are largely different. As discussed in the subsection 3.4, the shape parameters of productivity distribution affect the welfare effects mainly through volume of trade and productivity effect. The trade elasticity and productivity effects might be sensitive to the specification of productivity dispersion in model. In this section, we extend our model to incorporate the difference of productivity distribution across RCEP members and clarify how such extension affects welfare outcomes⁷. The detailed derivation of the extended model and its welfare formula can be seen in Appendix B. Considering differentiated dispersion parameters across countries brings about two main changes of welfare effects compared to the benchmark model.

Firstly, the volume of trade effects by country and sector now become

$$d \ln vot_n^j = -\tilde{\sigma}^j \sum_{i=1}^N \left[\frac{t_{ni}^j}{(1+t_{ni}^j)} \pi_{ni}^j \theta_i^j d \ln (1+t_{ni}^j) \right] + \epsilon_n^j \quad (45)$$

where θ_i^j is the shape parameter of productivity distribution of country i in sector j .

The derivative of volume of trade effect with respect to tariff reductions now is

$$\frac{d \ln \widetilde{vot_n^j}}{d \ln (1+t_n^j)} = - \sum_{i \neq n}^N \frac{t_{ni}^j \pi_{ni}^j}{1+t_{ni}^j} \tilde{\sigma}^j \theta_i^j. \quad (46)$$

As we can see, the trade elasticity is still correlated to the shape parameter of productivity distribution. But now it varies at the country-sector level rather than the sector level alone,

⁷We thank anonymous referee for pointing out that the Pareto dispersion parameter should vary across countries, which can better capture the heterogeneous productivity distribution among members within regional trade agreement.

depending on the shape parameters of origins where a country imports from. Conditional on tariff reductions, a greater share of imports sourced from countries with higher shape parameters implies a larger trade elasticity and, consequently, a stronger volume of trade effect of tariff liberalization.

Secondly, the productivity effect of tariff cuts with differentiated shape parameters is given by

$$d \ln prod_n^j = \frac{1}{\sum_{i=1}^N \pi_{ni}^j \theta_i^j} \left[\sum_{i=1}^N \pi_{ni}^j d \ln (1 + t_{ni}^j) - d \ln \frac{X_n^j}{w_n} \right] + \sum_{i=1}^N \pi_{ni}^j (1 - \iota_{ni}^j) \left[d \ln (1 + t_{ni}^j) c_i^j - \frac{1}{\sigma^j - 1} d \ln \frac{X_n^j}{w_n (1 + t_{ni}^j)} \right] \quad (47)$$

where $\iota_{ni}^j = \frac{\theta_i^j}{\sum_{l=1}^N \pi_{nl}^j \theta_l^j}$.

Conditional on the changes in market size and wage, the derivative of productivity effect with respect to tariff reductions in the heterogeneous shape parameter model can be expressed as

$$\frac{d \ln prod_n^j}{d \ln (1 + t_n^j)} = \frac{1 - \pi_{nn}^j}{\sum_{i=1}^N \pi_{ni}^j \theta_i^j} + \tilde{\sigma}^j \pi_{nn}^j \left(\frac{\theta_n^j}{\sum_{i=1}^N \pi_{ni}^j \theta_i^j} - 1 \right) \quad (48)$$

The relative shape parameter in second term is instrumental to the sign of the derivative. When $\theta_n^j > \sum_{i=1}^N \pi_{ni}^j \theta_i^j + \frac{1}{\tilde{\sigma}^j} \left(1 - \frac{1}{\pi_{nn}^j} \right)$, the derivative is positive, and negative otherwise. This newly added second term captures how the relative heterogeneity of productivity distribution between domestic and foreign firms affects productivity effects. If $\forall i, \theta_i^j = \theta_n^j$, the second term vanishes and the expression of productivity effect returns to the equation (43) and (44) without dispersion parameter heterogeneity. The intuition is that, in the benchmark model where the productivity distributions of domestic and foreign firms are identical, the productivity loss from entry of marginally less productive foreign firms induced by tariff reductions would outweigh the productivity gain from exit of less productive domestic firms, resulting in a net decline in average productivity. When the productivity distributions differ across countries, however, if domestic firms exhibit substantially greater productivity heterogeneity (smaller θ_n^j) than their foreign counterparts, the productivity gain from domestic exit would outweigh the productivity loss from foreign entry, resulting in an increase in average productivity. Conversely, if the productivity heterogeneity of domestic firms is equal to or lower (larger θ_n^j) than that of foreign firms, the productivity gain from domestic exit is smaller than the loss from foreign entry, leading to a decline in average productivity. Based on this, we propose the following proposition about a new factor that affects the welfare effects of tariff reductions:

Proposition 6 (Relative productivity heterogeneity between domestic and foreign firms)

When the shape parameters of the productivity distribution differ across countries, the effect of tariff reductions on average productivity hinges on the relative productivity heterogeneity between domestic and foreign firms. Specifically, tariff reductions tend to raise average productivity when domestic firms are relatively more heterogeneous, and lower it otherwise. Moreover, the larger the relative heterogeneity of domestic firms, the greater the productivity-enhancing effect of tariff reductions.

4 Trade Effects of RCEP

Section 4 and 5 evaluate the trade and welfare effects from RCEP tariff reductions. We consider the base year to be 2021, the year before RCEP came into force, and analyze the situation of full implementation of RCEP that is counterfactual 20 years after 2021. To maximize the number of sectors and countries conditional on solving the equilibrium system quickly and accurately, we choose a sample of $N = 31$, covering 15 RCEP countries, 15 non-RCEP countries and a constructed rest of the world (ROW), and $J = 19$, 2 non-manufacturing sectors (Agriculture and Mining), 16 manufacturing sectors and an aggregated service sector. To control changes of iceberg trade

cost and market entry cost, these two sections set $\hat{\tau}_{ni}^j = \hat{F}_{ni}^j = 1$. This section firstly quantifies aggregate trade effects according to equation (14) and then disaggregates the changes at the firm level using the simulation algorithm in section 3.2. Because of the limitation of other countries' firm-level data, we can only analyze the dynamic changes in Chinese manufacturing firms. But we think the core mechanism of the theoretical model is constant and applies to each country affected by RCEP.

4.1 Aggregate Changes

Figure 4a shows the extent of tariff reductions between RCEP members, measured as bilateral weighted average tariff changes across sectors, namely $\sum_{j=1}^J \frac{X_{ni}^j}{\sum_{k=1}^J X_{ni}^k} (t_{ni}^{j'} - t_{ni}^j)$. Since more than ten trade agreements already existed within the region before RCEP took effect, the magnitude of tariff reductions between most countries is relatively small. The larger tariff reductions are primarily concentrated between China, Japan, and South Korea, as well as between certain ASEAN countries (such as Malaysia, Vietnam, Cambodia, Indonesia, and Myanmar) and non-ASEAN countries. The average import tariffs imposed by China and South Korea on Japan have decreased by 3.74 percentage points. For these ASEAN countries, their average tariffs with non-ASEAN countries have fallen by 5.26 percentage points. Among them, Malaysia has made the greatest tariff concessions, with its average tariffs on imports from China, Australia, and New Zealand dropping by 6, 13.1, and 56.5 percentage points, respectively. In addition, Vietnam has also made notable tariff concessions to ASEAN countries, with an average reduction of 4.81 percentage points.

Figure 4b presents the aggregate trade effects from the tariff reductions. We can see that the RCEP has significant trade creation effect between some members and this effect is almost consistent with the magnitude of tariff reductions. For example, Malaysia's imports from China, Australia, and New Zealand will increase by 233%, 414%, and 1808%, respectively. RCEP meanwhile induces significant trade diversion effect. Specifically, Malaysia's and Cambodia's imports from other ASEAN countries decline 18% and 28.7% on average. And Singapore's import from other RCEP members, the members' domestic trade, and the members' exports to ROW also decrease by 6%, 2.9%, and 6.48% on average, respectively. Under the synthetic effect of trade creation and diversion, the total output and expenditure of members increase, with an average of 1.66% and 2.01%, respectively. The positive effect indicates that the agreement expands the members' production possibility frontiers and leads to Pareto superior outcomes.

Figure 5 decomposes the aggregate trade effects into intensive and extensive margin using equation (14). The decomposition result underscores the sources of the trade effects. It shows that the major sources of trade effects are boosts in extensive margin of trade. The extensive margin of trade contributes to about 84% of the increase in bilateral trade between members whereas the intensive margin only contributes to 16%. One way to understand such decomposition results is by looking at the turnovers of heterogeneous firms induced by tariff reductions. Tariff reductions raise the export sales of incumbent exporters, which tends to increase the average size of exporters (intensive margin). However, the set of exporters (extensive margin) varies at the same time. Tariff reductions lower the entry cutoff and attract the marginal firms to export. Compared to the existing ones, the new exporters have lower productivity levels and sales thus their entry tends to decrease the average size of exporters. When the productivity is Pareto distributed, the two opposite forces are offset and the intensive margin of trade is determined only by $\frac{\hat{w}_n \hat{F}_{ni}^j}{\hat{p}_{ni}^j}$.

Because this section fixes the market entry cost F_{ni}^j , the changes in intensive margin are thus subtle and almost all changes occur on the extensive margin.

Figure 5b also presents that the entry of foreign firms accompanies by the exit of domestic firms and the total measure of firms entering into each member increases. As shown in the last row of Figure 5b, Malaysia, Vietnam and Cambodia experience the largest net increase in the measure of entering firms, at 8.6%, 1.8% and 1.4%, respectively. Equation (14) indicates that the measure of firms entering into one country is determined by its market size, market entry cost and tariff reductions in the case of after-markup tariffs. Therefore the changes in number of total firms reflects the integrative action of the above factors.

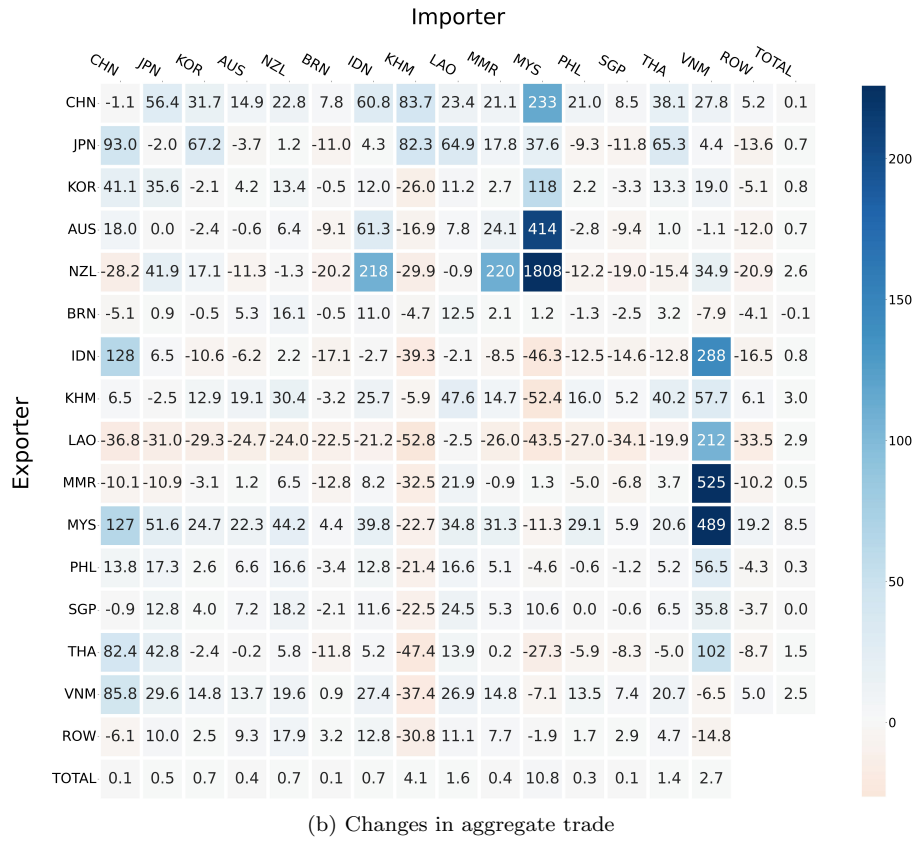
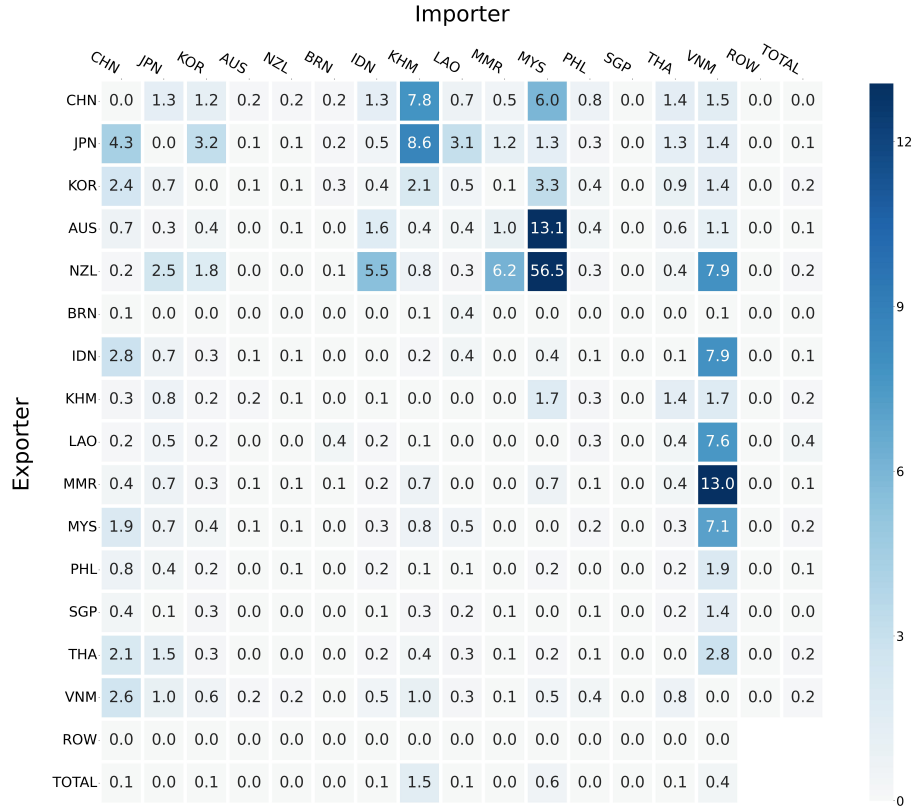


Figure 4 Aggregate trade effects from RCEP tariff reductions (%)
Note: The “TOTAL”s in the last column and row represent the exports to and imports from all countries respectively.

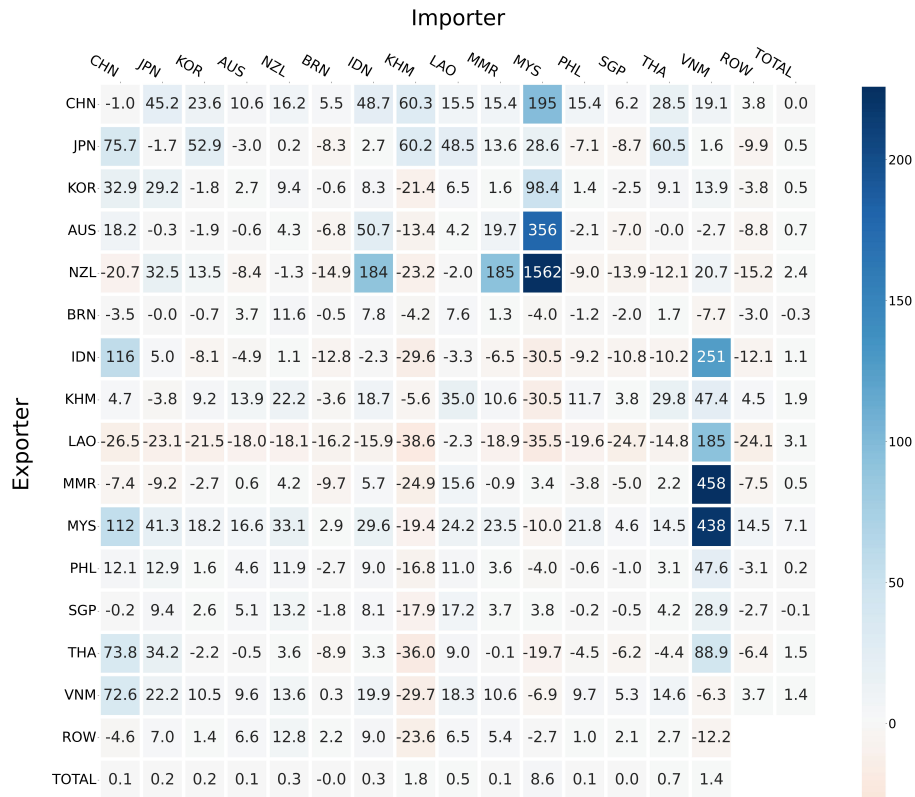
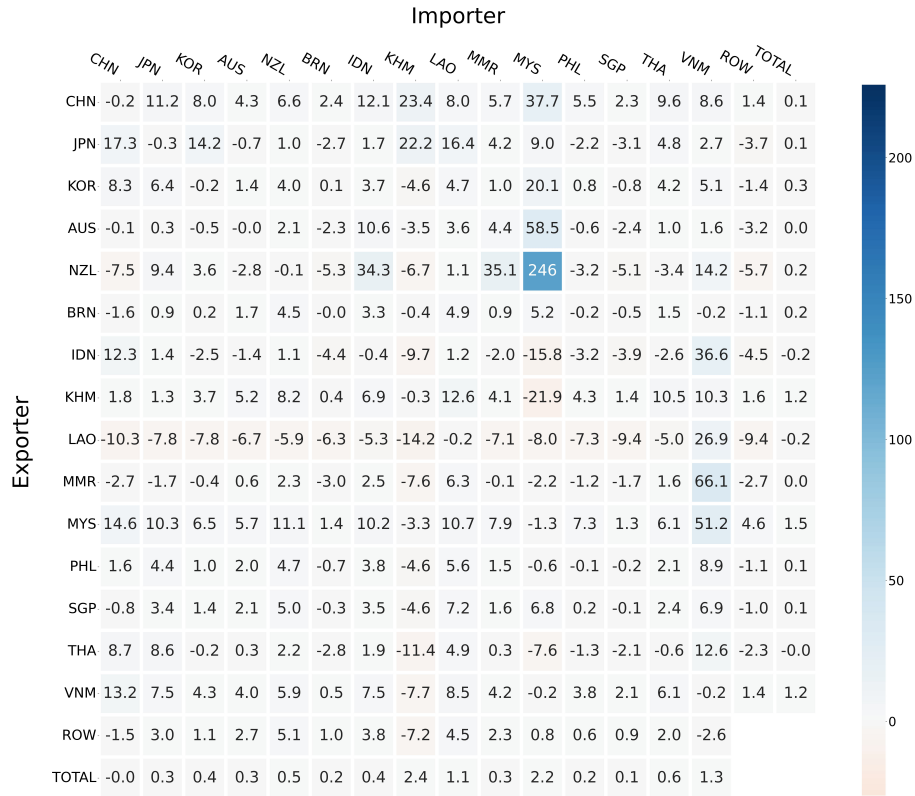


Figure 5 Trade decomposition by intensive margin and extensive margin (%)
Note: The “TOTAL”s in the last column and row represent the exports to and imports from all countries respectively.

4.2 Firm-level Changes: A Case of China

Table 2 and 3 present the impact of RCEP tariff reductions on individual Chinese firms. These figures are calculated via aggregating firm-level changes of each sector by different productivity decile. As in section 3.2, we denote the inverse of $v(s)$ as the fundamental productivity of simulated firms. The higher $v(s)$ corresponds to higher unit cost and lower productivity level.

The results in Table 2 show that 17,889 Chinese firms will become exporters after the agreement fully comes into effect. As predicted by Melitz (2003), more productive firms are more likely to export after trade liberalization. Firms in the top and second highest decile of productivity contribute to 10.7% and 15.4% of new exporters. Meanwhile, some least and less productive firms will also enter the export market. This pattern is driven by the presence of entry shocks η in export hurdles (18). For the less productive firms with favorable entry shocks, they face lower entry cutoffs than others and tariff reductions can more easily prompt their entrance. For example, if customers in members prefer some Chinese firms due to their idiosyncratic product appeal or marketing techniques, although the productivity levels of these firms are lower, entering the members' market can generate enough sales for these firms to cover the entry cost. Therefore, tariff reductions have further positive impact on the entry of these firms. And the more productive firms with adverse entry shocks are similarly affected for similar reasons, which explains why not all of the more productive firms export. Since the entry shocks are independently distributed across firms, the productivity is still dominant determinant for entry and the number of new exporters increases with the productivity level. RCEP also intensifies the domestic product and factor market competition and forces 6,852 Chinese firms to exit. Nearly all the firms in the bottom decile are wiped out, while others avoid attrition. The reason is that the entry cutoff of domestic market is far below that of foreign market. Only the productivity level determines firm's survival in domestic market and the entry shocks have almost no effect on it.

Table 3 shows that the RCEP will increase China's exports by \$70,620 millions. The exports in every decile boost and the extent of boost increases with the productivity level. Firms in the percentile of 50 to 100 account for about 80% of the increase and the top decile contributes to 38.4%. The firms with favorable entry shocks in lower decile also experience substantial export growth, although they account for a very small part of total increase. Despite the rise in export revenue, competition induced by RCEP contracts domestic sales by \$461,752 millions, which results in the decline of total sales by \$391,132 millions. Every decile suffers from total sales loss and the losses decrease with the productivity level. The greatest loss of sales emerges from the second lowest decile and the biggest percentage decrease in total sales is for firms in the bottom decile, whereas the total sales loss in the top decile is also considerable. The reason for such wide range of shrinkage is as follows. After RCEP takes effect, the export expansion of domestic more productive firms increases the factor demand and bids up the factor price. Meanwhile, the entry of foreign firms intensifies the product market competition and erodes domestic market shares. All firms incur a loss in their domestic sales under the dual pressure. The less productive firms who cannot export also incur a total sales loss. But the more productive firms who export can make up for the loss of their domestic sales with export sales. Nevertheless, the tariff concession arrangement under RCEP reallocates much of the Chinese market shares to other members since Chinese market size is much larger than other members' and even the same degree of bilateral tariff reductions implies loss of market shares for China. As a result, the export growth of Chinese firms is not enough to make up for their domestic sales loss. Therefore, the more productive firms also experience loss in total sales, although less than the less productive firms ⁸.

Figure 6 presents the changes in market shares of firms in different productivity decile by sector. The results are well informative and present considerable variation at the sector level. Depending on the relative magnitude of tariff reduction between China and its trading partners (see Figure D.11 in Appendix), only the Textile, Wood, Paper, and Other Transport sectors expands slightly after the tariff reductions while other sectors all shrink largely due to import competition.

⁸This pattern can also be supported by data of export intensity and domestic sales-to-exports ratio by different firms. Figure D.9 presents the relationship between export intensity and export changes of 100 firms across the productivity distribution (from the 1st to the 100th percentile) and Figure D.10 shows the relationship between domestic sales-to-export ratios and total sales changes of them. We can see that for firms with higher export intensity, their export growth is also larger. And for firms with higher domestic sales-to-exports ratio firms, the total sales loss is much heavier.

Table 2 Chinese firm entry and exit by productivity

Prod. Interval (percentile)	Exporters				All firms			
	Baseline (No.)	Change (No.)	Change (%)	Contribution (%)	Baseline (No.)	Change (No.)	Change (%)	Contribution (%)
0 to 10	1,607	1,066	66.3	6.0	31,350	-6,251	-19.9	91.2
10 to 20	1,927	1,179	61.2	6.6	31,361	-601	-1.9	8.8
20 to 30	2,337	1,294	55.4	7.2	31,360	0	0.0	0.0
30 to 40	2,869	1,479	51.5	8.3	31,359	0	0.0	0.0
40 to 50	3,598	1,672	46.5	9.3	31,359	0	0.0	0.0
50 to 60	4,663	1,885	40.4	10.5	31,360	0	0.0	0.0
60 to 70	6,221	2,160	34.7	12.1	31,359	0	0.0	0.0
70 to 80	8,786	2,488	28.3	13.9	31,360	0	0.0	0.0
80 to 90	13,542	2,750	20.3	15.4	31,358	0	0.0	0.0
90 to 100	23,777	1,918	8.1	10.7	31,361	0	0.0	0.0
Total	69,326	17,889	25.8	100.0	313,587	-6,852	-2.2	100.0

Table 3 Chinese firm growth by productivity

Prod. Interval (percentile)	Exports				Total sales			
	Baseline (\$ millions)	Change (\$ millions)	Change (%)	Contribution (%)	Baseline (\$ millions)	Change (\$ millions)	Change (%)	Contribution (%)
0 to 10	2,752	2,320	84.3	3.3	103,117	-33,543	-32.5	8.6
10 to 20	3,329	2,470	74.2	3.5	312,403	-54,644	-17.5	14.0
20 to 30	4,143	2,711	65.4	3.8	534,803	-52,426	-9.8	13.4
30 to 40	5,195	3,139	60.4	4.4	730,253	-43,383	-5.9	11.1
40 to 50	6,760	3,775	55.8	5.3	1,011,117	-46,309	-4.6	11.8
50 to 60	9,199	4,696	51.0	6.6	1,205,964	-36,162	-3.0	9.2
60 to 70	13,354	5,698	42.7	8.1	1,510,392	-37,333	-2.5	9.5
70 to 80	21,041	7,598	36.1	10.8	1,840,465	-31,294	-1.7	8.0
80 to 90	40,824	11,120	27.2	15.7	2,255,844	-28,207	-1.3	7.2
90 to 100	206,026	27,094	13.2	38.4	3,771,387	-27,831	-0.7	7.1
Total	312,622	70,620	22.6	100.0	13,275,745	-391,132	-2.9	100.0

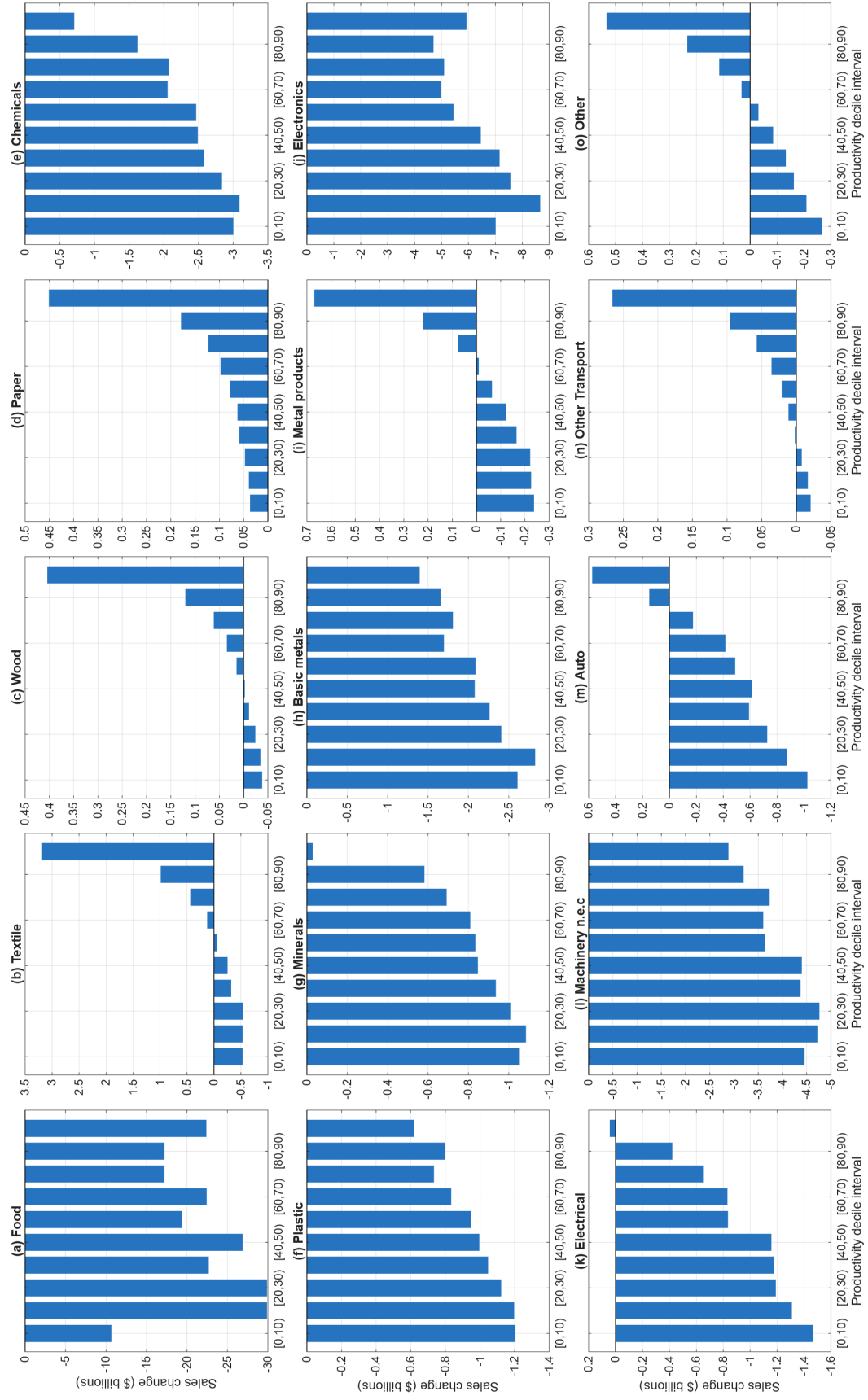


Figure 6 Reallocation of market shares among Chinese firms by sector

The firms in top productivity decile of Textile, Wood, Paper, Metal products, Auto and Other Transport sectors experience growth of total sales while the less productive firms in these sectors contract, which indicates that tariff reductions in these sectors improve the performance of the more productive firms at the expense of the rest. Other sectors exhibit the similar pattern in Table 3 that sales of each kind of firms decline and the magnitude of decline decreases with the productivity level.

5 Welfare Effects of RCEP

This section evaluates the welfare effects of RCEP, with a focus on the new welfare effects from the firm dynamics presented in section 4. The baseline welfare effects are calculated using equation (33), and the new welfare effects using equation (34). We also empirically examine the factors affecting these welfare changes proposed in 3.4 using quantitative results of country-sector-level welfare effects. We further perform two simple but instructive counterfactual experiment to guide the future negotiations of the agreement between members, that is, India’s accession into RCEP and decrease of market entry cost, respectively.

5.1 Baseline Analysis

Table 4 presents the welfare effects from RCEP tariff reductions. Column (1) shows that all members benefit from the RCEP and the members’ welfare will be improved. Some ASEAN countries such as Malaysia, Vietnam, and Cambodia benefit most with welfare gains of 22.2%, 5.73%, and 2.37%, respectively, while other countries benefit less. The reason Malaysia and Vietnam experience such large welfare gains is that their initial tariffs in the food sector against other RCEP members were exceptionally high — approximately 90% and nearly 30%, respectively. While this may partly reflect the use of simple average tariffs from the Teti (2024) database, we also conduct robustness checks using alternative tariff measures of that database, including trade-weighted and 95%-winsorized tariffs, and find that these three countries’ initial tariffs remain relatively high and their welfare gains from RCEP remain substantial (See Table C.1 - C.3 in Appendix). This confirms that their larger welfare gains are driven not by the choice of tariff measurement, but by their genuinely higher pre-RCEP tariff levels. Column (2) and (5) demonstrate that the welfare is improved through not only the “traditional” channels in forms of term of trade and volume of trade but also the “new” channels of the endogenous firm selection, namely variety and productivity effect. While the “traditional” gains still contribute to the majority of the positive welfare effects, the magnitude of “new” gains is considerable and account for about 23% of the average welfare changes.

There are also many differences of the sources of welfare gains between members. Column (4) shows that the volume of trade effect is the most important contributor to the welfare gains for members except Australia, New Zealand, Brunei and Singapore. As shown in Figure 4, the major determinant of trade volume is the degree of tariff reductions. Malaysia benefits from an enormous volume of trade effect of 17.23% from its considerable tariff cuts, mainly in the Food sector. For other members with greater tax concession commitment such as Vietnam, Cambodia, and Thailand, volume of trade effects are 4.71%, 1.69%, and 1.19% respectively, whereas almost zero for Australia, New Zealand, Brunei and Singapore with extremely low initial tariffs.

Another traditional source of welfare gains is terms of trade effect. From column (3) we can see that almost all members’ terms of trade improve while Malaysia’s, Vietnam’s, Cambodia’s, and China’s terms of trade deteriorate. As stated in Propositions 4 and 5, the terms-of-trade effect of tariff reductions is determined by the relative magnitude of tariff cuts and the relative market size across countries. Malaysia, Vietnam, and Cambodia receive relatively small tariff concessions from their trading partners while undertake substantial tariff reductions, and thus experience deterioration in their terms of trade. China, whose market size is far much larger than that of other RCEP members, even exchanges tariff cuts commitment reciprocally with its trading partners, still suffers from terms of trade worsening due to the uneven output changes. In contrast, Australia, New Zealand, Brunei, and Singapore makes few tariff concessions to but

Table 4 Welfare effects from RCEP tariff reductions (%)

Members	Welfare	“Traditional” gains			“New” gains		
		Total	Terms of trade	Volume of trade	Total	Variety	Productivity
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
China	0.37	0.29	-0.09	0.38	0.08	0.10	-0.03
Japan	0.50	0.38	0.18	0.21	0.11	0.15	-0.04
South Korea	0.49	0.35	0.12	0.23	0.14	0.18	-0.05
Australia	0.25	0.17	0.17	0.01	0.08	0.10	-0.03
New Zealand	0.48	0.34	0.32	0.01	0.14	0.19	-0.05
Brunei	0.14	0.16	0.16	-0.00	-0.01	-0.02	0.00
Indonesia	0.67	0.51	0.22	0.29	0.16	0.22	-0.05
Cambodia	2.37	1.52	-0.16	1.69	0.85	1.12	-0.27
Laos	1.02	0.80	0.72	0.07	0.23	0.30	-0.08
Myanmar	0.26	0.19	0.14	0.05	0.06	0.08	-0.02
Malaysia	22.20	16.59	-0.64	17.23	5.61	7.41	-1.79
Philippines	0.12	0.08	0.06	0.02	0.05	0.07	-0.02
Singapore	0.11	0.11	0.11	0.00	0.00	0.00	-0.00
Thailand	1.93	1.48	0.29	1.19	0.45	0.61	-0.15
Vietnam	5.73	4.34	-0.37	4.71	1.40	1.84	-0.44
Average	2.44	1.82	0.08	1.74	0.62	0.82	-0.20

Table 5 “New” welfare effects from RCEP tariff reductions (%)

Members	Variety			Productivity		
	Total	Domestic exit	Foreign entry	Total	Domestic exit	Foreign entry
	(1)	(2)	(3)	(4)	(5)	(6)
China	0.10	-0.83	0.94	-0.03	0.21	-0.23
Japan	0.15	-1.05	1.20	-0.04	0.26	-0.29
South Korea	0.18	-1.29	1.47	-0.05	0.33	-0.37
Australia	0.10	-0.37	0.48	-0.03	0.09	-0.12
New Zealand	0.19	-0.81	0.99	-0.05	0.20	-0.25
Brunei	-0.02	-0.27	0.25	0.00	0.07	-0.06
Indonesia	0.22	-1.38	1.60	-0.05	0.34	-0.39
Cambodia	1.12	-2.25	3.38	-0.27	0.56	-0.84
Laos	0.30	-1.10	1.41	-0.08	0.27	-0.35
Myanmar	0.08	-0.65	0.73	-0.02	0.16	-0.18
Malaysia	7.41	-6.30	13.71	-1.79	1.52	-3.32
Philippines	0.07	-0.33	0.39	-0.02	0.08	-0.10
Singapore	0.00	-0.27	0.27	-0.00	0.07	-0.07
Thailand	0.61	-2.71	3.32	-0.15	0.68	-0.83
Vietnam	1.84	-6.01	7.85	-0.44	1.47	-1.92
Average	0.82	-1.71	2.53	-0.20	0.42	-0.62

acquires considerable tariff reductions from RCEP members, so the improvement in terms of trade contributes the most to their welfare gains.

As for the “new” welfare effects, column (6) presents the welfare effects from changes in product varieties available to each member. The results show that each member realizes the variety gains from RCEP, which echoes the results of increases in extensive margin of trade in the last row of Figure 5b. Equation (41) shows that the variety effect depends primarily on the tariff reduction and, to a lesser extent, on relative changes of market size and entry costs. Again, members such as Malaysia, Vietnam, Cambodia, and Thailand benefit more from the variety effect. Due to the expansion in market size, Australia and New Zealand also gain from considerable variety effect even though their tariff reductions are very modest. Since China’s market size is shrunk under the import competition, the variety effect for China increases slightly by 0.08%.

Notice that the productivity effect in column (8) is negative for each member and is proportional to the variety effect, which can be seen in Equation (41) and (43). The economic intuition of such relation is that the additional foreign firms’ entrance into a member’s market is due to fall of market entry productivity cutoff. This implies that new entrants are less productive than incumbent ones, just as shown in table 2. As a result, aggregate productivity is pulled down by these new exporters and consumers bear welfare loss from these costly new varieties. As a result, members gain from more varieties at the expense of lower productivity. Furthermore, as we can see from Table 1, the $\tilde{\theta}^j = \frac{\theta^j}{\sigma^j - 1}$ is larger than 1, which means that the heterogeneity in productivity of firms is smaller than diversity of products they produce. Hence the variety gains from consuming more kinds of products outweigh their productivity losses and contribute to the positive “new” gains.

Table 5 decomposes the new welfare effects from RCEP tariff reductions by entry and exit of heterogeneous firms to shed light welfare implication of endogenous firm selection. The results show that for each member, exit of domestic firms leads to variety losses but productivity gains and entry of foreign firms brings about variety gains but productivity losses. The results show that the foreign entry effect outweighs the domestic exit effect. As a result, the import variety gains outweigh the domestic variety losses thereby generating the net variety gains and, when the productivity distribution between domestic and foreign firms is identical, the import productivity losses outweigh the domestic productivity gains thereby causing the net productivity losses. Just as reflected by equation (10) and (12), the dominant entry effect comes from two different channels through which tariff reductions affect market entry, that is, competitiveness channel and tax saving channel. Tariff reductions increase the competitiveness of foreign firms relative to the domestic firms thus attract foreign firms to enter while forcing an equal number of domestic firms to exit. Besides that the tariff reductions further promote foreign firms’ entry through saving the tax paid in destinations by them and increasing their profits, which is isomorphic to entry cost decrease. In aggregate, tariff reductions affect foreign firms’ entry to a larger extent than domestic firms’ exit thus bring about variety gains and productivity losses for each member.

As for the rest of world, we find that non-RCEP countries enjoy an average welfare gain of 0.003%. Since the agreement lowers the prices of RCEP-produced goods, non-RCEP countries import more from RCEP members and get a considerable positive volume of trade effect (0.005% on average) and improvement of terms of trade (0.002%). At the same time, trade diversion erodes the market share of non-RCEP countries within the RCEP region, driving down their output and wages contracting their market size, thereby reducing the variety of goods and resulting in loss of “new” welfare gains (-0.004% on average). It is worth noting that Brazil’s welfare declines by approximately 0.021%, as tariff cuts within RCEP undermine its export competitiveness in certain primary agricultural products and food sectors, substantially deteriorating its terms of trade. The detailed welfare effects for non-RCEP members are reported in Figure D.1 and D.2 of Appendix D.

Another concern is the cross-country heterogeneity of shape parameters. As discussed in section 3.5, if the shape parameters differ substantially across countries such that the productivity distribution of domestic firms is more dispersed than that of foreign firms, the productivity gains from the exit of domestic firms could outweigh the productivity losses from the entry of foreign firms, yielding a net productivity gain rather than a net loss. To assess this possibility, we recompute the welfare effects using a model that allows for country-specific shape parameters. The results show that, because country-level heterogeneity in the productivity distribution is very lim-

ited in the data, the productivity effect remains negative, and the relative welfare rankings across countries are largely unchanged under the alternative model. The baseline findings are therefore reasonably robust to the alternative model. The relevant results can be found in Appendix C.2.

5.2 Evidence on Determinants of Welfare Effects

As shown in the previous theoretical model, the welfare effects of RCEP tariff reductions are determined by various factors including magnitude of tariff reductions, shape parameter of productivity distribution, initial tariffs, relative magnitude of tariff reductions, relative market size, and relative productivity heterogeneity between domestic and foreign firms. To provide evidence on these factors affecting welfare effects, we first specify regression equations by channels of welfare change according to equations (39) - (47) and then exploit the variations of quantitative results in last subsection to examine the role of these factors.

According to equation (39) and proposition 1, 2, and 3 the volume of trade effect directly depends on magnitude of tariff reductions, and the larger shape parameter of productivity distribution or initial tariffs enhance the volume of trade effect from tariff cuts. Based on this, we construct the following regression model with interaction effects of shape parameter and initial tariffs respectively as ⁹

$$d \ln vot_n^j = \beta_0^{tot} + \beta_1^{tot} d \ln(1 + t_n^j) + \beta_2^{tot} \tilde{\theta}_n^j + \beta_3^{tot} d \ln(1 + t_n^j) \times \tilde{\theta}_n^j + \epsilon_n^j \quad (49)$$

$$d \ln vot_n^j = \gamma_0^{tot} + \gamma_1^{tot} d \ln(1 + t_n^j) + \gamma_2^{tot} t_n^{j,initial} + \gamma_3^{tot} d \ln(1 + t_n^j) \times t_n^{j,initial} + \epsilon_n^j \quad (50)$$

We aggregate the degree of tariff reductions, shape parameters and initial tariffs across origins weighted by trade share following equation (40)—specifically, the aggregated tariff change by country and sector is defined as $d \ln(1 + t_n^j) = \sum_{i \neq n}^N \frac{\pi_{ni}^j}{1 - \pi_{nn}^j} d \ln(1 + t_{ni}^j)$, the aggregated shape parameter of the productivity distribution as $\tilde{\theta}_n^j = \sum_{i \neq n}^N \pi_{ni}^j \theta_{ni}^j$ ¹⁰, and the aggregated initial tariff as $t_n^{j,initial} = \sum_{i \neq n}^N \frac{\pi_{ni}^j t_{ni}^j}{1 + t_{ni}^j}$. Notice that the shape parameters here are only sector specific, which may exhibit insufficient variation and the estimated coefficient may be heavily influenced by the weights of trade share. As a robustness check, we also run the interaction regressions using the extended model in section 3.5 that allows for heterogeneous shape parameters by country. The results are reported in the Appendix C.

As predicted by proposition 4 and 5, the terms of trade effect of tariff reductions is determined by the relative magnitude of tariff cuts between partner and importer and this effect is moderated by relative market size between them. We construct the following regression model with moderator variable.

$$d \ln tot_n^j = \beta_0^{tot} + \beta_1^{tot} d \ln(1 + t_n^j)^{rel.} + \beta_2^{tot} X_n^{j,rel.} + \beta_3^{tot} d \ln(1 + t_n^j)^{rel.} \times X_n^{j,rel.} + \epsilon_n^j \quad (51)$$

Since deriving an analytical expression for the terms-of-trade effect in a multi-sector quantitative model with intermediate inputs is practically infeasible (Bown et al., 2024), we construct the core regressors based on economic intuition. The relative tariff change is defined as $d \ln(1 + t_n^j)^{rel.} = \sum_i \left[\frac{X_{in}^j}{\sum_i X_{in}^j} d \ln(1 + t_{in}^j) - \frac{X_{ni}^j}{\sum_i X_{ni}^j} d \ln(1 + t_{ni}^j) \right]$, which is the difference between the tariff cuts country n receives from its trading partners and those it commits to them. And the relative market size between them is measured as $X_n^{j,rel.} = \frac{X_n^j}{\sum_i X_i^j - X_n^j}$.

In line with equation (41) and proposition 1, we specify the following econometric model to investigate whether tariff cuts expand the overall set of product varieties available to consumers.

$$d \ln var_n^j = \beta_0^{var} + \beta_1^{var} d \ln(1 + t_n^j) + \epsilon_n^j \quad (52)$$

⁹Because the initial tariffs and shape parameters are hardly correlated variables, when examining one interaction effect, we don't control another one.

¹⁰Notice that the parameter of elasticity of substitution is identical across all sectors and countries ($\sigma^j = 3.79$), and we therefore do not add it into moderating variable.

Guided by change in productivity effects (43) and proposition 1 and 2, the following econometric model is set up to empirically test the impact of tariff cuts on average productivity and the moderating role of shape parameter of productivity distribution.

$$d \ln prod_n^j = \beta_0^{prod} + \beta_1^{prod} d \ln (1 + \widetilde{t_n^j}) + \beta_2^{prod} \bar{\theta}_n^j + \beta_3^{prod} d \ln (1 + \widetilde{t_n^j}) \times \bar{\theta}_n^j + \epsilon_n^j \quad (53)$$

In particular, we define the aggregated inverse of shape parameter by country and sector as $\bar{\theta}_n^j = \frac{1 - \pi_{nn}^j}{\theta_n^j}$ following equation (44). To increase the variation in the shape parameters and mitigate the influence of trade share weights on the estimates, we further estimate the interaction model using the welfare results computed from the extended model that allows for country-specific heterogeneity in the shape parameter. The relevant robustness checks are presented in the Appendix C.

Lastly, as proposition 6 indicates, in the extended model with differentiated shape parameter by country, relative productivity heterogeneity between domestic and foreign firms determines the productivity effect of tariff cuts. Based on equation (47), we estimate the following model to examine the productivity effects of tariff cuts in the extended model and explore the role of relative productivity heterogeneity.

$$d \ln \widetilde{prod}_n^j = \beta_0^{\widetilde{prod}} + \beta_1^{\widetilde{prod}} d \ln (1 + \widetilde{t_n^j}) + \beta_2^{\widetilde{prod}} \theta_n^{j,rel.} + \beta_3^{\widetilde{prod}} d \ln (1 + \widetilde{t_n^j}) \times \theta_n^{j,rel.} + \epsilon_n^j \quad (54)$$

The explained variable $d \ln \widetilde{prod}_n^j$ now is computed using the extended model with differentiated shape parameter. The moderating variable follows $\theta_n^{j,rel.} = \frac{\theta_n^j}{\sum_i \pi_{ni}^j \theta_i^j}$, as shown in equation (48), which captures the relative shape parameter between domestic and foreign firms.

Using the welfare effects computed in Section 5.1 and regression model (49) - (54), we examine the factors influencing each component of welfare at the country-sector level. The regression results are reported in Tables 6 and 7. To mitigate the influence of extreme tariff cuts on the regression results, the top and bottom 1% of observations are winsorized.

Magnitude of Tariff Reductions Proposition 1 is validated by the results of column (1) in Table 6 and columns (1) and (2) in Table 7. The coefficients on tariff changes $d \ln (1 + \widetilde{t_n^j})$ in the column (1) of both Tables 6 and 7 are significantly negative, indicating that larger tariff reductions are associated with greater volume of trade and variety. On average, a one-standard-deviation (0.0084) larger tariff reduction raises the volume of trade effect by 0.41 percentage points and the variety effect by 0.32 percentage points. In contrast, the coefficient on tariff changes in the column (2) of Table 7 is significantly positive, suggesting that tariff reductions lead to a decline in average productivity — the larger the tariff cut, the more the productivity loss — with a one-standard-deviation tariff cut lowering the productivity effect by 0.08 percentage points. On net, a one-standard-deviation larger tariff reduction yields a welfare gain of approximately 0.65 percentage points.

Shape Parameters of Productivity Distribution The coefficient of interaction term between the aggregated shape parameter $\bar{\theta}_n^j$ and tariff changes in the column (2) of Table 6 is significantly negative, indicating that the volume of trade effect of tariff reductions grows larger as the shape parameter increases — that is, the shape parameter of the productivity distribution amplifies the volume of effect of tariff cuts. In the column (3) of Table 7, the interaction term between the aggregated inverse of shape parameter $\bar{\theta}_n^j$ and tariff changes is significantly positive, suggesting that the productivity-loss effect of tariff cuts weakens as the shape parameter increases ($\bar{\theta}_n^j$ decreases). Thus, the shape parameter amplifies the volume of trade effect and dampens the productivity loss effect of tariff reductions, thereby enhancing the net welfare effect of tariff cuts, consistent with the prediction of Proposition 2.

Initial Tariffs As we can see from column (3) in Table 6, the interaction term between initial tariffs variable $t_n^{j,initial}$ and tariff change is significantly negative. The result implies that as initial tariffs increase, the volume of trade effect of tariff reductions is strengthened, validating the

Table 6 Factors affecting volume of trade and terms of trade effect

	(1) $d \ln \widetilde{vot}_n^j$	(2) $d \ln vot_n^j$	(3) $d \ln vot_n^j$	(4) $d \ln tot_n^j$	(5) $d \ln tot_n^j$
$d \ln (1 + \widetilde{t}_n^j)$	-0.492*** (0.013)	-0.368*** (0.024)	-0.337*** (0.019)		
$\tilde{\theta}_n^j$		0.001 (0.000)			
$d \ln (1 + \widetilde{t}_n^j) \times \tilde{\theta}_n^j$		-0.020*** (0.004)			
$t_n^{j,initial}$			0.032*** (0.010)		
$d \ln (1 + \widetilde{t}_n^j) \times t_n^{j,initial}$			-3.226*** (0.430)		
$d \ln (1 + \widetilde{t}_n^j)^{rel.}$				-0.040*** (0.010)	-0.040*** (0.010)
$X_n^{j,rel.}$					-0.003*** (0.001)
$d \ln (1 + \widetilde{t}_n^j)^{rel.} \times X_n^{j,rel.}$					0.012 (0.124)
N	575	566	572	568	558
Adj. R^2	0.720	0.728	0.747	0.027	0.035

The sample consists of 31 countries and 19 sectors, with the top and bottom 1% of observations winsorized. Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The coefficient on $\tilde{\theta}_n^j$ is multiplied by 100.

Table 7 Factors affecting variety and productivity effect

	(1) $d \ln var_n^j$	(2) $d \ln prod_n^j$	(3) $d \ln prod_n^j$	(4) $d \ln \widetilde{prod}_n^j$	(5) $d \ln \widetilde{prod}_n^j$
$d \ln (1 + \widetilde{t}_n^j)$	-0.383*** (0.014)	0.090*** (0.004)	0.056*** (0.006)	0.219*** (0.011)	-0.532* (0.288)
$\bar{\theta}_n^j$			-0.004*** (0.001)		
$d \ln (1 + \widetilde{t}_n^j) \times \bar{\theta}_n^j$			0.781*** (0.124)		
$\theta_n^{j,rel.}$					0.000 (0.001)
$d \ln (1 + \widetilde{t}_n^j) \times \theta_n^{j,rel.}$					0.761*** (0.292)
N	577	575	566	576	567
Adj. R^2	0.552	0.478	0.562	0.424	0.429

The sample consists of 31 countries and 19 sectors, with the top and bottom 1% of observations winsorized. Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

moderating role of initial tariffs in Proposition 3. Evaluated at one standard deviation (0.0137) higher initial tariffs, a one-standard-deviation tariff cut generates an additional 0.04 percentage points of volume of trade effect. This increment is approximately one-tenth of the average effect of tariff cuts, suggesting a modest but non-negligible moderating role. The finding also helps explain why countries with relatively high initial tariffs – such as Malaysia, Cambodia, and Vietnam – experience systematically larger volume of trade gains from RCEP tariff reductions.

Relative Magnitude of Tariff Reductions The column (4) of Table 6 reports the results of regression of the terms of trade effect on relative tariff changes $d \ln \widetilde{(1 + t_n^j)}^{rel.}$, defined as the tariff reductions a country receives from its trading partners net of those it grants in return. The coefficient is significantly negative, consistent with Proposition 4: the more a country’s trading partners cut tariffs on its exports relative to its own tariff concessions, the greater its terms of trade improvement. A one-standard-deviation (0.0085) increase in this relative tariff reduction raises the terms of trade effect by 0.03 percentage points.

Relative Market Size The column (5) of Table 6 adds the interaction term with relative market size $X_n^{j,rel.}$. The interaction is neither statistically nor economically significant. This null result may stem from two sources. First, the terms of trade effect itself exhibits limited variation across country-sector observations. Second, unlike the volume of trade and productivity effects, the terms of trade effect lacks a closed-form analytical expression — it is determined by a system of general equilibrium variables, making it infeasible to derive theoretically guided regressors.

Relative Productivity Heterogeneity between Domestic and Foreign Firms In the column (4) of Table 7, we regress productivity changes $d \ln \widetilde{prod}_n^j$ — computed using the quantitative model with country-specific shape parameters — on tariff changes. We can see that the coefficient on tariff changes remains positive, indicating that tariff reductions still lead to a net productivity loss as in the baseline model without cross-country difference in the shape parameter. This result reflects that the productivity heterogeneity of domestic firms is not substantially greater than that of foreign firms. Thus the productivity gains from the exit of domestic firms do not outweigh the productivity losses from the entry of foreign firms and the productivity effect of tariff cuts remains negative, which aligns with the discussion in Proposition 6. The column (5) adds the interaction term with relative productivity heterogeneity between domestic and foreign firms $\theta_n^{j,rel.}$. The interaction coefficient is significantly positive, suggesting that as a country’s shape parameter becomes relatively larger — implying relatively lower productivity heterogeneity among its domestic firms — the productivity gains from the exit of domestic firms induced by tariff cuts decline, and the average productivity losses due to more foreign firms entry grow larger. This confirms the argument in Proposition 6 that the productivity gain from tariff reductions diminishes as domestic firm heterogeneity declines.

In summary, this section examines the factors influencing welfare effects of RCEP tariff reductions. We find that the magnitude of tariff reductions is the primary driver of welfare changes, and its effect is moderated by the shape parameter of the productivity distribution and initial tariffs. In addition, relative tariff reductions across countries, and relative productivity heterogeneity also have considerable impacts on welfare outcomes.

5.3 Further Analyses

India’s Rejoining to RCEP Due to concerns over trade deficits and pressure from domestic interest groups, India withdrew from the RCEP negotiations in 2019. Nevertheless, the agreement retains a “return clause,” under which India can rejoin at any time without renegotiation. To evaluate the implications of India’s accession, we conduct a counterfactual exercise in which, on basis of the full implementation of RCEP, tariffs between India and RCEP members are set to zero bilaterally, and compute the resulting welfare changes across countries. The results, reported in Table 8, indicate that India would experience a welfare gain of approximately 3.31%, attributable primarily to the volume of trade effect and the “new” welfare effect, while its terms of trade

would deteriorate. For other RCEP members, welfare would also rise on average, by roughly 1.21%, accompanied by an improvement in their terms of trade. Vietnam and Malaysia would enjoy relatively large welfare gains owing to their initially high tariffs with India. In contrast, Cambodia and Laos would suffer a net welfare loss: their own tariff reductions are modest given their low initial tariffs imposed on India, but the tariff concessions that other members extend to India undercut their competitiveness in the regional market, leading to output contraction and considerable deterioration in their terms of trade.

Table 8 Welfare effects from India’s accession to RCEP (%)

Members	Welfare	“Traditional” gains			“New” gains		
		Total	Terms of trade	Volume of trade	Total	Variety	Productivity
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
China	0.15	0.11	0.04	0.08	0.04	0.05	-0.01
Japan	0.03	0.02	0.01	0.01	0.01	0.01	-0.00
South Korea	0.12	0.11	-0.03	0.15	0.01	0.01	-0.00
Australia	0.03	0.03	0.03	0.01	-0.00	-0.00	0.00
New Zealand	-0.03	-0.00	-0.01	0.01	-0.03	-0.04	0.01
Brunei	-0.07	-0.05	-0.05	0.00	-0.03	-0.04	0.01
Indonesia	0.27	0.21	0.16	0.05	0.06	0.09	-0.02
Cambodia	-0.25	-0.17	-0.16	-0.00	-0.08	-0.11	0.03
Laos	-0.47	-0.37	-0.36	-0.01	-0.10	-0.13	0.03
Myanmar	-0.04	-0.02	-0.03	0.01	-0.03	-0.03	0.01
Malaysia	11.00	7.83	0.71	7.12	3.17	4.20	-1.03
Philippines	0.02	0.02	-0.00	0.02	0.01	0.01	-0.00
Singapore	0.18	0.16	0.16	0.00	0.02	0.03	-0.01
Thailand	0.94	0.70	-0.00	0.70	0.24	0.32	-0.08
Vietnam	6.26	4.86	-0.19	5.04	1.40	1.84	-0.44
Average	1.21	0.90	0.02	0.88	0.31	0.41	-0.10
India	3.31	2.56	-0.38	2.94	0.75	1.00	-0.24

Decrease in Entry Cost Table 9 presents the welfare changes if the members further decrease the bilateral market entry cost by 5% on the basis of current tariff reductions. The results indicate that a 5% reduction in market entry costs raises welfare by 0.41% on average across member countries, and this increase is almost entirely attributable to the “new” welfare effects generated by firm entry, while the volume of trade effect is negligible. The reason is that, unlike tariff reductions, a decline in entry costs induces only marginal firm entry, and the resulting increase in trade volume is too limited to exert a meaningful effect; welfare is affected solely through the variety and productivity channels. In addition, the welfare gains from lower entry costs are larger for ASEAN countries such as Vietnam, Malaysia, and Cambodia, reflecting their stronger trade linkages within the region. Because the market entry costs are always present in the form of barriers behind the border such as regulatory divergences in terms of technical barriers to trade or domestic commercial policies, this experiment appeals to reducing these forms of entry cost to release more “new” gains from trade.

6 Model Comparisons

The results in section 4 and 5 indicate that tariff reductions would induce entry and exit of heterogeneous firms simultaneously, thereby generating the variety and productivity effects and bringing about “new” welfare gains. These “new” gains from trade are ignored by the classical Ricardian model but highlighted in Melitz (2003). However, the inspirational discovery by the new generation of trade theories has been challenged by ACR, an influential work in the area of

Table 9 Welfare effects from RCEP tariff reductions and market entry cost decrease (%)

Members	Welfare	“Traditional” gains			“New” gains		
		Total	Terms of trade	Volume of trade	Total	Variety	Productivity
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
China	0.08	-0.01	-0.02	0.01	0.10	0.13	-0.03
Japan	0.16	0.03	0.03	0.00	0.13	0.17	-0.04
South Korea	0.32	0.07	0.03	0.04	0.25	0.34	-0.09
Australia	0.14	0.00	0.01	-0.00	0.14	0.18	-0.05
New Zealand	0.18	0.01	0.02	-0.01	0.17	0.22	-0.06
Brunei	0.40	0.11	0.11	-0.00	0.29	0.39	-0.10
Indonesia	0.19	0.01	0.01	-0.00	0.18	0.25	-0.06
Cambodia	0.74	-0.01	0.00	-0.02	0.75	1.00	-0.25
Laos	0.33	0.00	0.02	-0.01	0.33	0.44	-0.11
Myanmar	0.19	-0.05	-0.05	-0.00	0.23	0.31	-0.08
Malaysia	1.12	0.19	0.13	0.05	0.93	1.25	-0.32
Philippines	0.24	-0.01	-0.00	-0.01	0.25	0.34	-0.09
Singapore	0.24	0.09	0.09	0.00	0.15	0.20	-0.05
Thailand	0.58	0.07	0.08	-0.00	0.50	0.68	-0.18
Vietnam	1.19	0.06	0.08	-0.01	1.12	1.51	-0.39
Average	0.41	0.04	0.04	0.00	0.37	0.49	-0.13

quantitative trade. They argue that conditional on the change in the domestic trade share and the estimated trade elasticity with respect to the variable trade cost, the quantitative results for gains from trade are equivalent regardless of whether researchers use Ricardian or Melitz-Chaney model.

The reason for the equivalent result is that, in Melitz-Chaney model, domestic productivity gains from exit of domestic firms are counteracted exactly by import productivity losses due to entry of foreign firms, and in a similar way, import variety gains are offset completely by domestic variety losses. As a result, the net “new” gains vanish in figures when accounting for all these gains and losses, which left only the “traditional” gains. In Ricardian model represented by [Eaton and Kortum \(2002\)](#), there are no “new” welfare gains because the market is perfectly competitive and the set of consumption is fixed. Therefore, ACR derived that trade liberalization generates gains from trade only through the “traditional” channel and there are no extra gains in Melitz-Chaney model compared to Ricardian model.

However, ACR focused on the changes in iceberg trade cost only and omitted the case of tariff reductions. As analyzed in equation (10), the impact of tariff reductions on firm’s entry decision is quite distinct from the iceberg trade cost. Tariff reductions affect entry of firms through both competitiveness channel and tax saving channel, whereas decrease in iceberg trade cost only through competitiveness channel. Decrease in variable trade cost, regardless of tariffs or iceberg trade costs, strengthens relative competitiveness of foreign firms and motivates their entry, meanwhile it also weakens domestic firms and forces them to exit. Overall, the competitiveness channel contributes no changes to total amount of firms serving a market and the net “new” welfare remains unchanged, just as ACR argued. Nevertheless, import tariff reductions further promote entry of foreign firms by means of saving tax paid by them in destinations and increasing their export profits, while domestic firms are not affected in this channel. Consequently, tariff reductions induce more entry of foreign firms and generate the net “new” gains through the tax saving channel, which can be reflected by equation (41) and (43) upon aggregated across domestic and foreign origins.

To clarify the mechanism through which tariff reductions affect market entry and the “new” welfare, we distinguish two types of tariff under the Melitz-Chaney model. The first is to assume that tariffs are imposed before markups. In this case they are isomorphic to the iceberg trade cost in modelling and affect entry only through the competitiveness channel. The second is the

Table 10 Trade and welfare effects of RCEP across different models (%)

Members	Welfare			Growth in varieties consumed		
	Ricardian	Melitz-Chaney		Ricardian	Melitz-Chaney	
		Before markup	Benchmark		Before markup	Benchmark
	(1)	(2)	(3)	(4)	(5)	(6)
China	0.18	0.11	0.37	0.00	-0.05	0.11
Japan	0.27	0.28	0.50	0.00	0.06	0.22
South Korea	0.26	0.30	0.49	0.00	0.05	0.22
Australia	0.12	0.18	0.25	0.00	0.10	0.15
New Zealand	0.31	0.43	0.48	0.00	0.23	0.26
Brunei	0.12	0.12	0.14	0.00	-0.02	-0.03
Indonesia	0.32	0.35	0.67	0.00	0.09	0.32
Cambodia	1.19	1.32	2.37	0.00	-0.36	1.77
Laos	0.65	0.83	1.02	0.00	0.34	0.51
Myanmar	0.15	0.19	0.26	0.00	0.05	0.10
Malaysia	15.04	11.52	22.20	0.00	0.30	8.56
Philippines	0.05	0.09	0.12	0.00	0.03	0.10
Singapore	0.07	0.07	0.11	0.00	0.00	0.00
Thailand	0.82	0.79	1.93	0.00	0.07	0.73
Vietnam	3.25	2.34	5.73	0.00	-0.30	1.43
Average	1.52	1.26	2.44	0.00	0.04	0.96

Table 11 Welfare effects of RCEP using the Melitz-Chaney model with before-markup tariff (%)

Members	“Traditional” gains	“New” gains					
		Variety			Productivity		
		Total	Domestic exit	Foreign entry	Total	Domestic exit	Foreign entry
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
China	0.15	-0.05	-0.57	0.52	0.01	0.14	-0.13
Japan	0.25	0.04	-0.74	0.78	-0.01	0.18	-0.19
South Korea	0.27	0.04	-0.97	1.01	-0.01	0.25	-0.26
Australia	0.13	0.07	-0.26	0.33	-0.02	0.07	-0.08
New Zealand	0.31	0.17	-0.73	0.90	-0.04	0.18	-0.22
Brunei	0.13	-0.01	-0.21	0.20	0.00	0.05	-0.05
Indonesia	0.30	0.06	-0.90	0.96	-0.01	0.22	-0.24
Cambodia	1.49	-0.23	-1.83	1.60	0.06	0.46	-0.40
Laos	0.67	0.21	-0.90	1.11	-0.05	0.22	-0.27
Myanmar	0.16	0.04	-0.50	0.55	-0.01	0.13	-0.14
Malaysia	11.33	0.26	-5.49	5.75	-0.07	1.32	-1.39
Philippines	0.07	0.02	-0.24	0.26	-0.00	0.06	-0.06
Singapore	0.07	0.00	-0.16	0.16	-0.00	0.04	-0.04
Thailand	0.74	0.06	-1.75	1.81	-0.02	0.44	-0.46
Vietnam	2.63	-0.39	-4.68	4.28	0.10	1.15	-1.05
Average	1.25	0.02	-1.33	1.35	-0.00	0.33	-0.33

practical case analyzed in the benchmark model, where tariffs are imposed after markups and affect entry through both competitiveness channel and tax saving channel. Then this section compares the results from the benchmark model of our analysis (the Melitz-Chaney model with tariff imposed after markup) to the Ricardian model and the Melitz-Chaney model with before-markup tariff. We calibrate each of two models to the same Input-Output data as the one used to calibrate benchmark model by setting $\delta = 0$ and $\delta = 1$, $\zeta = 0$, respectively and use the same trade elasticity estimated in Table 1.¹¹ Table 10 presents the trade and welfare effects implied by these models.

As ACR argued, when the tariffs are imposed before markups, the welfare effects for the Melitz-Chaney model are very similar to the Ricardian model for each member, with an average of 1.26% compared to 1.52%. Table 11 shows the decomposition of the welfare effects implied by the Melitz-Chaney model in this case. The results indicate that almost all of the welfare effects arise from the “traditional” gains, with 1.25% on average. The import variety gains of 1.35% are offset by the domestic variety losses of -1.33%, while the domestic productivity gains are almost entirely counteracted by the import productivity losses. As a result, the net “new” gains are only 0.02% on average and substantially lower than the results from the benchmark model in our analysis. This reflects that tariff reductions bring about the net “new” welfare gains wholly through the tax saving channel, which works only in the Melitz-Chaney model with tariffs imposed after markups.

Trade effects are also similar across the Ricardian model and the Melitz-Chaney model with tariffs imposed before markup. The last three columns in table 10 present the growth of varieties consumed in each member implied by these models. As we can see, the varieties each member consumes are unchanged in the Ricardian model because it assumes that the set of goods access to consumers is fixed. The varieties growth in the Melitz-Chaney model where tariffs are imposed before markup is irregular, positive for some members while negative for others, but the average growth among all members is only 0.04%, much lower than the growth of 0.96% implied by the benchmark model. Overall, the above results unmask the implication of the entry effects in the Melitz-Chaney model and the importance of the after-markup tariffs on such effects.

7 Conclusion

We develop and apply an extended Melitz–Chaney framework calibrated with firm-level data and embedded in a multi-country, multi-sector general equilibrium model with input–output linkages to quantify the trade and welfare effects of RCEP’s tariff reductions from the perspective of heterogeneous firms. The model jointly captures aggregate trade responses and micro-level firm dynamics, allowing a clear decomposition of welfare impacts into “traditional” and “new” components.

The results show that these new welfare gains—arising from endogenous firm selection, entry, and variety expansion—are quantitatively and economically significant and should be considered alongside conventional measures of trade gains. We also find that the magnitude of tariff reductions primarily drives these “new” welfare effects while shape parameter of productivity distribution and even relative productivity heterogeneity between domestic and foreign firms also have considerable impacts on it.

Using Chinese firm-level data, the analysis quantifies RCEP’s effects on Chinese exporters and welfare. Although microdata from other member economies are unavailable, the observed firm reallocation patterns and sectoral heterogeneity in China provide useful insights for other participants. The findings indicate that tariff reductions under RCEP stimulate foreign entries that exceed domestic exits, leading to net expansion of product varieties and positive “new” welfare gains.

The model’s results imply that heterogeneous adjustment across firms underlies the aggregate welfare effects of trade liberalization. Tariff reductions encourage marginal exporters to enter regional markets and exploit their product or marketing advantages, contributing to variety-induced welfare gains. At the same time, less productive firms exit under heightened competition, raising

¹¹In the Ricardian model, productivity is drawn from a Fréchet distribution and the trade elasticity equals the shape parameter of that distribution; in the Melitz model, productivity follows a Pareto distribution and the trade elasticity is likewise given by the Pareto shape parameter. To ensure comparability across models, we set the trade elasticity to a common value — the Pareto shape parameter estimated in our analysis — for all specifications.

aggregate productivity but possibly displacing workers. Governments should therefore complement liberalization with targeted training programs and reemployment support to ensure smooth labor reallocation from exiting to more productive firms. For sustained gains, regional members should also strengthen technological upgrading so that new entrants become more productive and competitive internationally.

From China’s perspective, increased import competition—especially from sectors such as electronics—compresses domestic market shares, but these effects can be offset by reducing internal trade barriers and deepening domestic integration. Furthermore, while RCEP has achieved tariff liberalization for over 90% of intra-regional trade, numerous non-tariff and regulatory barriers still constrain foreign entry. Reducing these behind-the-border restrictions and advancing regulatory convergence would magnify the welfare benefits and promote sustainable regional economic integration.

While we examine the trade and welfare effects of RCEP tariff reductions through the lens of heterogeneous firms, several critical issues deserve future exploration. Firstly, we only simulate the impact of RCEP tariff cuts on firm entry, exit, and the consequent “new” welfare gains *ex ante*, but provide no empirical evidence on firms’ actual responses to RCEP. If richer firm-level data can be found, future work could incorporate observed changes in firm dynamics into the formula of “new” welfare effects to account for the actual “new” gains from RCEP (Hsieh et al., 2020). Secondly, modern trade agreements are governed by rules of origin (RoO): preferential tariffs apply only to products whose regional content satisfies a specified threshold. Compliance with RoO is costly for firms, which may deter them from adjusting sourcing decisions and obtaining preferential tax, creating a gap between the tariffs firms actually face and the statutory tariffs set out in the agreement (Teti, 2024). However, data on effectively applied tariffs at the firm level is unavailable. Future research on trade agreement evaluation should therefore compile more granular tariff data to mitigate the bias in quantitative results. Another promising direction for future research is to quantify the optimal level of RCEP’s regional content requirement, for instance by applying the “Laffer curve” for rules of origin developed by Head et al. (2024) to assess whether the current 40% threshold is optimal. This would, however, require detailed product-level data on regional sourcing and a model of firms’ sourcing behavior under RoO. Thirdly, RCEP extends beyond tariff cuts—it also includes investment facilitation and regulatory cooperation—suggesting that other liberalization channels may generate additional unmeasured welfare improvements, which merit future research.

Disclosure statement

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Appendix

A Factors Affecting Welfare Effects

A.1 Volume of Trade

According to the definition of trade volume $M_{ni}^j = \frac{\pi_{ni}^j}{1+t_{ni}^j} X_n^j$, we have

$$d \ln M_{ni}^j = d \ln \pi_{ni}^j + d \ln X_n^j - d \ln (1 + t_{ni}^j) \quad (55)$$

From equation (13), we can derive the change in trade share as

$$d \ln \pi_{ni}^j = - \left(\theta^j + \tilde{\theta}^j - 1 \right) d \ln \left(1 + t_{ni}^j \right) - \theta^j d \ln c_i^j - \left(\tilde{\theta}^j - 1 \right) d \ln \left(w_n F_{ni}^j \right) - d \ln \Phi_n^j \quad (56)$$

where $\Phi_n^j = \sum_{l=1}^N T_l^j \kappa_1^j \left[\tilde{\sigma}^j \tau_{nl}^j \left(1 + t_{nl}^j \right) c_l^j \right]^{-\theta^j} \left[\sigma^j E_{nl}^j \left(1 + t_{nl}^j \right)^\zeta \right]^{-(\tilde{\theta}^j - 1)}$.

Substituting the above equation into $d \ln M_{ni}^j - d \ln c_i^j$, we can obtain the change in the bilateral volume of trade:

$$d \ln M_{ni}^j - d \ln c_i^j = -\tilde{\sigma}^j \theta^j d \ln \left(1 + t_{ni}^j \right) - (\theta^j + 1) d \ln c_i^j + d \ln X_n^j - \left(\tilde{\theta}^j - 1 \right) d \ln \left(w_n F_{ni}^j \right) - d \ln \Phi_n^j \quad (57)$$

Substituting the above equation into the expression of volume of trade effect (36), we have

$$d \ln \text{vot}_n^j = -\tilde{\sigma}^j \sum_{i=1}^N \left[\frac{t_{ni}^j}{(1+t_{ni}^j)} \pi_{ni}^j \theta^j d \ln \left(1 + t_{ni}^j \right) \right] + \epsilon_n^j \quad (58)$$

where ϵ_n^j is the residual term that captures the effects of changes in domestic prices, foreign prices, and other factors on the volume of trade¹².

A.2 Variety

The number of firms from country i that sell in country n is given by equation (14). Totally differentiating it we have

$$d \ln V_{ni}^j = -\theta^j d \ln \left[\frac{(1+t_{ni}^j) c_i^j}{P_n^j} \right] + \tilde{\theta}^j d \ln \left[\frac{X_n^j}{E_{ni}^j (1+t_{ni}^j)} \right] \quad (59)$$

Totally differentiating the price index in equation (12), we have

$$d \ln P_n^j = \sum_{i=1}^N \pi_{ni}^j \left[d \ln \left(1 + t_{ni}^j \right) c_i^j + \left(\frac{1}{\theta^j} - \frac{1}{\sigma^j - 1} \right) d \ln \frac{X_n^j}{E_{ni}^j (1+t_{ni}^j)} \right] \quad (60)$$

Substituting the above equation into the expression of $d \ln V_{ni}^j$ and aggregating it across origins as equation (37), we can obtain

$$d \ln \text{var}_n^j = \frac{1}{\sigma^j - 1} \left[- \sum_{i=1}^N \pi_{ni}^j d \ln \left(1 + t_{ni}^j \right) + d \ln \frac{X_n^j}{w_n} \right] \quad (61)$$

¹²

$$\epsilon_n^j = \sum_{i=1}^N \left\{ \frac{t_{ni}^j \pi_{ni}^j}{(1+t_{ni}^j)} \left[-(\theta^j + 1) d \ln c_i^j + d \ln X_n^j - (\tilde{\theta}^j - 1) d \ln \left(w_n F_{ni}^j \right) - d \ln \Phi_n^j \right] \right\}$$

A.3 Productivity

With Pareto distribution, the average productivity of firms from country i that sell in country n is given by

$$\tilde{\phi}_{ni}^j = \kappa_3^j \frac{\theta^j}{\theta^j - (\sigma^j - 1)} \frac{\tilde{\sigma}^j \tau_{ni}^j (1 + t_{ni}^j) c_i^j}{P_n^j} \left[\frac{X_n^j}{\sigma^j E_{ni}^j (1 + t_{ni}^j)} \right]^{\frac{1}{1 - \sigma^j}} \quad (62)$$

where $\kappa_3^j = \int \eta^{\frac{1}{1 - \sigma^j}} g^j(\eta) d\eta$.

Totally differentiating it we have

$$d \ln \tilde{\phi}_{ni}^j = d \ln \left[\frac{(1 + t_{ni}^j) c_i^j}{P_n^j} \right] - \frac{1}{\sigma^j - 1} d \ln \left[\frac{X_n^j}{E_{ni}^j (1 + t_{ni}^j)} \right] \quad (63)$$

Substituting the differentiated price index (60) into above expression and aggregating it across origins as equation (38), we can obtain

$$d \ln prod_n^j = \frac{1}{\theta^j} \left[\sum_{i=1}^N \pi_{ni}^j d \ln (1 + t_{ni}^j) - d \ln \frac{X_n^j}{w_n} \right] \quad (64)$$

B Differentiating Shape Parameters of Productivity Distribution by Country

B.1 Aggregation

The basic setting of the extended model is the same as the benchmark model except the specification of the productivity distribution, which is now allowed to vary by country. Specifically, we assume that the productivity level is independent across firms and drawn from a Pareto distribution, G_i^j , with different location and dispersion parameter specific to country and sector, T_i^j and θ_i^j :

$$G_i^j(\phi) = 1 - T_i^j \phi^{-\theta_i^j} \quad (65)$$

With the CES demand function, the price index for each country and sector satisfies the following equation:

$$\sum_{i=1}^N \left\{ T_i^j \kappa_{1,i}^j \left[\frac{\tilde{\sigma}^j \tau_{ni}^j (1 + t_{ni}^j) c_i^j}{P_n^j} \right]^{-\theta_i^j} \left[\frac{X_n^j}{\sigma^j E_{ni}^j (1 + t_{ni}^j)^\zeta} \right]^{(\tilde{\theta}_i^j - 1)} \right\} = 1 \quad (66)$$

where $\kappa_{1,i}^j = \kappa_{0,i}^j \int \alpha \eta^{\tilde{\theta}_i^j - 1} g^j(\alpha, \eta) d\alpha d\eta$, $\kappa_{0,i}^j = \frac{\tilde{\theta}_i^j}{\tilde{\theta}_i^j - 1} - \frac{\tilde{\theta}_i^j}{\tilde{\theta}_i^j + (\lambda^j - 1)}$ and $\tilde{\theta}_i^j = \frac{\theta_i^j}{\sigma^j - 1}$. $g^j(\alpha, \eta)$ is the joint probability density function of the preference shock α and entry shock η .

The share of expenditure in country n on sector j goods from country i is given by

$$\pi_{ni}^j = \frac{T_i^j \kappa_{1,i}^j \left[\frac{\tilde{\sigma}^j \tau_{ni}^j (1 + t_{ni}^j) c_i^j}{P_n^j} \right]^{-\theta_i^j} \left[\frac{X_n^j}{\sigma^j E_{ni}^j (1 + t_{ni}^j)^\zeta} \right]^{(\tilde{\theta}_i^j - 1)}}{\sum_{l=1}^N T_l^j \kappa_{1,l}^j \left[\frac{\tilde{\sigma}^j \tau_{nl}^j (1 + t_{nl}^j) c_l^j}{P_n^j} \right]^{-\theta_l^j} \left[\frac{X_n^j}{\sigma^j E_{nl}^j (1 + t_{nl}^j)^\zeta} \right]^{(\tilde{\theta}_l^j - 1)}} \quad (67)$$

Total measure of firms from country i that sell in n is expressed as

$$V_{ni}^j = \frac{\kappa_{2,i}^j}{\kappa_{1,i}^j} \frac{\pi_{ni}^j X_n^j}{\sigma^j E_{ni}^j (1 + t_{ni}^j)^\zeta} \quad (68)$$

where $\kappa_{2,i}^j = \int \eta^{\tilde{\theta}_i^j} g^j(\eta) d\eta$ and $g^j(\eta)$ is the marginal probability density of $g^j(\alpha, \eta)$.

B.2 Equilibrium Using the Exact Hat Algebra

The full hat algebra representation of the equilibrium system is given by:

$$\hat{c}_n^j = \hat{w}_n^{\gamma_n^j} \prod_{k=1}^J \hat{P}_n^k \gamma_n^{k,j} \quad (69)$$

$$\sum_{i=1}^N \left\{ \pi_{ni}^j \left[\frac{\hat{\tau}_{ni}^j (1 + t_{ni}^j) \hat{c}_i^j}{\hat{P}_n^j} \right]^{-\theta_i^j} \left[\frac{\hat{X}_n^j}{\hat{w}_n \hat{F}_{ni}^j (1 + t_{ni}^j)^\zeta} \right]^{\delta(\tilde{\theta}_i^j - 1)} \right\} = 1 \quad (70)$$

$$\hat{\pi}_{ni}^j = \frac{\pi_{ni}^j \left[\frac{\hat{\tau}_{ni}^j (1 + t_{ni}^j) \hat{c}_i^j}{\hat{P}_n^j} \right]^{-\theta_i^j} \left[\frac{\hat{X}_n^j}{\hat{w}_n \hat{F}_{ni}^j (1 + t_{ni}^j)^\zeta} \right]^{\delta(\tilde{\theta}_i^j - 1)}}{\sum_{l=1}^N \pi_{nl}^j \left[\frac{\hat{\tau}_{nl}^j (1 + t_{nl}^j) \hat{c}_l^j}{\hat{P}_n^j} \right]^{-\theta_l^j} \left[\frac{\hat{X}_n^j}{\hat{w}_n \hat{F}_{nl}^j (1 + t_{nl}^j)^\zeta} \right]^{\delta(\tilde{\theta}_l^j - 1)}} \quad (71)$$

$$R_n^{j'} = \sum_{i=1}^N \left(1 + \frac{t_{in}^{j'}}{\sigma^j} \right)^{\delta(1-\zeta)} \frac{\pi_{in}^{j'}}{1 + t_{in}^{j'}} X_i^{j'} \quad (72)$$

$$X_n^{j'} = \beta_n^j (\hat{w}_n w_n L_n + T'_n + D_n) + \sum_{k=1}^J \gamma_n^{j,k} R_n^{k'} \quad (73)$$

$$\hat{w}_n w_n L_n = \sum_{k=1}^J \gamma_n^{k,j} R_n^{k'} \quad (74)$$

As we can see, the equilibrium system using the exact hat algebra is the same as the one in the benchmark model, except the expressions of change in price index (70) and trade share (71).

B.3 Welfare Effects from Tariff Reductions

The welfare decomposition is also the same as the benchmark model:

$$\begin{aligned}
d \ln W_n = & \frac{1}{I_n} \sum_{j=1}^J \sum_{i=1}^N \underbrace{\left[\left(1 + \frac{t_{ni}^j}{\sigma^j} \right)^{\delta(1-\zeta)} E_{ni}^j d \ln c_n^j - M_{ni}^j d \ln c_i^j \right]}_{\text{Terms of trade}} \\
& + \frac{1}{I_n} \sum_{j=1}^J \sum_{i=1}^N \underbrace{t_{ni}^j M_{ni}^j \left[\left(\frac{\sigma^j - 1}{\sigma^j} \right)^{\delta(1-\zeta)} d \ln M_{ni}^j - d \ln c_i^j - \frac{\delta(1-\zeta)}{\sigma^j} d \ln t_{ni}^j \right]}_{\text{Volume of trade}} \\
& + \frac{\delta}{I_n} \sum_{j=1}^J \sum_{i=1}^N \underbrace{\left[\frac{(1+t_{ni}^j) M_{ni}^j}{\sigma^j - 1} d \ln V_{ni}^j \right]}_{\text{Variety}} + \frac{\delta}{I_n} \sum_{j=1}^J \sum_{i=1}^N \underbrace{\left[(1+t_{ni}^j) M_{ni}^j d \ln \tilde{\phi}_{ni}^j \right]}_{\text{Productivity}}
\end{aligned} \tag{75}$$

Volume of Trade From equation (67), we can derive the change in trade share as

$$d \ln \pi_{ni}^j = - \left(\theta_i^j + \tilde{\theta}_i^j - 1 \right) d \ln \left(1 + t_{ni}^j \right) - \theta_i^j d \ln \left(\frac{c_i^j}{P_n^j} \right) + \left(\tilde{\theta}_i^j - 1 \right) d \ln \left(\frac{X_n^j}{w_n F_{ni}^j} \right) - d \ln \Phi_n^j \tag{76}$$

$$\text{where } \Phi_n^j = \sum_{l=1}^N T_l^j \kappa_{1,l}^j \left[\frac{\tilde{\sigma}^j \tau_{nl}^j (1+t_{nl}^j) c_l^j}{P_n^j} \right]^{-\theta_l^j} \left[\frac{X_n^j}{\sigma^j E_{nl}^j (1+t_{nl}^j)^\zeta} \right]^{(\tilde{\theta}_l^j - 1)}.$$

Substituting the above equation into the expression of $d \ln M_{ni}^j - d \ln c_i^j$, we can obtain the change in the volume of trade by country and sector:

$$d \ln M_{ni}^j - d \ln c_i^j = -\tilde{\sigma}^j \theta_i^j d \ln \left(1 + t_{ni}^j \right) - \left(\theta_i^j + 1 \right) d \ln c_i^j + \theta_i^j d \ln P_n^j + \tilde{\theta}_i^j d \ln X_n^j - \left(\tilde{\theta}_i^j - 1 \right) d \ln \left(w_n F_{ni}^j \right) - d \ln \Phi_n^j \tag{77}$$

Substituting the above equation into the expression of $d \ln \text{vot}_n^j$, we can obtain

$$d \ln \text{vot}_n^j = -\tilde{\sigma}^j \sum_{i=1}^N \left[\frac{t_{ni}^j}{(1+t_{ni}^j)} \pi_{ni}^j \theta_i^j d \ln \left(1 + t_{ni}^j \right) \right] + \epsilon_n^j \tag{78}$$

where ϵ_n^j is the residual term that captures the effects of changes in domestic prices, foreign prices, and other factors on the volume of trade¹³.

Differentiating the trade volume effect with respect to tariff changes, we have

$$\frac{d \ln \text{vot}_n^j}{d \ln (1 + t_{ni}^j)} \approx -\tilde{\sigma}^j \frac{t_{ni}^j}{(1 + t_{ni}^j)} \pi_{ni}^j \theta_i^j \tag{79}$$

As we can see, the volume of trade effect is positively related to the magnitude of tariff reductions. Countries committed to higher tariff reductions will experience larger welfare gains through the expanded volume of trade. Moreover, this positive relationship is also affected by the shape parameter of productivity distribution θ_i^j . Because larger θ_i^j implies less firm heterogeneity, which further means higher trade elasticity under Melitz-Chaney model. So if a country has more import from origins with larger shape parameters, the trade elasticity between them will also be higher, and tariff reduction can thus create more trade and generate larger gains. In a word, the dispersion parameter of productivity distribution θ_i^j has positively moderating role in the volume of trade effects of tariff reduction.

¹³

$$\epsilon_n^j = \sum_{i=1}^N \left\{ \frac{t_{ni}^j \pi_{ni}^j}{(1+t_{ni}^j)} \left[- \left(\theta_i^j + 1 \right) d \ln c_i^j + \theta_i^j d \ln P_n^j + \tilde{\theta}_i^j d \ln X_n^j - \left(\tilde{\theta}_i^j - 1 \right) d \ln \left(w_n F_{ni}^j \right) - d \ln \Phi_n^j \right] \right\}$$

Variety The number of firms from country i that sell in country n is given by equation (68). Totally differentiating it we have

$$d \ln V_{ni}^j = -\theta_i^j d \ln \left[\frac{(1+t_{ni}^j) c_i^j}{P_n^j} \right] + \tilde{\theta}_i^j d \ln \left[\frac{X_n^j}{E_{ni}^j (1+t_{ni}^j)} \right] \quad (80)$$

Totally differentiating the price index in equation (66), we have

$$d \ln P_n^j = \sum_{i=1}^N \pi_{ni}^j \iota_{ni}^j \left[d \ln (1+t_{ni}^j) c_i^j + \left(\frac{1}{\theta_i^j} - \frac{1}{\sigma^j - 1} \right) d \ln \frac{X_n^j}{E_{ni}^j (1+t_{ni}^j)} \right] \quad (81)$$

where $\iota_{ni}^j = \frac{\theta_i^j}{\sum_{l=1}^N \pi_{nl}^j \theta_l^j}$.

Substituting the above equation into the expression of $d \ln V_{ni}^j$ and aggregating it across origins as equation (37), we can obtain

$$d \ln var_n^j = \frac{1}{\sigma^j - 1} \left[- \sum_{i=1}^N \pi_{ni}^j d \ln (1+t_{ni}^j) + d \ln \frac{X_n^j}{w_n} \right] \quad (82)$$

The above equation indicates that the variety effect rises positively with the magnitude of tariff liberalization — countries undertaking more cuts reap larger welfare gains from expanded import variety. But unlike the volume of trade effect, the productivity dispersion parameter θ_i^j exerts no explicit moderating role on the variety gains from tariff reductions. Intuitively, the variety effect is directly determined by the number of firms that can enter the market, which is only affected by the entry cost and the market size, just as (82). The shape parameter θ_i^j affects the distribution of productivity among firms, but it does not directly influence the number of firms that can enter the market. Therefore, while θ_i^j may indirectly affect the variety effect through its impact on market size, it does not have a direct moderating role in the variety effects of tariff reduction.

Productivity With Pareto distribution, the average productivity of firms from country i that sell in country n is given by

$$\tilde{\phi}_{ni}^j = \kappa_3^j \frac{\theta_i^j}{\theta_i^j - (\sigma^j - 1)} \frac{\tilde{\sigma}^j \tau_{ni}^j (1+t_{ni}^j) c_i^j}{P_n^j} \left[\frac{X_n^j}{\sigma^j E_{ni}^j (1+t_{ni}^j)} \right]^{\frac{1}{1-\sigma^j}} \quad (83)$$

where $\kappa_3^j = \int \eta^{\frac{1}{1-\sigma^j}} g^j(\eta) d\eta$.

Totally differentiating it we have

$$d \ln \tilde{\phi}_{ni}^j = d \ln \left[\frac{(1+t_{ni}^j) c_i^j}{P_n^j} \right] - \frac{1}{\sigma^j - 1} d \ln \left[\frac{X_n^j}{E_{ni}^j (1+t_{ni}^j)} \right] \quad (84)$$

Substituting the differentiated price index (81) into above expression and aggregating it across origins, we can obtain

$$\begin{aligned} d \ln prod_n^j &= \frac{1}{\sum_{i=1}^N \pi_{ni}^j \theta_i^j} \left[\sum_{i=1}^N \pi_{ni}^j d \ln (1+t_{ni}^j) - d \ln \frac{X_n^j}{w_n} \right] \\ &+ \sum_{i=1}^N \pi_{ni}^j (1 - \iota_{ni}^j) \left[d \ln (1+t_{ni}^j) c_i^j - \frac{1}{\sigma^j - 1} d \ln \frac{X_n^j}{w_n (1+t_{ni}^j)} \right] \end{aligned} \quad (85)$$

where $\iota_{ni}^j = \frac{\theta_i^j}{\sum_{l=1}^N \pi_{nl}^j \theta_l^j}$.

Suppose that the magnitudes of tariff reductions are the same across all origins, that is, $\forall i \neq$

$n, d \ln (1 + t_{ni}^j) \equiv d \ln (\widetilde{1 + t_n^j})$. Taking the derivative of the productivity effect with respect to this uniform tariff changes $d \ln (\widetilde{1 + t_n^j})$, we have

$$\frac{d \ln prod_n^j}{d \ln (\widetilde{1 + t_n^j})} = \frac{1 - \pi_{nn}^j}{\sum_{i=1}^N \pi_{ni}^j \theta_i^j} + \tilde{\sigma}^j \pi_{nn}^j \left(\frac{\theta_n^j}{\sum_{i=1}^N \pi_{ni}^j \theta_i^j} - 1 \right) \quad (86)$$

It is evident that the sign of the derivative depends on the relative value of θ_n^j to the weighted average of θ_i^j across all origins $\sum_{i=1}^N \pi_{ni}^j \theta_i^j$. If θ_n^j is equals to or greater than the weighted average, then the derivative must be positive, indicating that a uniform tariff reduction will lead to decrease in productivity. Conversely, if θ_n^j is less than the weighted average heavily, the derivative might be negative, indicating a increase in productivity. More precise expression for the sign of the derivative is given by:

$$\frac{d \ln prod_n^j}{d \ln (\widetilde{1 + t_n^j})} \begin{cases} > 0, & \text{if } \theta_n^j > \sum_{i=1}^N \pi_{ni}^j \theta_i^j + \frac{1}{\tilde{\sigma}^j} \left(1 - \frac{1}{\pi_{nn}^j} \right), \\ < 0, & \text{if } \theta_n^j < \sum_{i=1}^N \pi_{ni}^j \theta_i^j + \frac{1}{\tilde{\sigma}^j} \left(1 - \frac{1}{\pi_{nn}^j} \right). \end{cases} \quad (87)$$

The sign of the derivative shows that the relative heterogeneity of domestic versus foreign firms' productivity moderates the productivity effect of tariff liberalization. Specifically, tariff reductions will raise average productivity when domestic firms are relatively more heterogeneous (smaller θ_n^j), and lower it when they are relatively less so (larger θ_n^j). If the productivity distributions of domestic and foreign firms are identical, that is, $\forall i, \theta_i^j = \theta^j$, the second term of productivity effect (85) and the derivative expression (86) both equal zero, reducing to the benchmark case as equation (64).

The moderating role of relative heterogeneity of productivity distribution is insightful. Intuitively, import tariff reductions induce exit of domestic firms and entry of foreign firms. Since the productivity levels of both marginally entering and exiting firms lie below incumbents', the exit of domestic firms raises average productivity while the entry of foreign firms lowers it. These two opposing forces jointly determine the direction of the change in average productivity. In our baseline analysis the tariffs are imposed after markups, so the effect of tariff reductions is isomorphic to that of a reduction in entry costs: the number of foreign entrants exceeds the number of domestic exits. When the productivity distributions of domestic and foreign firms are identical—that is, the degree of heterogeneity is the same across countries ($\forall i, \theta_i^j = \theta^j$)—entry of these extra less productive foreign firms leads to a decline in average productivity, just as shown in (64). When the productivity distributions differ across countries, however, if domestic firms exhibit substantially greater productivity heterogeneity (smaller θ_n^j) than their foreign counterparts, the productivity gain from domestic exit outweighs the productivity loss from foreign entry, resulting in an increase in average productivity—corresponding to the case where the derivative is negative. Conversely, if the productivity heterogeneity of domestic firms is equal to or lower (larger θ_n^j) than that of foreign firms, the productivity gain from domestic exit is smaller than the loss from foreign entry, leading to a decline in average productivity.

C Robustness Check

C.1 Welfare Effects with Alternative Tariffs from Teti (2024)

The Global Tariff Database from Teti (2024) provides four alternative sector-level tariff measures: (i) unweighted, (ii) trade-weighted using HS6-level trade flows, (iii) 95%-winsorized unweighted, and (iv) 95%-winsorized and trade-weighted using HS6-level trade flows. Our baseline results use the unweighted measure; this section reports welfare effects under the remaining three measures as robustness checks (Tables C.1 - C.3).

Figure C.1 shows that, across all three alternative specifications, the qualitative patterns remain consistent with the baseline: Malaysia, Vietnam, and Cambodia consistently experience the largest welfare gains, averaging 13.8%, 0.86%, and 1.06%, respectively. These welfare effects are

also driven primarily by the volume-of-trade effect and “new” welfare effects from their substantial tariff reductions. The “new” welfare effects—variety and productivity—of all members continue to account for a considerable share of total welfare changes, about 22.4%, 34.8%, and 37.5%, respectively.

Table C.1 Welfare effects from RCEP tariff reductions using weighted tariffs as baserates (%)

Members	Welfare	“Traditional” gains			“New” gains		
		Total	Terms of trade	Volume of trade	Total	Variety	Productivity
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
China	0.23	0.18	0.03	0.15	0.04	0.06	-0.01
Japan	0.32	0.24	0.13	0.11	0.08	0.10	-0.03
South Korea	5.69	4.89	0.28	4.61	0.80	1.08	-0.27
Australia	0.11	0.07	0.07	0.01	0.03	0.04	-0.01
New Zealand	0.30	0.20	0.18	0.02	0.10	0.13	-0.03
Brunei	-0.03	0.01	0.01	0.00	-0.04	-0.06	0.01
Indonesia	0.40	0.30	0.12	0.18	0.11	0.14	-0.03
Cambodia	14.21	10.48	0.11	10.37	3.73	4.96	-1.23
Laos	1.37	1.09	1.01	0.08	0.28	0.38	-0.09
Myanmar	1.05	0.72	0.58	0.13	0.34	0.44	-0.11
Malaysia	16.40	12.80	-0.82	13.62	3.60	4.73	-1.13
Philippines	0.77	0.65	-0.20	0.85	0.12	0.15	-0.04
Singapore	0.10	0.08	0.08	0.00	0.03	0.04	-0.01
Thailand	2.54	2.00	-0.04	2.04	0.54	0.71	-0.17
Vietnam	10.71	8.35	-0.71	9.06	2.36	3.09	-0.73
Average	3.61	2.80	0.06	2.75	0.81	1.07	-0.26

Table C.2 Welfare effects from RCEP tariff reductions using unweighted 95% winsorized tariffs as baserates (%)

Members	Welfare	“Traditional” gains			“New” gains		
		Total	Terms of trade	Volume of trade	Total	Variety	Productivity
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
China	0.04	0.03	-0.07	0.10	0.02	0.02	-0.01
Japan	0.31	0.24	0.18	0.06	0.07	0.10	-0.02
South Korea	0.19	0.12	0.13	-0.01	0.08	0.11	-0.03
Australia	0.07	0.05	0.05	0.00	0.03	0.03	-0.01
New Zealand	-0.13	-0.10	-0.09	-0.00	-0.03	-0.04	0.01
Brunei	0.03	0.04	0.04	0.00	-0.01	-0.01	0.00
Indonesia	0.10	0.07	0.02	0.05	0.03	0.05	-0.01
Cambodia	1.41	0.86	-0.37	1.23	0.55	0.72	-0.17
Laos	0.30	0.21	0.20	0.01	0.09	0.11	-0.03
Myanmar	-0.03	-0.03	-0.03	0.01	-0.00	-0.00	0.00
Malaysia	0.84	0.59	-0.20	0.79	0.25	0.33	-0.08
Philippines	0.01	-0.01	-0.01	-0.00	0.02	0.03	-0.01
Singapore	-0.02	-0.02	-0.02	0.00	-0.00	-0.00	0.00
Thailand	-0.07	-0.10	0.06	-0.17	0.04	0.05	-0.01
Vietnam	0.33	0.19	-0.18	0.36	0.15	0.19	-0.05
Average	0.23	0.14	-0.02	0.16	0.08	0.11	-0.03

Table C.3 Welfare effects from RCEP tariff reductions using weighted and 95% winsorized tariffs as baserates (%)

Members	Welfare	“Traditional” gains			“New” gains		
		Total	Terms of trade	Volume of trade	Total	Variety	Productivity
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
China	0.13	0.10	0.03	0.08	0.03	0.04	-0.01
Japan	0.24	0.18	0.11	0.07	0.06	0.09	-0.02
South Korea	0.14	0.07	0.09	-0.02	0.07	0.09	-0.02
Australia	0.10	0.07	0.06	0.01	0.03	0.04	-0.01
New Zealand	-0.22	-0.17	-0.17	0.01	-0.06	-0.07	0.02
Brunei	-0.02	-0.01	-0.01	0.00	-0.01	-0.01	0.00
Indonesia	0.23	0.16	0.07	0.10	0.07	0.09	-0.02
Cambodia	1.74	0.93	-0.32	1.24	0.81	1.08	-0.26
Laos	0.26	0.20	0.18	0.02	0.06	0.08	-0.02
Myanmar	0.41	0.26	0.18	0.08	0.16	0.21	-0.05
Malaysia	1.12	0.80	-0.26	1.06	0.31	0.41	-0.10
Philippines	0.20	0.17	-0.12	0.30	0.03	0.04	-0.01
Singapore	-0.00	-0.01	-0.01	0.00	0.01	0.01	-0.00
Thailand	0.12	0.05	0.04	0.01	0.07	0.10	-0.02
Vietnam	0.32	0.25	-0.17	0.42	0.07	0.08	-0.01
Average	0.32	0.20	-0.02	0.22	0.12	0.15	-0.04

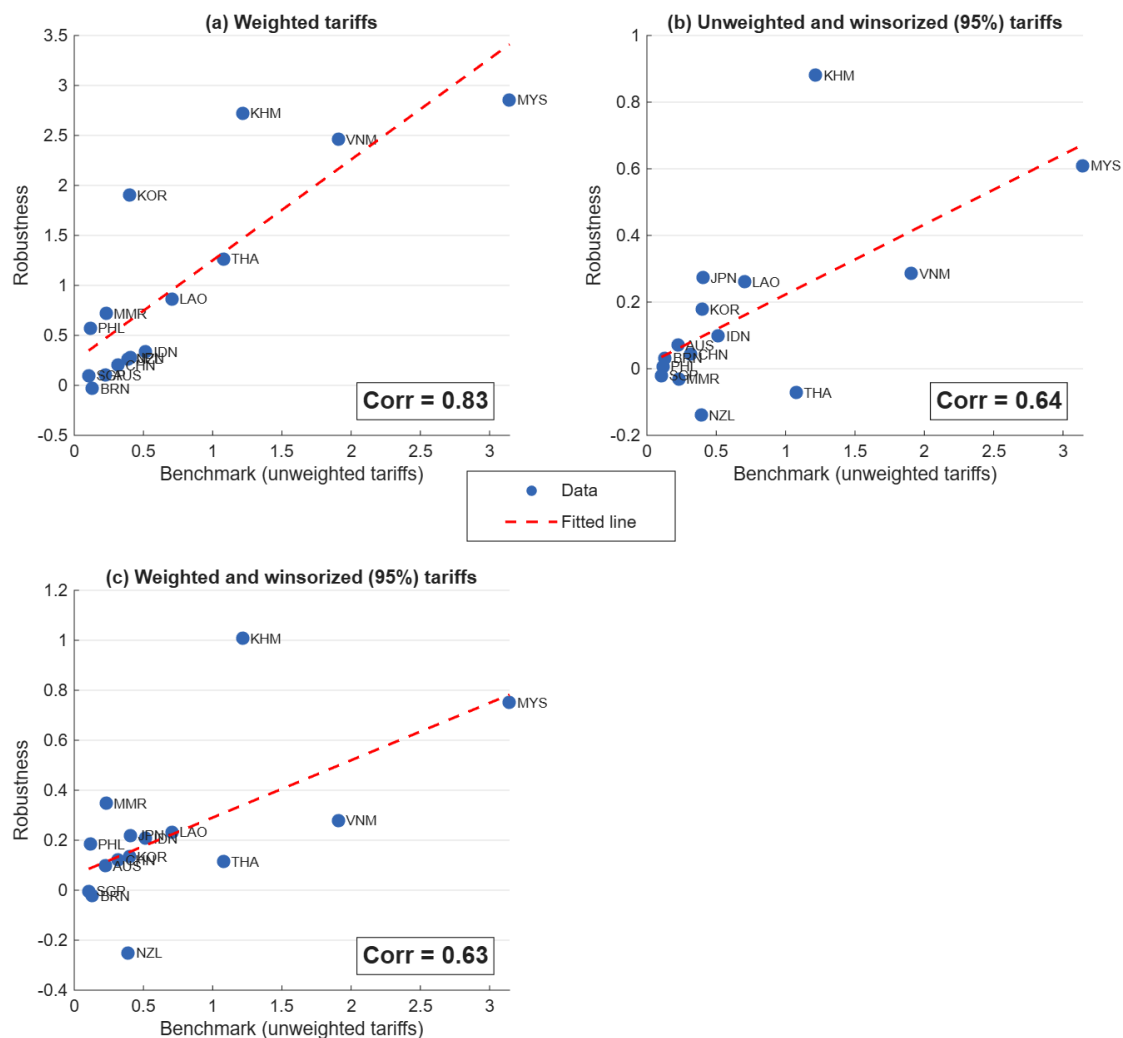


Figure C.1 Welfare effects of RCEP across different types of initial tariffs

Note: In all subfigures, the horizontal axis represents the baseline welfare effects computed using simple average tariffs unweighted by HS-6 digit trade flows as initial tariffs. The vertical axis of panel (a) represents welfare effects using weighted average tariffs by HS-6 digit product trade flows as initial tariffs. The vertical axis of panel (b) represents welfare effects using unweighted simple average tariffs with 95% winsorization at the HS-6 tariff line. The vertical axis of panel (c) represents welfare effects using weighted average tariffs with 95% winsorization at the HS-6 tariff line. Both the horizontal and vertical axes are presented on a log scale as $\ln(1 + x)$.

C.2 Welfare Effects of RCEP with Country-Specific Shape Parameters

As stated in section 3.5, RCEP encompasses a diverse set of countries at different stages of development, and the productivity distribution may differ across them. Proposition 6 argues that, allowing for country-specific shape parameter of productivity distribution affects the quantitative results primarily through two channels: the volume of trade effect and the productivity effect. If a country imports a large share of goods from partners with larger productivity heterogeneity, tariff reductions would amplify the volume of trade effect. Moreover, if a country’s productivity distribution is more heterogeneous than that of its import origins, the productivity gains from the exit of domestic firms may outweigh the productivity losses from the entry of foreign firms, yielding a net productivity gain rather than the productivity loss predicted by the benchmark model without shape parameter heterogeneity. Hence, when the shape parameters of the productivity distribution differ substantially across countries, the baseline results may no longer be robust.

To address this concern, we recalibrate the model of Appendix B.2 with country-specific shape parameter of productivity distribution. Specifically, we employ the country-sector level shape parameters estimated by Spearot (2016)¹⁴. We set $\sigma^j = 2.53$ following Fajgelbaum et al. (2020) to ensure that $\theta_i^j > \sigma_i^j - 1$, which is required for solution to firm’s optimization problem under CES preferences. We recalibrate this alternative model with the same parameters as in the benchmark model and recompute the welfare effects of RCEP tariff reductions using it.

Table C.4 shows that the cross-members variations in shape parameters within sectors are very small. The coefficient variance of θ_i^j across members is 0.19 on average—suggesting that the common-parameter assumption is not severely at odds with the data. Tables C.5 and C.6 report the welfare effects of RCEP tariff reductions for each member country when the shape parameter is allowed to differ across countries. Overall, the magnitude of the welfare effects is smaller than in the benchmark model, because both the trade elasticity and the substitution elasticity are lower in alternative model. Nevertheless, the relative gains across countries remain largely unchanged except Cambodia: the volume of trade effect continues to dominate the welfare changes, and Malaysia and Vietnam — who undertake the largest tariff reductions — still experience substantial volume of trade effects and variety gains. As for the productivity effect, for all members productivity gains from exit of domestic firms remain smaller than productivity losses from entry of foreign firms, so the net productivity effect keeps negative. Meanwhile, the net productivity loss does not exceed the variety gain, and member countries still experience an increase in the “new” welfare effects, with the exception of Cambodia. Cambodia’s productivity loss is so large that the new welfare effect turns negative and outweighs the volume of trade effect, leading to a net welfare loss.

The reason is that Cambodia commits substantial tariff reductions in the Coke and Petroleum sectors to China, Japan and South Korea (See Figures D.1 - D.3), where the productivity distributions differ markedly between Cambodia and these countries: Cambodia’s θ is 5.03, compared to 3.61, 3.26, and 2.48 for China, Japan, and South Korea, respectively. This implies that the firm productivity heterogeneity in Cambodia is far lower than in China, Japan, and South Korea. Consequently, for Cambodia, the productivity gains from the exit of domestic firms are substantially smaller than the productivity losses from the entry of foreign firms from China, Japan, and South Korea, leading to a net productivity loss so large that it outweighs the variety gain, resulting in an enormous loss from the “new” welfare effects.

Figure C.2 displays the correlations between benchmark model and the alternative model for each of four welfare components. We exclude Malaysia, Vietnam, and Cambodia from the figures due to their extreme value. We can see that the correlations for all different welfare effects all exceed 85%. The correlation pattern regarding productivity effect confirms the conclusion of Proposition 6: when the productivity distribution differs little between domestic and foreign

¹⁴Spearot (2016) is the first paper estimating country-sector-level shaper parameter of Pareto distribution. He combines the quadratic utility function of Melitz and Ottaviano (2008) with a Pareto productivity distribution and derives a structural equation linking trade flows, the vector of shape parameters of the productivity distribution, and tariff shocks. He then estimates the country-sector level Pareto shape parameters for 87 countries and 39 sectors, using his structural trade flow equation and data on tariff reductions and aggregate trade flows during the Uruguay Round. Since the estimates of Spearot (2016) for Brunei, Cambodia, Laos, and Myanmar are missing, we use the sectoral averages of the other ASEAN countries as substitutes. The mean values and standard deviations of θ_i^j across RCEP members is reported in Table C.4.

firms. Specifically, when the productivity heterogeneity of domestic firms relative to foreign firms is not sufficiently large, the productivity effect remains dominated by the entry of marginally less-productive foreign firms, thereby pulling down average productivity. Figure C.3 further reports the relative magnitude of the shape parameters of the productivity distribution between domestic and foreign firms $\theta_n^{j,rel.} \equiv \frac{\theta_n^j}{\sum_i \pi_{ni}^j \theta_i^j}$. We can see that most of values are concentrated around one, indicating that the shape parameters differ negligibly between domestic and foreign firms.

In summary, the above results show that observed country-level heterogeneity of productivity distribution is very small in general, and the quantitative results based on the benchmark model — which allows members share common shape parameters — are reasonably robust.

Table C.4 Summary statistics of Pareto shape parameters in [Spearot \(2016\)](#)

Sector	Mean (1)	Std. Dev. (2)	C.V. (3)
Food	3.41	0.76	0.22
Textile	1.60	0.13	0.08
Wood	1.57	0.12	0.08
Paper	1.79	0.17	0.09
Coke and Petroleum	4.93	2.31	0.47
Chemicals	1.84	0.07	0.04
Plastic	1.84	0.07	0.04
Minerals	1.89	0.26	0.14
Basic metals	2.69	1.14	0.42
Metal products	2.65	2.26	0.85
Electronics	1.83	0.14	0.08
Electrical	1.83	0.14	0.08
Machinery n.e.c	1.77	0.19	0.10
Auto	1.70	0.13	0.07
Other Transport	2.48	0.21	0.08
Other	1.61	0.19	0.12
Average	2.22	0.52	0.19

Note: Std. Dev. and C.V. denote the standard deviation and coefficient of variation of the RCEP member-level shape parameters within each sector, respectively.

Table C.5 Welfare effects from RCEP tariff reductions with country-specific shape parameters (%)

Members	Welfare	“Traditional” gains			“New” gains		
		Total	Terms of trade	Volume of trade	Total	Variety	Productivity
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
China	0.09	0.03	-0.02	0.06	0.06	0.07	-0.01
Japan	0.14	0.11	0.07	0.04	0.03	0.08	-0.05
South Korea	0.16	0.14	0.09	0.05	0.02	0.17	-0.15
Australia	0.06	0.05	0.05	0.00	0.01	0.04	-0.03
New Zealand	0.10	0.10	0.09	0.00	0.00	0.06	-0.06
Brunei	0.09	0.08	0.08	0.00	0.01	0.02	-0.01
Indonesia	0.31	0.26	0.21	0.05	0.05	0.23	-0.18
Cambodia	-0.12	0.13	-0.23	0.36	-0.25	1.08	-1.34
Laos	0.61	0.55	0.52	0.03	0.06	0.23	-0.17
Myanmar	0.10	0.09	0.08	0.01	0.01	0.08	-0.07
Malaysia	6.55	5.30	-0.15	5.45	1.25	3.72	-2.47
Philippines	0.02	0.01	0.00	0.00	0.01	0.06	-0.05
Singapore	0.01	0.00	0.00	0.00	0.01	0.01	-0.00
Thailand	0.27	0.24	0.14	0.10	0.04	0.27	-0.23
Vietnam	1.83	1.06	-0.17	1.22	0.77	1.09	-0.31
Average	0.68	0.54	0.05	0.49	0.14	0.48	-0.34

Table C.6 New welfare effects from RCEP tariff reductions with country-specific shape parameters (%)

Members	Variety			Productivity		
	Total	Domestic exit	Foreign entry	Total	Domestic exit	Foreign entry
	(1)	(2)	(3)	(4)	(5)	(6)
China	0.07	-0.25	0.32	-0.01	0.19	-0.20
Japan	0.08	-0.23	0.31	-0.05	0.14	-0.19
South Korea	0.17	-0.32	0.49	-0.15	0.24	-0.39
Australia	0.04	-0.03	0.07	-0.03	0.02	-0.05
New Zealand	0.06	-0.10	0.16	-0.06	0.05	-0.11
Brunei	0.02	-0.17	0.18	-0.01	0.06	-0.07
Indonesia	0.23	-0.31	0.54	-0.18	0.16	-0.34
Cambodia	1.08	-0.92	2.00	-1.34	0.61	-1.95
Laos	0.23	-0.27	0.50	-0.17	0.16	-0.34
Myanmar	0.08	-0.09	0.17	-0.07	0.07	-0.14
Malaysia	3.72	-3.32	7.04	-2.47	1.65	-4.12
Philippines	0.06	-0.06	0.12	-0.05	0.05	-0.10
Singapore	0.01	-0.01	0.02	-0.00	0.01	-0.01
Thailand	0.27	-0.51	0.78	-0.23	0.34	-0.57
Vietnam	1.09	-3.03	4.12	-0.31	1.55	-1.87
Average	0.48	-0.64	1.12	-0.34	0.35	-0.69

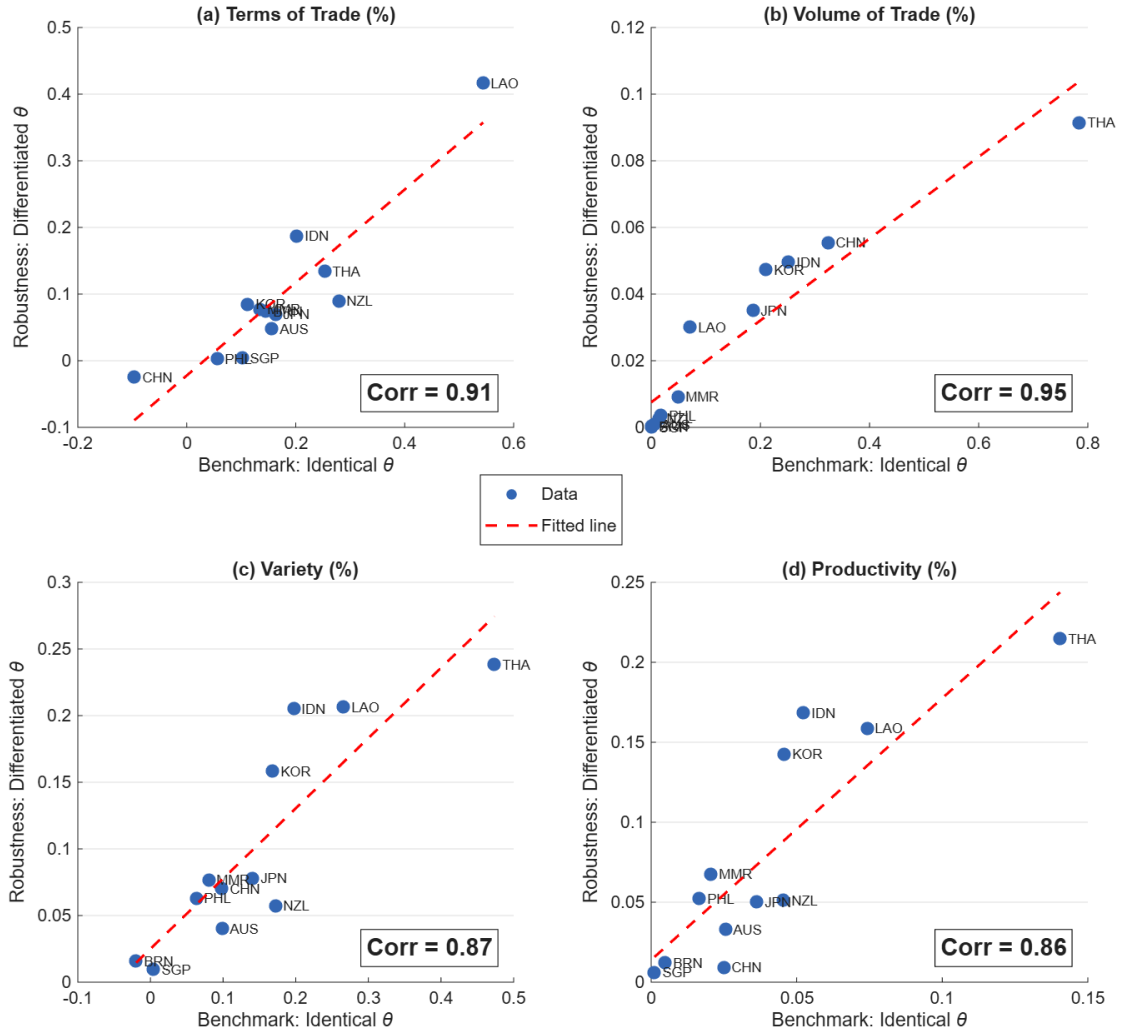


Figure C.2 Welfare effects between different models

Note: The horizontal axis represents welfare effects computed from the baseline model, where the shape parameter of productivity distribution θ is identical across countries, and the vertical axis represents those computed from the model with country-specific shape parameters. To mitigate the influence of outliers driven by the large tariff reductions of Malaysia, Vietnam, and Cambodia, these three countries are excluded from the figures. Both the horizontal and vertical axes are presented on a log scale as $\ln(1+x)$, where for the productivity effect, which is uniformly negative, the transformation is $\ln(1+|x|)$.

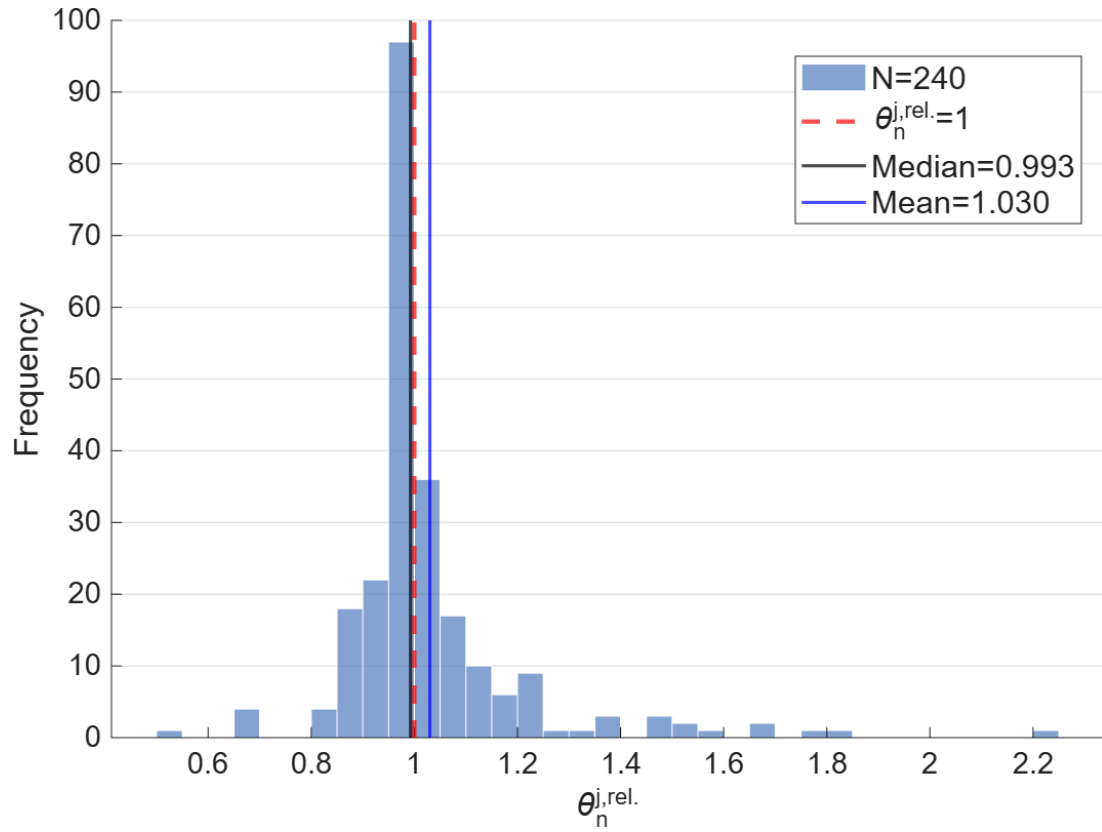


Figure C.3 Distribution of relative productivity heterogeneity of RCEP countries
Note: We measure the relative productivity heterogeneity between domestic and foreign firms by country and sector as $\theta_n^{j,rel.} \equiv \frac{\theta_n^j}{\sum_i \pi_{ni} \theta_i^j}$. The sample size is 240, including 15 RCEP countries and 16 manufacturing sectors.

C.3 Examining Factors Affecting Welfare Effects with Country-Specific Shape Parameters

Table C.7 reestimates the regression model (49) and (53) using welfare effects computed by extended model from Section 3.5, in which the shape parameter varies across countries and exhibits greater heterogeneity and variation. The estimated coefficients of interaction effects are therefore more reliable. $d \ln \widetilde{vot}_n^j$ and $d \ln \widetilde{prod}_n^j$ is the volume of trade and productivity effect under the extended model. The aggregated shape parameter follows the benchmark forms, that is, $\tilde{\theta}_n^j = \sum_{i \neq n}^N \pi_{ni}^j \theta_i^j$ and $\bar{\theta}_n^j = \frac{1 - \pi_{nn}^j}{\sum_i \pi_{ni}^j \theta_i^j}$. The results show that tariff reductions continue to increase trade volume but reduce productivity, and the interaction coefficients are consistent with the baseline regressions: a larger shape parameter of the productivity distribution amplifies the trade volume effect of tariff cuts and dampens the productivity effect.

Table C.7 Replacing with model allowing for country-specific shape parameters

	(1) $d \ln \widetilde{vot}_n^j$	(2) $d \ln \widetilde{vot}_n^j$	(3) $d \ln \widetilde{prod}_n^j$	(4) $d \ln \widetilde{prod}_n^j$
$d \ln (1 + t_n^j)$	-0.080*** (0.003)	-0.041*** (0.004)	0.219*** (0.011)	-0.035*** (0.012)
$\tilde{\theta}_n^j$		0.001 (0.000)		
$d \ln (1 + t_n^j) \times \tilde{\theta}_n^j$		-0.043*** (0.003)		
$\bar{\theta}_n^j$				-0.001 (0.000)
$d \ln (1 + t_n^j) \times \bar{\theta}_n^j$				0.955*** (0.037)
N	576	566	576	568
Adj. R^2	0.593	0.757	0.424	0.769

The sample consists of 31 countries and 19 sectors, with the top and bottom 1% of observations winsorized. Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The coefficient on $\tilde{\theta}_n^j$ is multiplied by 100.

D Figures and Tables

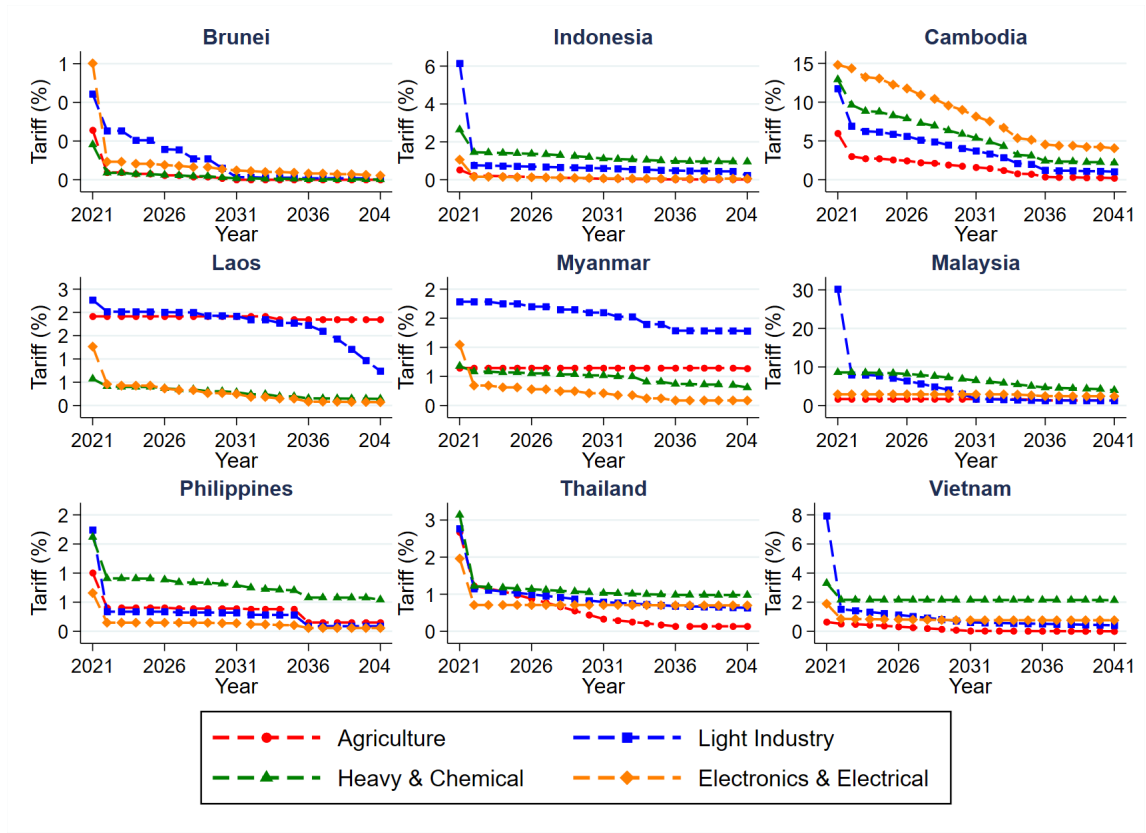


Figure D.1 Tariff reduction schedule committed by ASEAN members to China

Note: The baseline tariffs in 2021 come from Teti (2024) and the commitment levels of tariffs in following 20 years are based on the RCEP agreement text. Sectors are classified into four categories: Agriculture (ISIC 01-03), Light Industry (ISIC 10-18), Heavy & Chemical Industry (ISIC 05-09, 19-25, 28-30), and Electronics & Electrical Equipment (ISIC 26-27).

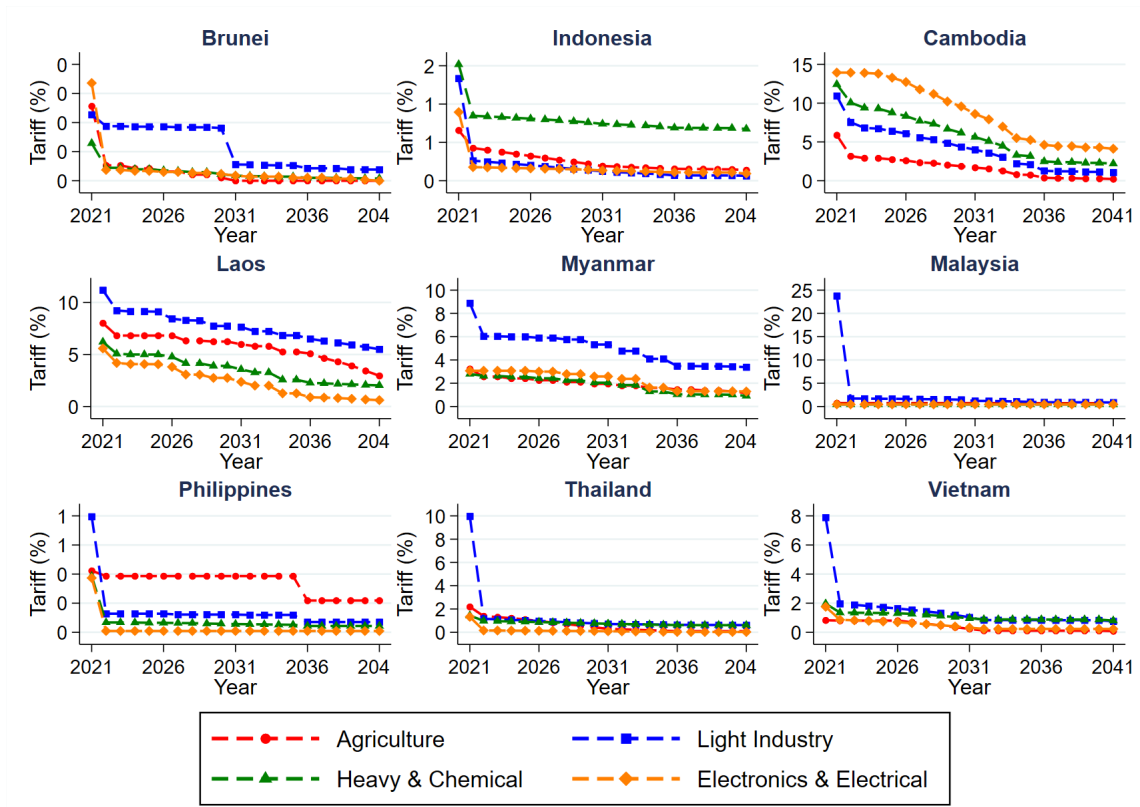


Figure D.2 Tariff reduction schedule committed by ASEAN members to Japan
Note: The baseline tariffs in 2021 come from Teti (2024) and the commitment levels of tariffs in following 20 years are based on the RCEP agreement text. Sectors are classified into four categories: Agriculture (ISIC 01-03), Light Industry (ISIC 10-18), Heavy & Chemical Industry (ISIC 05-09, 19-25, 28-30), and Electronics & Electrical Equipment (ISIC 26-27).

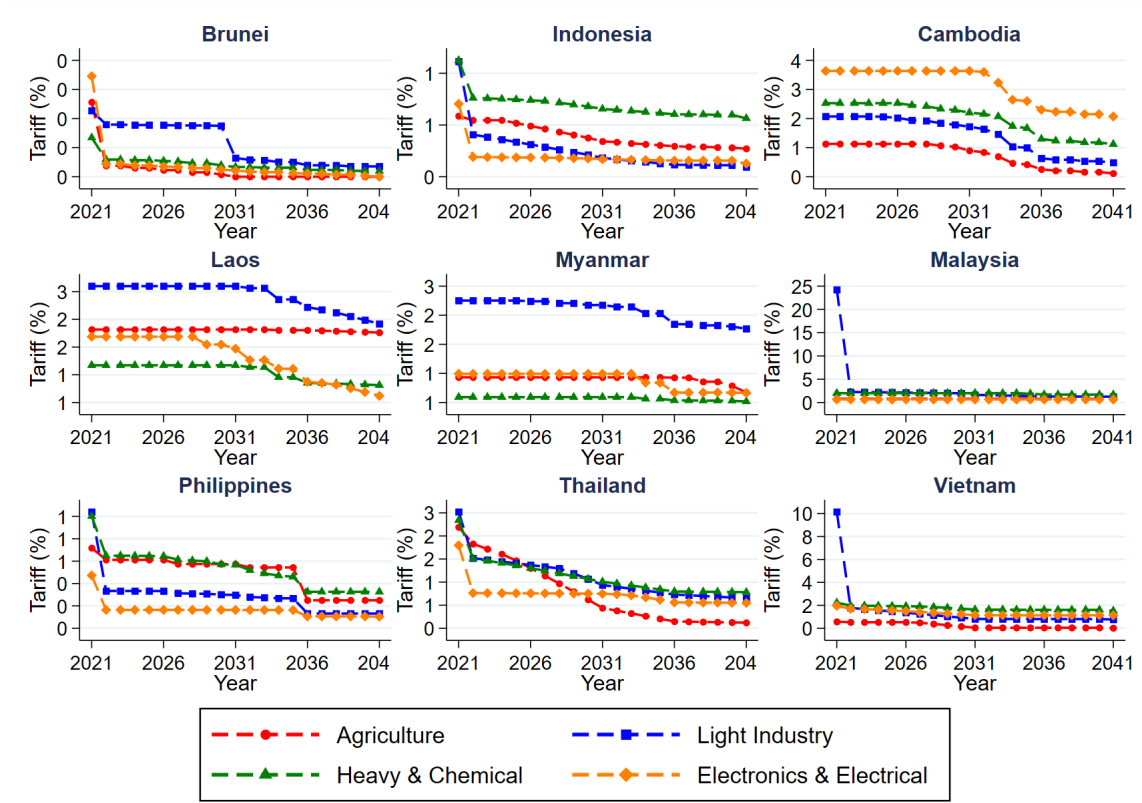


Figure D.3 Tariff reduction schedule committed by ASEAN members to South Korea
Note: The baseline tariffs in 2021 come from Teti (2024) and the commitment levels of tariffs in following 20 years are based on the RCEP agreement text. Sectors are classified into four categories: Agriculture (ISIC 01-03), Light Industry (ISIC 10-18), Heavy & Chemical Industry (ISIC 05-09, 19-25, 28-30), and Electronics & Electrical Equipment (ISIC 26-27).

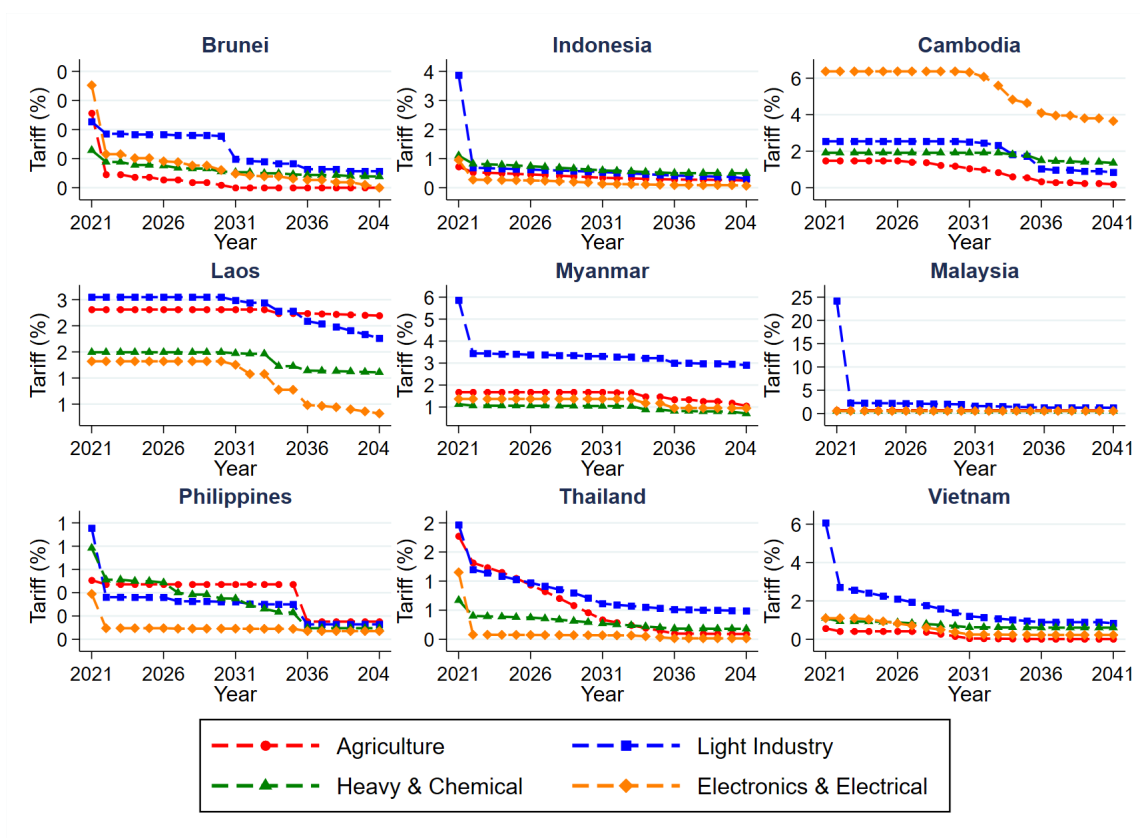


Figure D.4 Tariff reduction schedule committed by ASEAN members to Australia
Note: The baseline tariffs in 2021 come from Teti (2024) and the commitment levels of tariffs in following 20 years are based on the RCEP agreement text. Sectors are classified into four categories: Agriculture (ISIC 01-03), Light Industry (ISIC 10-18), Heavy & Chemical Industry (ISIC 05-09, 19-25, 28-30), and Electronics & Electrical Equipment (ISIC 26-27).

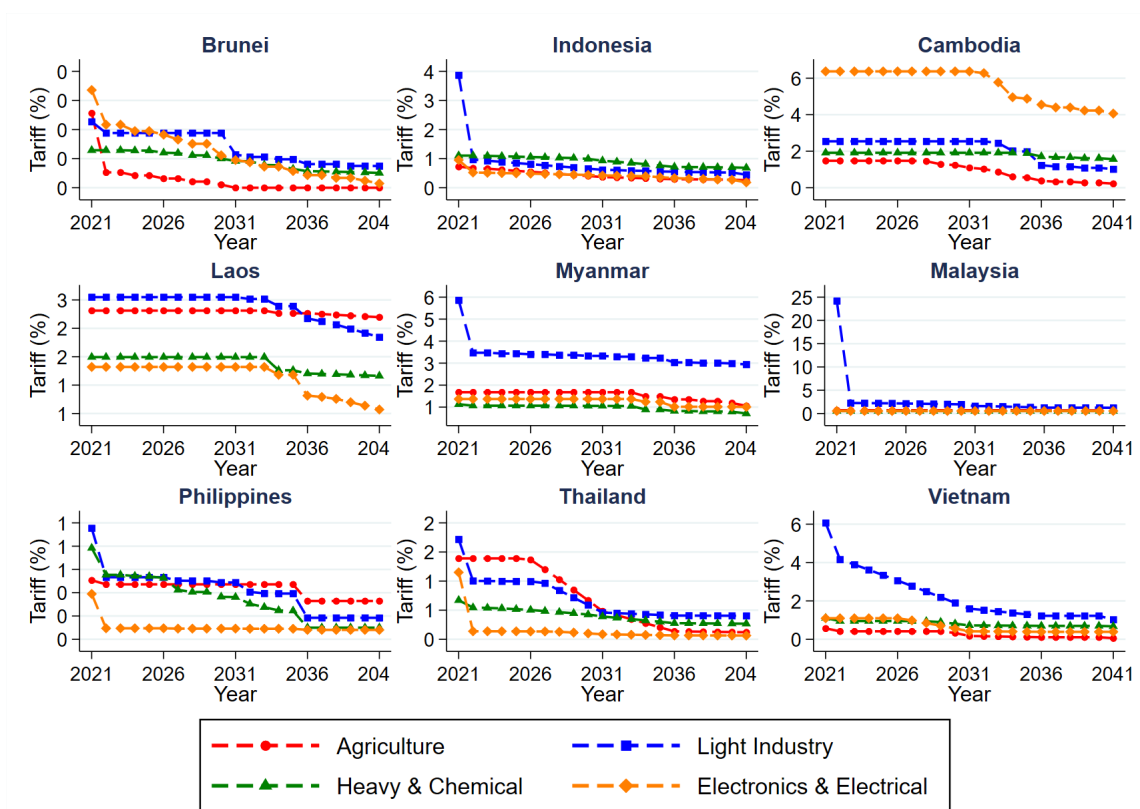


Figure D.5 Tariff reduction schedule committed by ASEAN members to New Zealand
Note: The baseline tariffs in 2021 come from Teti (2024) and the commitment levels of tariffs in following 20 years are based on the RCEP agreement text. Sectors are classified into four categories: Agriculture (ISIC 01-03), Light Industry (ISIC 10-18), Heavy & Chemical Industry (ISIC 05-09, 19-25, 28-30), and Electronics & Electrical Equipment (ISIC 26-27).

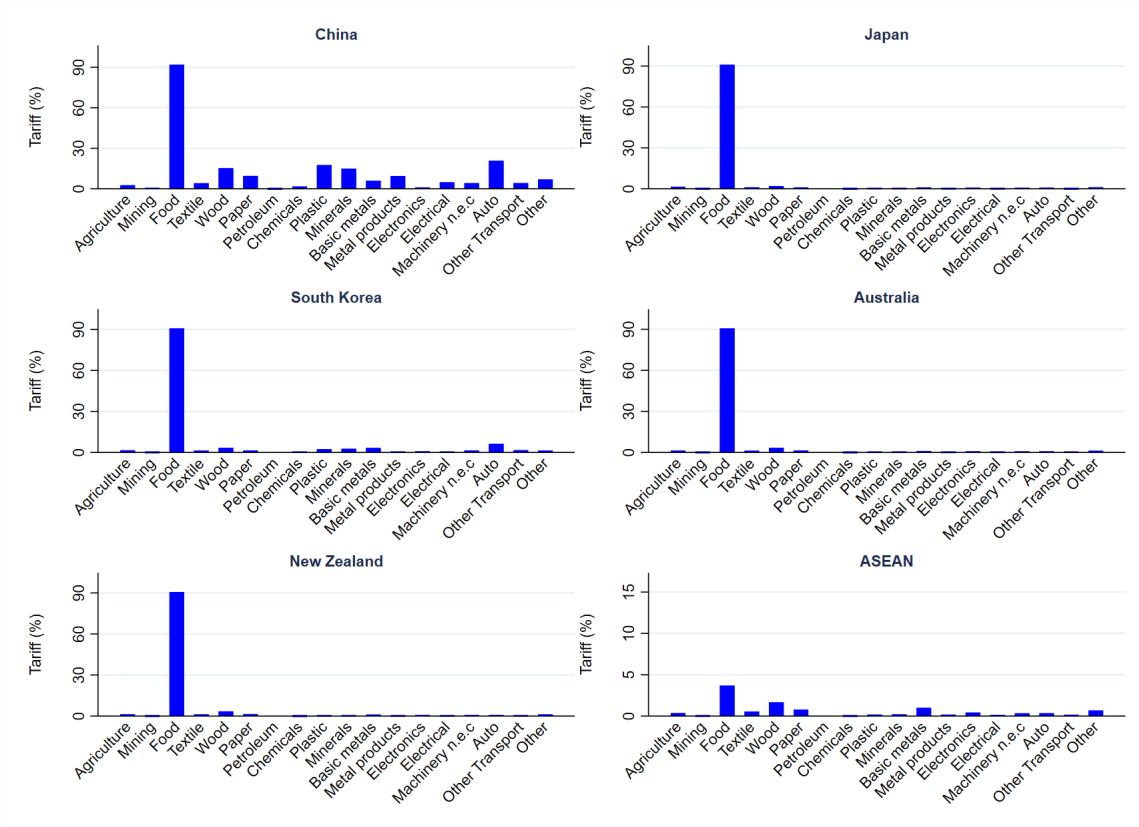


Figure D.6 Applied tariff rates Malaysia to members
Note: The baseline tariffs in 2021 come from Teti (2024).

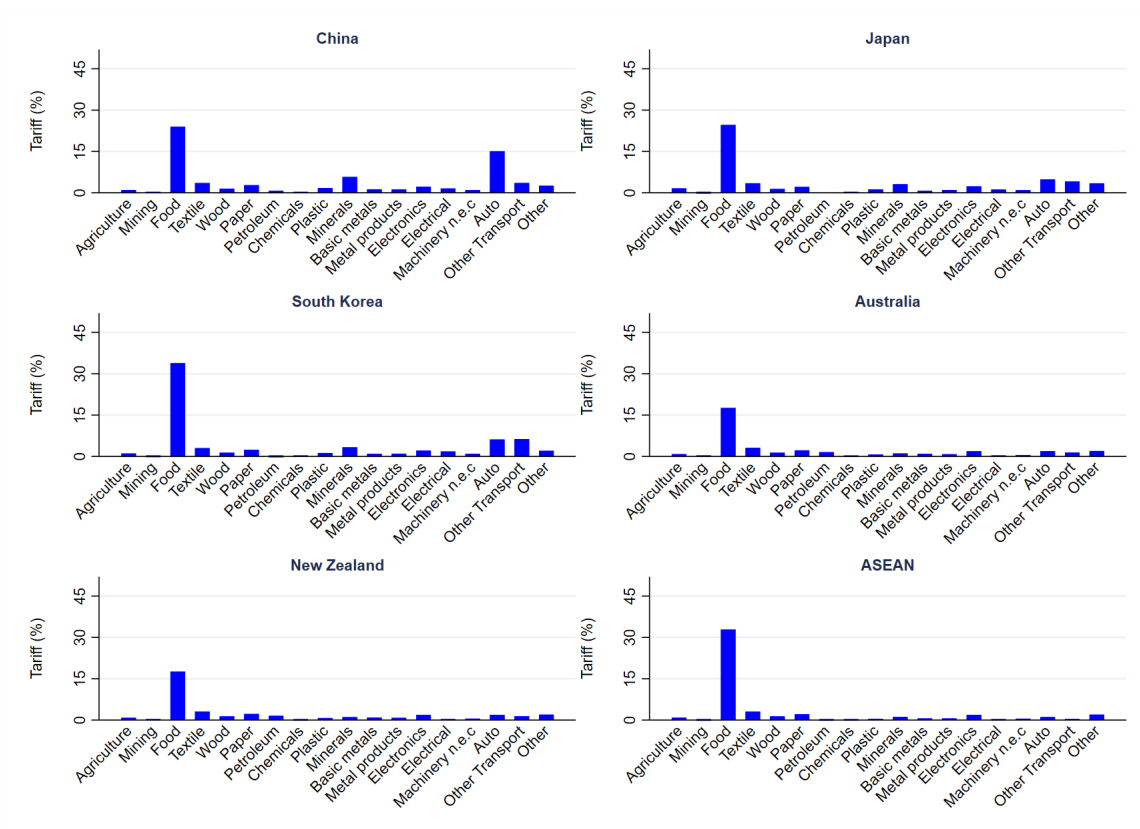


Figure D.7 Applied tariff rates Vietnam to members
Note: The baseline tariffs in 2021 come from Teti (2024).

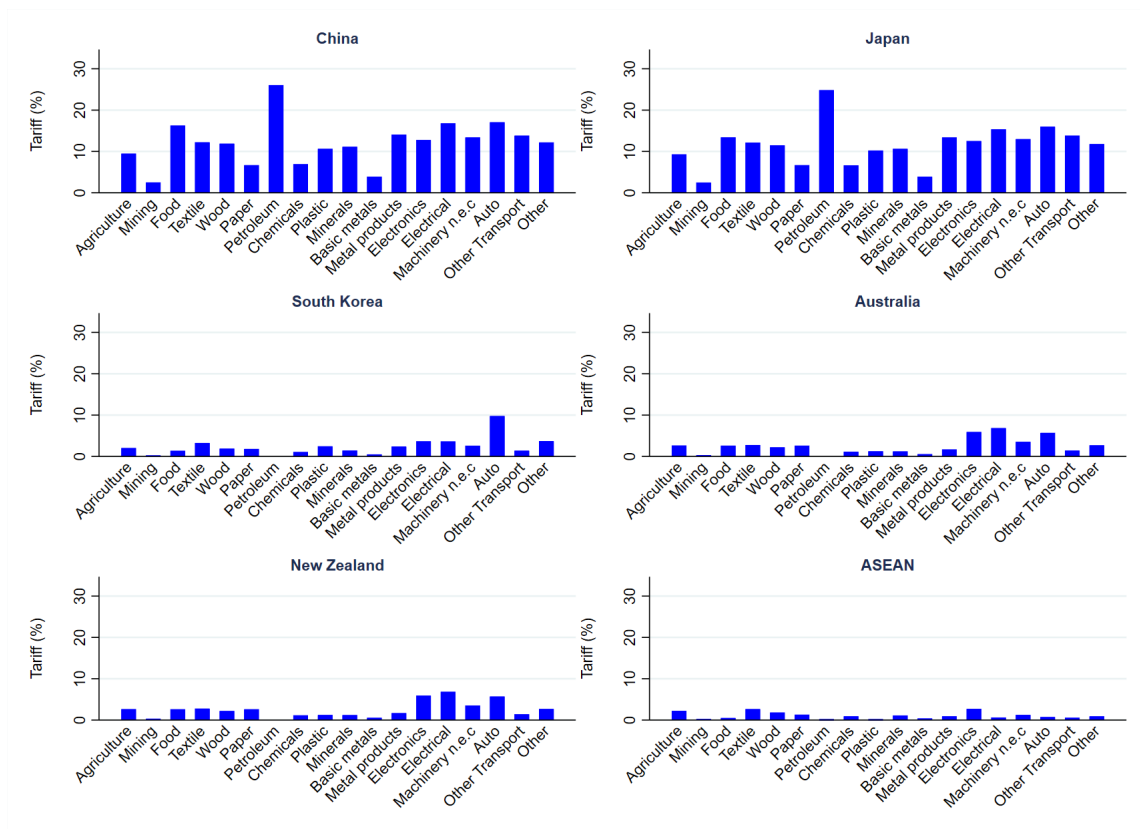


Figure D.8 Applied tariff rates Cambodia to members
Note: The baseline tariffs in 2021 come from Teti (2024).

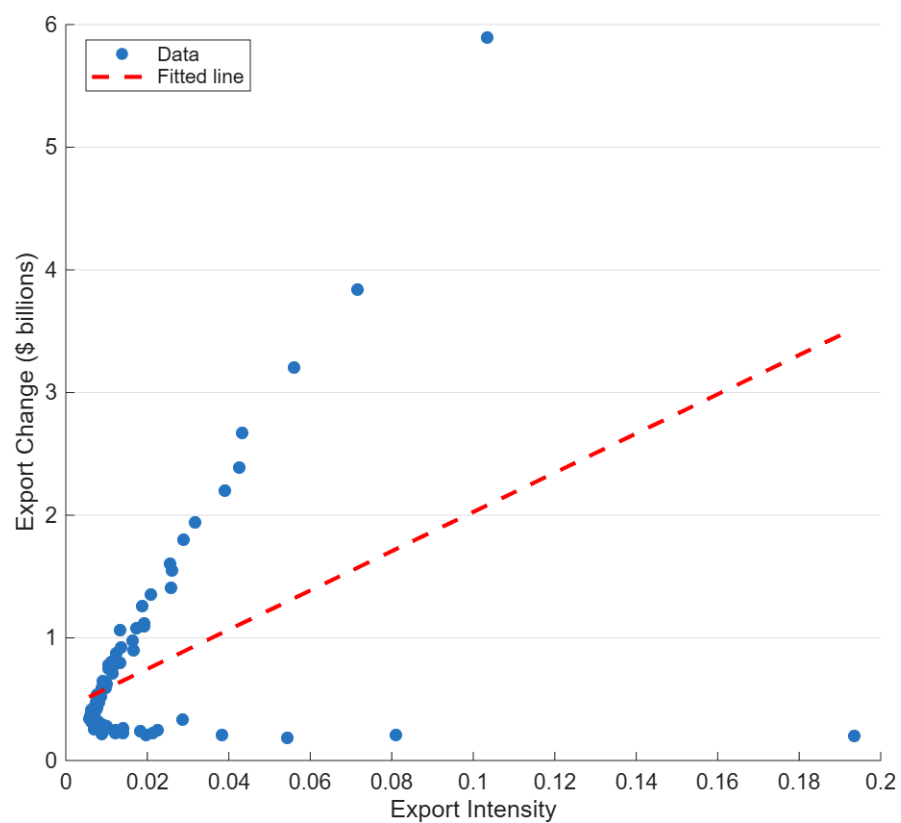


Figure D.9 Export intensity v.s export change
Note: Each dot represents a firm from the 1st to the 100th productivity percentile.

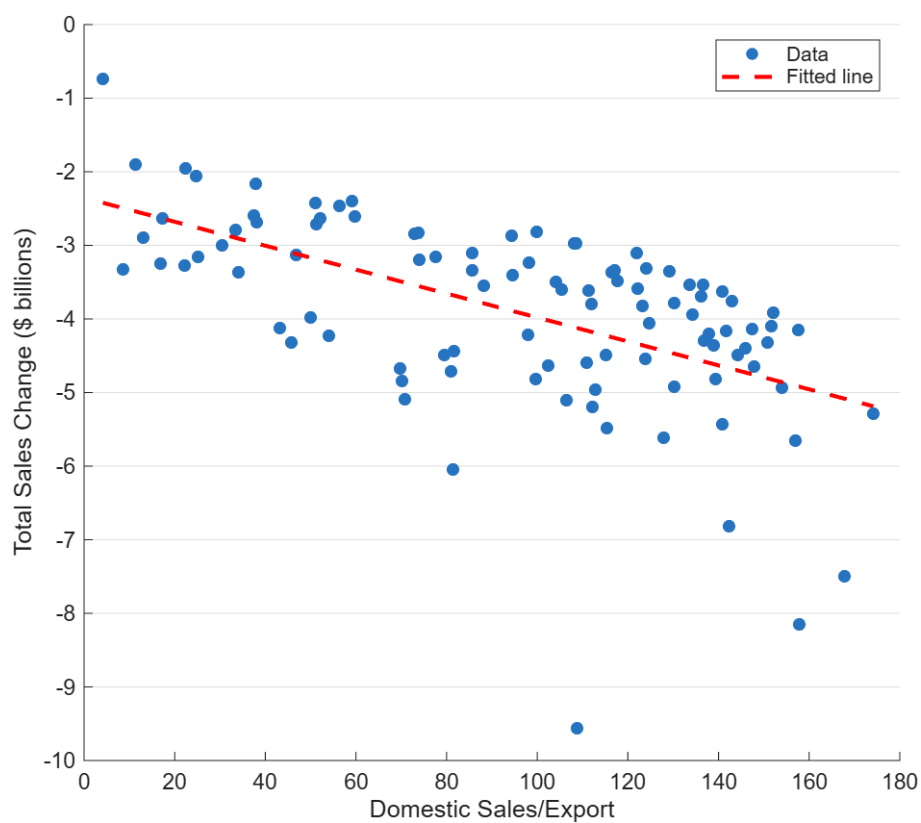
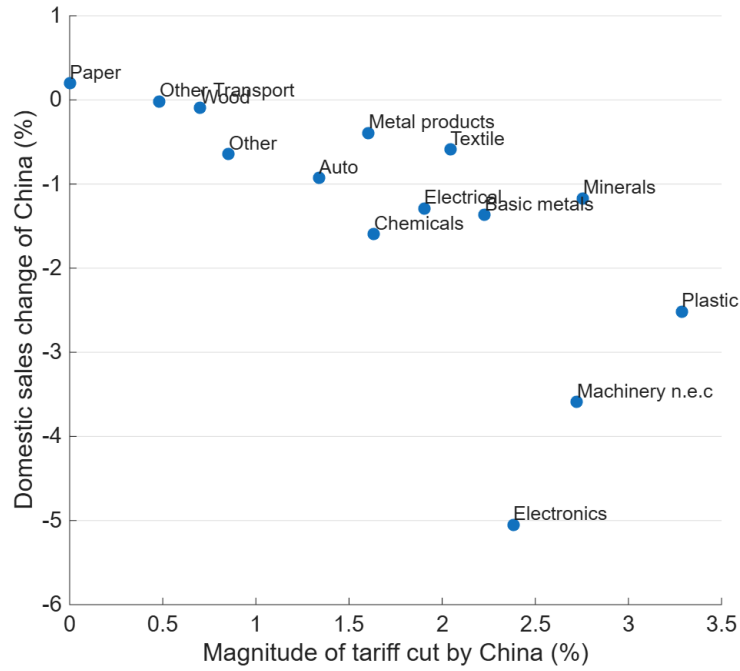
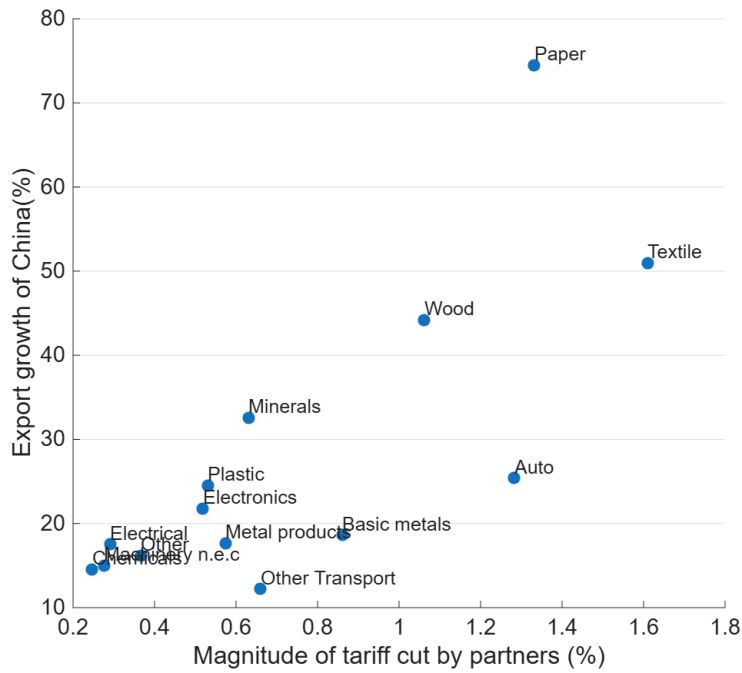


Figure D.10 Domestic sales-to-export ratio v.s total sales change
Note: Each dot represents a firm from the 1st to the 100th productivity percentile.



(a) Domestic sales change and tariff cuts by China



(b) Export change and tariff cuts by China's partners

Figure D.11 Tariff cuts and China's sales change by sector

Note: We exclude the extreme values of the Food sector. As importer, China's tariff for the Food sector is reduced by 9.75 percentage points, with domestic sales declining by 3%. As exporter, China's Food sector faces an average tariff reduction of 8.36 percentage points, with exports increasing by 179%. After RCEP took effect, Malaysia's tariff on food imports from China dropped from 91% in 2021 to zero, leading to a substantial increase in China's food exports.

Table D.1 Welfare effects from RCEP tariff reductions for non-RCEP countries (%)

Country	Welfare	“Traditional” gains			“New” gains		
		Total	Terms of trade	Volume of trade	Total	Variety	Produc- tivity
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
United States	0.007	0.007	0.003	0.004	-0.000	-0.000	0.000
Germany	0.002	0.005	0.000	0.005	-0.003	-0.004	0.001
India	0.006	0.022	-0.001	0.023	-0.016	-0.021	0.005
Russia	-0.005	-0.002	0.004	-0.006	-0.003	-0.004	0.001
United Kingdom	0.009	0.010	0.008	0.002	-0.001	-0.001	0.000
Brazil	-0.021	-0.011	-0.010	-0.001	-0.010	-0.013	0.003
Netherlands	0.008	0.013	0.001	0.012	-0.006	-0.007	0.002
Italy	0.000	0.004	-0.001	0.005	-0.003	-0.004	0.001
Canada	0.006	0.008	0.005	0.003	-0.002	-0.003	0.001
France	0.000	0.002	-0.000	0.002	-0.001	-0.002	0.000
Mexico	0.020	0.026	0.012	0.014	-0.006	-0.008	0.002
Spain	0.002	0.004	0.001	0.003	-0.002	-0.003	0.001
Turkey	0.000	0.004	-0.000	0.005	-0.004	-0.006	0.001
Saudi Arabia	0.007	0.010	0.007	0.003	-0.003	-0.005	0.001
South Africa	0.008	0.010	0.004	0.006	-0.002	-0.003	0.001
Average	0.003	0.008	0.002	0.005	-0.004	-0.006	0.001

Table D.2 Decomposition of “new” welfare effects for non-RCEP countries (%)

Country	Variety			Productivity		
	Total	Domestic exit	Foreign entry	Total	Domestic exit	Foreign entry
	(1)	(2)	(3)	(4)	(5)	(6)
United States	-0.000	-0.009	0.008	0.000	0.002	-0.002
Germany	-0.004	0.008	-0.012	0.001	-0.002	0.003
India	-0.021	-0.034	0.013	0.005	0.008	-0.003
Russia	-0.004	-0.005	0.001	0.001	0.001	-0.000
United Kingdom	-0.001	-0.012	0.011	0.000	0.003	-0.003
Brazil	-0.013	0.021	-0.034	0.003	-0.005	0.008
Netherlands	-0.007	0.006	-0.013	0.002	-0.001	0.003
Italy	-0.004	-0.001	-0.004	0.001	0.000	0.001
Canada	-0.003	-0.008	0.005	0.001	0.002	-0.001
France	-0.002	0.004	-0.006	0.000	-0.001	0.001
Mexico	-0.008	-0.015	0.007	0.002	0.004	-0.002
Spain	-0.003	-0.003	-0.000	0.001	0.001	0.000
Turkey	-0.006	-0.009	0.003	0.001	0.002	-0.001
Saudi Arabia	-0.005	-0.019	0.015	0.001	0.005	-0.004
South Africa	-0.003	-0.003	0.001	0.001	0.001	-0.000
Average	-0.006	-0.005	-0.000	0.001	0.001	-0.000